106.1 A fixed regular domain is available in the original map

Write $\mathcal{U}_j(X;\alpha_0,\alpha_1)$ for the original paired domain of I (1.11)–(1.16). The second entry is an independent current in the definition of this space. When a pair is explicitly written as

$$(U_j(D_0,e^{iB}),J_j(D_0,e^{iB})),\tag{RG1}$$

the second entry is the **actual Wilson current of the displayed minimum**. This distinction is used below.

The original I Lemma 4, (3.36)–(3.53), is an admission theorem for RG1. The outer configuration belongs to an enlarged current-scale core with constants $(1+2\beta)\alpha_0,(1+2\beta)\alpha_1$; the relative field A satisfies precisely I (3.31); and an additional determining-field displacement lies in the fixed disc $|B'|<\alpha_3$. The resulting old pair belongs to $\mathcal{U}_j(X;\alpha_0,\alpha_1)$. The proof explicitly uses $(1+7\beta)L^{-2}\le 1$, together with restrictions such as

$$B_3\alpha_3<\tfrac{1}{2}\alpha_1,\qquad 2B_3\alpha_3\le\beta L^{-2}\alpha_0.\tag{RG2}$$

This is not the unsupported inference $C_L L^{-2} < 1$ excluded in DA. The coefficient close to one and the surplus input core are obtained in the original argument.

The outer fluctuation response in II (1.17) obeys, on its whole source-wall polydisc,

$$\|H_{k,s}(B')\|_{\mathrm{I}(3.31)} \le 4B_0C_1e^{16\kappa_1}g_k\|B\|_\infty. \tag{RG3}$$

II (1.21)–(1.22) chooses the original physical fluctuation radius ϵ_1 so that this quantity, after multiplication by the prescribed smooth cutoffs and the distinguished Cauchy variable, lies in I (3.31). In particular one retains the actual restriction

$$4B_0C_1e^{16\kappa_1}\epsilon_1 \le \tfrac{1}{2}\alpha_2. \tag{RG4}$$

The important order is visible at the beginning of II: the function $E(D_0, U, J, A)$, containing the inner minimum, is already local in D_0; only the outer expression inserted for A is weakened in this step. Therefore the inner function RG1 remains a genuine physical minimum on every one of these source walls. It is not replaced by an arbitrary random-walk-weakened critical map.

These statements apply by evaluation to **any** analytic carrying function on the original old paired core. No special form of an unmarked effective action is used in that evaluation. They settle the regular core inclusion for this literal completed source slot. They do not settle a different composition which inserts an additional reconstruction or changes the boundary problem.

106.1.1 Joint source and unmarked walls

There is an exact way to avoid an artificial joint-domain problem. Perform the source localization in the original real fluctuation coordinates B. Retain its coefficient functions of (U, J, B). Subsequently localize the Gaussian and the unmarked vertices in those same coordinates. The later walls multiply or alter kernel factors; they do not substitute another value of B into a source coefficient already constructed.

The resulting product is analytic on the product of the original source-wall and kernel-wall discs. Source admission is RG1–RG4; the Gaussian comparison and its one remaining quadratic reserve are the original II bounds used in S9. There is no inference from separate analyticity for a composition: there is no extra composition in this order.

If a later operation does change fluctuation coordinates, this argument no longer applies as stated. Its exact pullback and the joint range of the substituted source must be checked. The corresponding cutoffs and determinant factors remain in the same original Wilson integral.

106.2 Why the physical norm alone is not an invariant incoming norm

Let $E_{j,X}$ be an actual paired coefficient, with its full anchor vector retained. MT40 controls its physical pullback and physical quadratic/cubic kernels. It need not control the same function on independently varied (U, J).

For example, in a finite-dimensional admissible chart consider

$$N_g(U, J) = g^{-3}\langle J - J_{\mathrm{phys}}(U), J - J_{\mathrm{phys}}(U)\rangle. \tag{RG5}$$

Its physical pullback and all physical field jets vanish. Its full fixed-core bound is of the allowed order g^{-3}. On a canonical independent pair $U = e^{gA}, J = gj$, however, it has size g^{-1} for generic (A, j). Thus a physical version of MT40 cannot be used on a far-tail readout such as the independently extended pair in I (3.13).

The needed additional component is a canonical **paired** norm. On a fixed smaller flat paired chart $\mathcal{B}$, retain

$$\begin{aligned} b^{\mathrm{ph}}_{j,X} &:= (g_j/\alpha)^d \sup_{\mathcal{U}_j(X;\alpha_0,\alpha_1)} \|E_{j,X}(U,J)\|_{\ell^1(\text{anchors})}, \\ b^{\mathrm{can}}_{j,X} &:= \sup_{(A,j')\in\mathcal{B}} \|E_{j,X}(e^{g_j A}, g_j j')\|_{\ell^1(\text{anchors})}, \qquad d \in \{2,3\}. \end{aligned} \tag{RG6}$$

Here the exponential includes the correct bond-length factor; A is measured in the original own-unit first-order chart. The number α is a fixed positive radius comparable to the chosen clean paired constants. The chart includes every determining datum, and its reference and boundary data are flat when a canonical estimate is taken. No norm at a generic nonflat frozen exterior is being called a canonical norm.

The canonical ball is a Banach ball of field configurations. It is not a coefficient ℓ^1 norm over microscopic bonds. Directional Cauchy in this ball has constants independent of the dimension of that microscopic space.

All zeroth, first, lower, current and contact coordinates are retained separately. Their inclusion in RG6 does not authorize discarding them in a physical quotient. In a higher finite-source degree one must use the higher grade described in Section 9.

106.3 An elementary two-radius envelope

This is the analytic part of the new norm which can be proved without any diagrammatic assertion. Let F be Banach-valued and holomorphic on a fixed physical ball of radius α. Suppose

$$\|F\|_\alpha \le A(\alpha/g)^d, \qquad \sup_{\|z\|\le c_0} \|F(gz)\| \le A, \qquad d = 2 \text{ or } 3. \tag{RG7}$$

Let P_m be its ordinary homogeneous Taylor polynomial of degree m. Directional Cauchy on the two balls, with fixed smaller radii, gives

$$\|P_m(z)\| \le C_m A \min\{g^{-m}, \alpha^{d-m} g^{-d}\} \, \|z\|^m, \quad 0 \le m \le 3. \tag{RG8}$$

The bound for the Taylor remainder is

$$\|F(z) - \sum_{m=0}^{3} P_m(z)\| \le C A \alpha^{d-4} g^{-d} \|z\|^4, \qquad \|z\| \le \alpha/2. \tag{RG9}$$

These statements hold equally in the anchor ℓ^1 space. They do not require a microscopic coefficient count.

Suppose a literal source readout $\Psi_{j,k,s}$ lies in the original physical ball and has the canonical range estimate

$$\|\Psi_{j,k,s}(g_kZ, g_kB)\|_{\mathrm{pair}} \le Cg_kq_Y, \qquad q_Y \le C(1+\mathscr{E}_Y(B))^{1/2}. \tag{RG10}$$

Then, writing $\rho_{j,k} = g_k/g_j$, RG8–RG9 give

$$\|E_{j,X}\circ\Psi_{j,k,s}\| \le CA_{j,X}\left\{\sum_{m=0}^{3}\rho_{j,k}^m q_Y^m + \alpha^{d-4}g_k^{4-d}\rho_{j,k}^d q_Y^4\right\}. \tag{RG11}$$

The fourth-order term is $O(g_k)$ for $d=3$, and $O(g_k^2)$ for $d=2$, apart from a relative coupling ratio. This calculation is valid on the actual physical disc first; the energy polynomial is substituted afterward. It is the same favorable cancellation as MT44, now also available for an originally paired tail map.

106.3.1 Verification of the range estimate in a regular slot

For the inner completed minimum RG1, the common physical disc follows from Lemma 4. At the fully flat determining data and zero fluctuation the minimum is the identity and its physical current is zero. The holomorphic response bounds of the original variational construction, or radial Cauchy on a strictly smaller common disc, therefore give RG10 when all active determining data are scaled together. The same statement holds on the full source-wall disc because RG3 supplies the same admitted A-class there.

For the far paired extension I (3.13), its current is not identified with the current of its first argument. Its original analytic pair range, with the same flat zero and the original $A, \nabla A, D^*DA, \Delta A$ bounds, gives RG10 as a **paired** bound. This is where RG6 is necessary. A primitive physical kernel norm alone would not suffice by RG5.

The constants are fixed by the original clean core and localization parameters. They are independent of the ultraviolet cutoff and of the number of old microscopic bonds.

106.5 Ward kernels, symmetries, and the torus qualification

Each original paired coefficient under consideration is gauge invariant, and the genuine minimum is gauge covariant. Therefore the complete anchored function RG12 is gauge invariant. Its global quadratic Ward kernel is closed in each potential leg. After subtracting the whole declared quadratic covariant completion, the cubic Ward kernel has the homogeneous closedness needed in its three legs.

One may apply the coarse cochain homotopy to these **completed anchored kernels**, using RG16–RG17 for the coefficient sums. The resulting path lengths and fixed moments cost polynomial factors only. This does not require asserting gauge invariance of each auxiliary wall coefficient of a subsequently localized physical pullback. A Ward identity is used after the relevant coefficient family has been completed.

On a torus, the cochain homotopy has a harmonic or winding part. It must be retained. A divergence-free exponentially localized kernel can be decomposed into a curl part and representatives of fundamental cycles. The flux determining a fundamental-cycle coefficient can be read through a cut placed a fixed fraction of the torus diameter away from the source root. RG16 bounds this flux exponentially in that distance, with the finite cut-size polynomial included. This is an admitted tail term, not zero. The construction on contractible protected carriers supplies the local quadratic/cubic kernels; winding members stay in the original tail representation.

106.5.1 Completing lattice covariance without changing the insertion

A universal pure-clean marked functional is covariant under the lattice symmetries preserving the actual torus and hierarchy. Its localized representation can be completed equivariantly. If $F = \sum_X F_X$, define for a finite symmetry group G

$$F_X^{\mathrm{sym}}(U; f) = |G|^{-1} \sum_{r \in G} F_{r^{-1}X}(r^{-1}U; r^{-1}f). \tag{RG18}$$

The total is exactly F. The rooted norm, with its supremum over roots, does not increase under this average. On a finite torus the full translation average is harmless: its normalization cancels the number of translations. If the original coefficient system is periodic with a fixed localization-cell period, averaging over the finite cell offsets gives the same completion.

Reflection acts on the carrier, tensor labels, orientation, chosen paths and the original anchor offset. Those data cannot be held fixed while only one numerical coefficient is averaged. After RG18, the shifts used in CD are exact reindexings of the full coarse-root sum. Smooth current-scale filing uses $f\chi_Q$ with $\sum_Q \chi_Q = 1$; a sharp truncation of the root sum is not substituted.

A selected large-field history, or a coefficient with a prescribed nonsymmetric exterior, does not satisfy this universal covariance statement. Averaging would then change the object unless its entire transformed history and exterior family were carried with it.

106.6 Quartic tails need only four local coarse derivatives

For the same completed physical pullback subtract its full lower packet, its whole quadratic covariant completion, and its whole cubic completion. Denote the remaining function by $R_{\geq 4}$. It is gauge invariant and has vanishing ordinary physical jets through degree three.

An integral Taylor remainder avoids a potential factorial loss from an infinite coefficient expansion:

$$R_{\geq 4}(v) = \frac{1}{3!} \int_0^1 (1-t)^3 D^4 R_{\geq 4}(tv)[v, v, v, v], dt. \tag{RG19}$$

Take a current regular chart on a protected neighborhood of a source root. Let its physical potential/curvature regularity size be q, and let $r = L^{j-k}$. A root gauge, continued with a fixed current-scale cutoff, gives the following useful form of the old coarse-potential bound:

$$|v(h)| \leq Cr^2 q(1 + \operatorname{diam} X + \operatorname{dist}(h, X)), \qquad \|v\|_\infty \leq Crq. \tag{RG20}$$

Near the root this is the root-gauge estimate DA together with the actual centered-coarse comparison CB. Outside the fixed protected neighborhood, the weaker global potential bound is enough: the old coarse distance there is at least a constant times r^{-1}. The formula is used only where such a valid global small potential bound or the retained original tail representation is available.

The segment tv must stay in the same coarse polydisc. RG16 at order four, RG20, and four fixed polynomial moments give the following bound after the extra $(1 + \operatorname{diam} X)^4$ factor has been paid from carrier locality:

$$|R_{\geq 4, j \to k}| \leq C A_j r^2 \alpha^{d-4} g_j^{-d} q^4. \tag{RG21}$$

The factor is $r^8 r^{-4} r^{-2} = r^2$: four fields, the old-root count, and the D6 normalization. On a canonical current readout $q = g_k q_Y$, it is

$$C A_j r^2 \alpha^{d-4} g_k^{4-d} (g_k/g_j)^d q_Y^4. \tag{RG22}$$

Only four derivative kernels have been used. No assumption that an arbitrary background has spatial derivatives of every order has entered. The locality constant in RG16 also controls long coarse tails after a valid global argument has been retained; it never permits evaluation outside the analytic domain.

For boundary or mixed-history data lacking the global argument in RG20, retain the original admitted tail or joint-operator representation. The present argument does not replace it by a periodic minimum.

106.7 Formation preserves the two amplitude grades at a regular step

Use the exact clean recurrence XP40 and the deterministic/proper split HI9–HI10. For one incoming scalar mark let $D_s(B) = F_s(B) - F_s(0)$. Its zero at $B = 0$ holds for every source wall. On the common original physical disc, Schwarz therefore gives

$$\|D_s(B)\|_{\ell^1} \le C \, \frac{g_k \|B\|_\infty}{\epsilon_1} A_j (\alpha/g_j)^d. \tag{RG23}$$

At a fixed canonical external pair, the stronger polynomial envelope is RG11. It applies also to $F_s(0)$. Use the finite-mark moment estimate on the same conditioned Gaussian, with its auxiliary exterior cancellation performed before absolute norms. One remaining positive quadratic reserve absorbs the **total** polynomial degree. The actual formed coefficient consequently has both a full paired-core bound and a canonical paired bound:

$$\begin{aligned} (g_{k+1}/\alpha)^d \|E_{k+1}^{\mathrm{new}}\|_{\mathrm{ph}} &\le C \frac{g_k}{\epsilon_1} (g_{k+1}/g_j)^d A_j, \\ \|E_{k+1}^{\mathrm{new}}\|_{\mathrm{can}} &\le C A_j \left[\sum_{m=0}^{3} (g_k/g_j)^m + \alpha^{d-4} g_k^{4-d} (g_k/g_j)^d \right]. \end{aligned} \tag{RG24}$$

The fixed one-step ratios between the canonical external normalization and g_k are included explicitly in the constant or retained as relative factors. RG24 is a **one-step norm estimate**, not by itself a uniform all-depth bound. The deterministic old tag is not counted as a new birth. Its direct age estimate is kept separately.

The output is the actual original paired coefficient because all source factors were carried in the unchanged original integration coordinates and the source wall expansion has its correct all-one endpoint. It remains analytic on the same original output paired core; Section 1 supplies the incoming inclusion from the enlarged outer core. This is the concrete reason a new physical radius need not be multiplied by a loss factor at every clean birth.

The finite projection into lower, D6 and stable coordinates is bounded in this enlarged norm. Physical kernels use RG17; compact completions use both sides of that minimum. A canonical completion does not replace the original full paired-core bound by a larger fictitious g_j^{-3} bound for an H label. Every complement is kept.

106.7.1 A physical compact completion has a bounded original-pair extension

This extension can be specified directly. Let $\mathcal{O}_K(v)$ be the declared coarse curvature/path completion of a kernel. Define

$$\mathcal{C}_K(U,J) := \mathcal{O}_K(C_jU), \tag{RG24a}$$

where C_j is the **actual own-coarse raw determining map**, with its complete prefix paths, endpoint convention and read set. This expression ignores the independent J. On the genuine minimum, target preservation gives $C_jP_j(v) = e^v$, up to the prescribed exact gauge, so its physical pullback has precisely the required completion jets.

For a uniform clean hierarchy, the original paired domain and the raw-mean estimates bound the own-coarse plaquette flux by $C(\alpha_0+\alpha_1)$. On the flat canonical pair the bound is Cg_j. Differences at this coarse spacing cost only fixed constants; no microscopic derivative of curvature is taken. A path of length D has matrix norm at most $e^{C\alpha_1 D}$, relative to its unitary reference transport. Choose the fixed complex radius small enough that this cost fits within a declared carrier-locality margin.

If the degree-m kernel obeys RG17, its full paired completion has bound

$$C\alpha^m A_j \alpha^{d-m} g_j^{-d} = CA_j\alpha^d g_j^{-d}, \tag{RG24b}$$

while its canonical bound follows from $g_j^m\|K_m\| \le CA_j$. The whole quadratic completion, including its cubic terms, is treated this way before the cubic Ward kernel is extracted. The physical and paired remainder are both retained after subtraction. Thus completion does not force another source reconstruction or a new analytic-radius loss.

This argument requires that the chosen C_j is defined on the complete carrier in the actual hierarchy. A mixed boundary mean with missing prefix rows is not substituted for it. Noncontractible path members retain their winding/tail filing.

106.8 Locality is recycled by actual coarsening

It is insufficient to spend a fixed exponential-locality margin every time a marked coefficient is formed. The original covering inequality S10 provides the relevant recycling mechanism. Let the output carrier size satisfy

$$d_c(X) \le C_{\mathrm{cov}}\left(n + L^{-1}\sum_{\nu=1}^{n} d_f(Y_\nu) + |P| + h\right). \tag{RG25}$$

The selected-vertex count is n, the failed tests are P, and h is the wall count. Suppose the absolute local term has the S9 envelope with $e^{C_0 d_c(X)}$, input factors $a_Y e^{\mu_0 d_f(Y)}$, failed-test factor p_*, and wall factor $(R_c-1)^{-h}$.

To output the same exponent μ, it is sufficient to retain a positive margin δ in

$$\mu_0 + C_{\mathrm{cov}}(\mu + C_0)/L + \delta < \mu. \tag{RG26}$$

Indeed multiply the local bound by $e^{\mu d_c(X)}$ and use RG25. Each input factor is then weighted by at most

$$a_Y \exp\{[\mu_0 + C_{\mathrm{cov}}(\mu + C_0)/L] d_f(Y)\},$$

while each selected vertex, failed test and wall receives only the fixed extra factor $e^{C_{\mathrm{cov}}(\mu+C_0)}$. The original small activity and wall-radius choices absorb those factors and the connected-series entropy. The spare δ pays the finite pointing, anchor and coarse-Cauchy moment powers.

Thus the output exponent can return to μ; it is not $\mu - \delta$ after every age. One must check RG26 with the actual covering constants and parameter choice. If a representation adds a collar, its complete collar enters RG25; a fixed collar per selected vertex changes the vertex factor, while a carrier-size-dependent collar changes the covering inequality itself.

The coarse derivative argument in Section 4 may use a smaller auxiliary exponent ν. It is rederived from the paired μ-norm at every birth and is used for a fixed number of moments; it is not itself repeatedly decremented along the RG trajectory.

For the fixed current-packet filing of Sections 13–15, the geometric slope can be separated from its extra collar cost. Traverse a fine spanning tree of length N, with each edge traversed at most twice, and split the resulting walk into pieces of length at most L. Each piece meets at most a dimension-dependent number of coarse cubes. Its coarse image is connected, so the coarse tree length is at most $C_{\mathrm{dim}}(1 + N/L)$, with C_{dim} independent of the microscopic cutoff and of the later fixed packet radius. A current Q^H carrier or its CE carrier adds at most C_H cubes **per marked vertex**. It does not replace C_{dim}/L by C_H/L on the old tree length. Long carriers use their existing original complete carrier instead of being padded anew around every old cell. Thus this filing has

$$d_c(X) \le C_{\mathrm{dim}} \left(n + L^{-1} \sum_Y d_f(Y) + |P| + h \right) + C_H n_{\mathrm{marked}}. \tag{RG26a}$$

Choose the odd blocking factor once with $L > 2C_{\mathrm{dim}}$. Then choose the common exponent μ so that $\mu_0 + C_{\mathrm{dim}}(\mu + C_0)/L + \delta < \mu$. The extra factor $e^{(\mu+C_0)C_H}$ per marked vertex is fixed; it changes the marked formation constant and the required weak ceiling, not the incoming tree slope. The original wall-radius and Gaussian choices can be made afterward as in marked-bridges Section 5.3. This verifies recycling for this specific clean filing. A different carrier enlargement, particularly one made along a live boundary, needs its own inequality.

106.10 The free companion must be retained before claiming an extra coupling

An $O(1)$ canonical paired norm alone does not imply that a coefficient has an $O(g)$ free comparison. For example the version of RG5 with coefficient g^{-2} has an $O(1)$ paired canonical limit and vanishes physically. An arbitrary off-graph localization could display nonzero free terms which cancel only when its full physical coefficient is completed.

For the actual primitive minimum source the leading canonical readout is a polynomial in the free physical coarse field. The independent canonical current enters the original propagators through a physical factor g; consequently it does not enter that leading primitive polynomial. At a regular step the flat Gaussian and the leading localized response are independent of the independent canonical current. This property passes to the completed scalar free companion under the same fixed-coordinate Gaussian integration. Any explicit differential-current/contact rows which do have a free current dependence must instead be kept as exact triangular rows.

For a polynomial of ordinary field degree at most three and a centered Gaussian fluctuation,

$$\mathbb{E}_0[P(MA+B)] - P(MA) \tag{RG29}$$

has ordinary field degree at most one. Thus its full D6 projection is zero after its vacuum and lower packet is retained. This statement is about the actual free mean/covariance tower MT49–MT53, including the exact lower quadratic family. It does not replace that tower by a nearest-neighbor Gaussian.

A useful stronger norm therefore stores the exact free companion and a canonical discrepancy, for example

$$\|E_j(g_j A, g_j j') - E_j^{(0)}(A, j')\|_{\mathrm{can}} \tag{RG30}$$

with its declared coupling/tail weight. To propagate an $O(g)$ discrepancy, use IV's actual $O(gB)$ inherited vertex, the fresh nonlinear response estimate, and the Gaussian score MT31 on the same

cutoff. The physical remainder contributes g^{4-d}, which is at least g for $d \le 3$. No free current-null coefficient is simply discarded on an independently weakened wall.

This proves the structural free cancellation used in a regular center comparison. A claimed $O(g)$ birth estimate must specify whether it concerns a proper Gaussian birth after the full lower packet, or a stable remainder produced by refiling a deterministic free tag. Those are different operations.

106.10.1 The cancellation can be checked on each original source wall

At the flat canonical reference, a completed source response has the form $M_sA + L_sB$, where the zero-fluctuation baseline M_sA is independent of B. The flat source-zero Gaussian on each unmarked wall is centered and color invariant. Its covariance need not be a nearest-neighbor covariance. If a fixed real cutoff remains, its exact contribution is retained, and parity/color statements are used only when its actual selector has these symmetries.

For a polynomial P of degree at most three, evenness in B shows directly that

$$\mathbb{E}_{0,w}\{P(M_sA + L_sB) - P(M_sA)\} \tag{RG30a}$$

is a constant plus a linear function of A. A scalar invariant linear color tensor is zero for a compact semisimple gauge group: it would be an invariant vector in its adjoint representation. Thus the proper free birth has no D6 row on each such pair of source and kernel walls. The constant is retained with its anchor/test dependence. Explicit current and contact rows remain outside this scalar argument.

For a quantitative comparison, suppose the paired free companion has degree at most $d \le 3$, is independent of the independent canonical current, and the incoming discrepancy jets through degree three are bounded by g_jA_j. The physical remainder is still bounded by RG9. The actual source response differs from its linear part by $O(g_k^2 q_Y^2)$ in physical units on its fixed disc. In the proper difference the constant discrepancy cancels. Each degree-$m \ge 1$ discrepancy term therefore has the factor

$$g_j(g_k/g_j)^m = g_k(g_k/g_j)^{m-1}. \tag{RG30b}$$

No inverse relative coupling is needed. The high-degree discrepancy contributes $g_k^{4-d}(g_k/g_j)^d$, and the nonlinear response contributes g_k times a fixed polynomial of relative ratios and q_Y. IV and the same-coordinate Gaussian score give the same extra g_k for the interaction and density comparisons. After the one joint moment reserve, this yields an $O(g_k)$ proper scalar canonical birth, with explicit relative ratios and the original cutoff-tail terms.

This is a local propagation lemma for the free companion and discrepancy grade. It requires the declared incoming discrepancy jets; it does not infer them from an ungraded physical norm. The primitive comparison MT32 supplies the initial scalar column, and the present lemma explains the additional row which an all-depth induction must retain.

106.14 A fixed extension and a residual comparison for the same old coefficient

The next comparison makes RG38 usable with the physical coarse kernels of RG12. It avoids interpolating between unrelated boundary problems or topological sectors. It applies to the pure-clean current packet whose full short read sets lie in the protected region of Section 13.

Choose one fixed nested collection of current windows, with inner member containing the packet and complete read sets, and outer member inside the allowed D_0. Let H denote their fixed separation. Apply CE to the actual field W, with its fixed tree, root, cutoff and extension box. Insert the resulting field W^{ext} into the **same global old action graph** used in RG12, flat outside its contractible extension window. The tree extension and its undoing are part of this definition. CE14 proves its exact equivariance. A comparison real reference is used only in estimates and is not chosen anew to define the function.

On the complete inner read sets, $W^{\mathrm{ext}} = W$. On the extension graph CE gives a controlled weak chart with small $W^{1,p}$ potential and bounded curvature. A sufficiently small ceiling is chosen after H and the extension geometry. In current units the resulting constants are C_H, and in old units the coarse logarithmic target has size at most $C_H r\epsilon$. There is therefore a fixed integer m_0, depending only on the chosen geometry and radii, such that

$$k - j \geq m_0 \quad \Longrightarrow \quad v_j^{\mathrm{ext}} := \log C_j W^{\mathrm{ext}} \text{ lies in a fixed spare old coarse disc.} \tag{RG45}$$

The logarithm is taken in the actual covariant coarse frame. Gauge invariance permits estimating it in CE's canonical frame. C_j is the original composed determining map, not a substituted direct average. The finitely many recent ages use the original admitted representative and a larger fixed constant. No estimate $C_L L^{-2} < 1$ is being assumed.

106.14.1 Nested multipliers and a remote residual

Inside the common inner window, the true current minimum W is also critical for its own old determining data. This follows either by the nested variational identity or by differentiating the exact factorization of the original means. In current physical normalization its old multiplier is the physical adjoint of that factorization applied to λ_k. **Before estimating this adjoint**, use $\delta_{V_k}\lambda_k = 0$ to replace the raw suffix differential by its endpoint-compatible product. The vertical endpoint term pairs to zero. Only that compatible product is bounded by CL17. Raw gauge covariance of the exact composition then gives the inherited old Ward equation on the protected rows. The complete-prefix bounds, followed by the change to old units, give

$$\lambda_j^{\mathrm{old}} = r^3\,(DC_{k:j})^{\mathsf{t}}_{\mathrm{physical}}\lambda_k, \qquad \|\lambda_j^{\mathrm{old}}\|_{\ell^p_{\mathrm{uloc}}\cap\ell^\infty} \le Cr^3\epsilon. \tag{RG46}$$

The factor r^3 follows from the physical tangent and volume weights: an old target variation is r times the current physical potential variation, while the old coarse cell has current volume r^4. Adjacent physical tangent factors telescope in the actual composed derivative, so a separate chart constant is not multiplied once per intermediate scale. The inherited multiplier obeys the old coarse Ward equation on the protected rows.

Let χ be a smooth old coarse cutoff equal to one near the packet and zero outside the region where the nested equation holds. Its transition is at old distance at least cH/r from every short read set. Include all complete old prefix stencils in that separation. Consider MC9 on the same global old graph, based at reference W^{ext}, with zero relative target. The exact solution reconstructs the old physical minimum for the actual target C_jW^{ext}. The approximate five-type vector is

$$x_{\mathrm{app}} = (0,0,0,0,\chi\lambda_j^{\mathrm{old}}). \tag{RG47}$$

With the sign convention RG39, its only nonzero equation residual is

$$\mathcal{R} = J(W^{\mathrm{ext}}) + A^{\mathsf{t}}_{W^{\mathrm{ext}}}(\chi\lambda_j^{\mathrm{old}}). \tag{RG48}$$

It vanishes identically near the packet. There $W^{\mathrm{ext}} = W$, $\chi = 1$ on the complete row stencils, and RG42 converts the compatible adjoint into the genuine nested KKT row. The cutoff need not preserve the coarse Ward equation in the transition annulus; every resulting endpoint term remains in that remote annulus with its complete prefix collar. The other four equations vanish because their variables and relative target vanish. In MC10 the nonlinear remainder at RG47 is zero: a multiplier enters only the first equation and its nonlinear coefficient difference contains b.

The weak norm of $\mathcal{R}$ is bounded by a fixed multiple of the admitted small reference norm. The Wilson current is estimated in its proper negative row from the small weak connection and curvature, not by postulating a current supremum at the CE seam. Multiplication of the coarse multiplier by the smooth cutoff is bounded in its unit old row space, and the compatible full-space adjoint then bounds the second term of RG48. In particular, do not rewrite the annular discrepancy as the raw thin-path endpoint transpose applied to a Ward commutator; FI does not supply an unrestricted negative-norm bound for that expression. Inner vanishing uses Ward only where the cutoff is exactly one on every complete stencil. Thus no microscopic derivative of a sharp force is being introduced.

106.14.2 The comparison uses one inverse and one small ball

Let $\widehat{x}$ be the exact augmented solution. Subtract its equation from RG47 using the **same** global inverse $\Gamma_{W^{\mathrm{ext}}}$:

$$\widehat{x} - x_{\mathrm{app}} = -\Gamma_{W^{\mathrm{ext}}}\mathcal{R} - \Gamma_{W^{\mathrm{ext}}}\{N(\widehat{x}) - N(x_{\mathrm{app}})\}. \tag{RG49}$$

Use the uniformly local old norm weighted by $e^{-\tau \operatorname{dist}_{\mathrm{old}}(\cdot,Q)}$, with τ below the inverse decay rate. The residual is remote, and the nonlinear Lipschitz constant on this same small ball is below one half. Absorbing the second term proves

$$\|\widehat{x} - x_{\mathrm{app}}\|_{X,\mathrm{near}\,Q} \le C\epsilon_{\mathrm{old}} e^{-cH/r}. \tag{RG50}$$

This argument has no interpolation of exterior sectors, no comparison of different torus coefficient functions and no sum of unweighted fine boundary columns. It uses an approximate solution of one actual global equation.

The exact solution is a physical old minimum. MC's endpoint elimination removes its auxiliary force and restores the literal raw target by the bounded nodal gauge. Gauge invariance of $R_{j,X}$ and joint current covariance then identify its value exactly with $f_{j,X}(C_j W^{\mathrm{ext}})$. For the comparison one may evaluate before that restoration, in the common W^{ext} frame: the original paired domain is a union of the declared gauge orbits. Thus one is not subtracting two unrelated gauge-fixed minima.

Near the packet the approximate field is the actual current minimum W, whose nested old KKT equation holds. Curvature comparison follows by the covariant weak/strong interpolation in Section 12.2. Current comparison uses the physical equation and the close old multipliers in RG50; both auxiliary scalars vanish for these physical solutions. The compatible adjoint has the CL17 supremum bound and analytic field-variation bound on the common small chart. Hence the two physical pairs are exponentially close on the old read sets, with a possibly reduced positive exponent. Original local gauges or their explicitly controlled common alignment supply any required potential component.

Evaluate the **same** local coefficient $R_{j,X}$ on these two pairs and use Banach Cauchy in its spare old paired core. The exact difference

$$\mathcal{D}^{Q}_{j,k,X} = R_{j,X}(W, J(W)) - f_{j,X}(C_j W^{\mathrm{ext}}) \tag{RG51}$$

is a local current function of D_0: both terms depend only on the declared D_0 data, even though the second evaluates the fixed global old-minimum recipe. It retains the old birth label. It is not declared to be a newly generated $O(g_k)$ interaction.

The same comparison on the full physical ball and on the canonical ball gives two bounds for RG51. The latter admission has a separate check. MT34 and a once-chosen weak floor give $\sup_{j\le k} r(g_k/g_j) < C$: the geometric factor $r = L^{j-k}$ dominates the allowed relative coupling growth. On a fixed current canonical ball, the old target and old pair have size at most $C_H r g_k$ times its radius. Choose that radius once smaller than the old canonical spare radius divided by $C_H \sup r(g_k/g_j)$. Both old arguments then lie on the same spare canonical ball. This check is required; a physical-ball residual estimate alone would retain the unwanted g_j^{-d} factor.

Apply field Cauchy only to its low jets and use the physical fourth-order remainder for the rest. Thus its real-cutoff evaluation has an exponentially small version of RG11, including the coefficient $\alpha^{d-4} g_k^{4-d} (g_k/g_j)^d$. One does not assume that the entire real fluctuation cutoff fits inside the fixed canonical ball. Fixed powers of r^{-1} from engineering, root count, or derivative conversion are absorbed by $e^{-cH/r}$. This proves an age-summable difference in the two-amplitude graded row without an absolute inverse power of g_j.

The smaller canonical comparison ball is used to bound finitely many Taylor coefficients. The output canonical norm is recovered on its once-fixed ball from those coefficients and the full physical remainder, exactly as in RG8–11. Its radius is not divided by the response constant again at each birth; that constant enters the finite coefficient bound and then the formation constant.

106.14.3 Only the protected part needs the fractional upgrade

CE proves a weak extension, not a strong $W^{2+s,p}$ estimate for an axial cutoff. No such stronger seam estimate is needed here. Apply the physical-kernel calculation to $f_{j,X}(C_j W^{\mathrm{ext}})$, then separate kernels whose full original means, curvature paths and source offsets remain in the protected inner window. There $W^{\mathrm{ext}} = W$, and RG44 together with the actual CB coarse bridge gives the precise regularity used by CD. The entire short family sees one common field. The exact dimension-four family is retained; the $O(r^s)$ dimension-six comparison in CB is not applied to that family after multiplying by r^{-2}.

Long derivative kernels and long original carriers retain their exponential tail reserve and the two-radius envelope. Winding and wrapping representatives stay in their original admitted filing. No flat extension is made through a noncontractible packet. The same split is made before absolute values so reflection, root phases, source offsets and the area-commutator cubic cancellation are preserved.

For the fourth-order analytic remainder one may use a root-adapted gauge **only in the estimate**. Compare the root frames in CE's already small canonical representative, aligning their constant root conjugations first. Their transition is then near the identity and its first difference is controlled by their bounded link potentials. Two tree gauges of an arbitrary original real field need not have this property; a rough pure gauge would disprove that assertion. Cut off the small transition in the outer fixed collar, leaving the already defined W^{ext} unchanged as a gauge field. This gives the global coarse potential estimate RG20 relative to each old root. All gauge-invariant values and the one common field remain the same. It does not define a different extension for each kernel term.

These steps prove the physical-kernel representation and strict-age readout for the completed contractible pure-clean scalar packet. They do not assign that representation to a selected nonsymmetric history or to a read set crossing an unhealed component.

106.15 A test-functional norm closes the clean scalar source row

One norm change is necessary. CD's strict-age estimate is a smooth-test estimate. It is generally false in the raw sum of absolute source-anchor coefficients. The original anchor vector remains exact algebraic data, but after a completed regular projection it should be estimated in the following weaker norm.

For a completed carrier X, let $\mathcal{T}^{2,s}(X)$ be the restrictions of scalar $C^{2,s}$ tests to its complete local test window, with the usual own-unit norm. Define

$$\mathfrak{T}(X) = \mathcal{T}^{2,s}(X)^*, \qquad \|E\|_{\mathfrak{T}(X)} = \sup_{\|f\|_{C^{2,s}} \le 1} |E(f)|. \tag{RG52}$$

At finite cutoff, $E(f)$ is still its exact finite sum over original anchors. Neither the labels nor their locations have been discarded. An initial anchor ℓ^1 estimate implies RG52, and every displayed original contact identity remains an identity of those finite sums. What changes is the norm used after regular cancellation.

Take the physical and canonical paired suprema of RG6 with values in $\mathfrak{T}(X)$. The response/Cauchy proof RG15–17 is Banach-valued and applies verbatim. It bounds each coarse field coefficient in $\mathfrak{T}(X)$; the sum over field coordinates and their path moments is unchanged. Cochain homotopy acts on field legs, not on the test variable. The relevant anchor moments below are exact signed source moments, bounded by this operator norm and a fixed power of carrier diameter. We no longer claim that the stricter sum of absolute microscopic anchor coefficients has acquired CD's age gain.

106.15.1 Source moments prevent a derivative loss at each birth

A bare instruction to apply CD to a $(C^{2,s})^*$-valued kernel would be insufficient: if such a kernel acts like a second derivative of a point mass, applying an ungraded second-order argument again would appear to require four derivatives of the test. The following joint source/field grading removes that error.

Let $\Lambda \in \mathfrak{T}(X)$ be supported within old distance D of its chosen root, and let x be that root in current coordinates. For $0 < r \le 1$, Taylor's exact finite remainder gives

$$\begin{aligned}\Lambda[z \mapsto f(x+rz)] = M_0(\Lambda)f(x) + r\sum_a M_a(\Lambda)\partial_a f(x) \\ + \frac{r^2}{2}\sum_{a,b} M_{ab}(\Lambda)\partial_{ab}f(x) + \Lambda(\mathcal{R}_{x,r}),\end{aligned} \tag{RG53}$$

where $M_0 = \Lambda(1)$, $M_a = \Lambda(z_a)$ and $M_{ab} = \Lambda(z_a z_b)$. Moreover

$$\begin{aligned}|M_\nu(\Lambda)| &\le C(1+D)^2\|\Lambda\|_{\mathfrak{T}(X)}, \qquad |\nu| \le 2, \\ \|\mathcal{R}_{x,r}\|_{C^{2,s}(X)} &\le Cr^{2+s}(1+D)^{2+s}\|f\|_{C^{2,s}}.\end{aligned} \tag{RG54}$$

To check the second inequality, differentiate the ordinary second-order Taylor remainder in the old variable z. Its derivatives of order $m \le 2$ cost $r^{2+s}D^{2+s-m}$; the Hölder seminorm of its second derivative costs exactly r^{2+s}. This proves the bound in the full test norm, not only a supremum bound. The carrier polynomial is paid by the reserved locality moment.

Apply this identity to each completed quadratic curvature kernel. Its source moment of order m already carries r^m. Only $2-m$ additional spatial orders remain before the dimension-six projection:

- For $m = 0$, use the completed CD quadratic formula with its full pointed descendants and path-area cubic term.
- For $m = 1$, use the first-order field/path expansion CD8. Its remainder is $O(r^{1+s})$, and the displayed source factor makes the total $O(r^{2+s})$. Retain the resulting $(\partial f)F\,DF$ term and any lower row.
- For $m = 2$, retain the coincident quadratic term. Moving its fields or a test root costs a positive Hölder power; since $\beta > s$, its remainder is again $O(r^{2+s})$.
- The last term of RG53 already has r^{2+s} and needs only the elementary quadratic field bound.

Thus the total number of test derivatives used is always at most two, with the same Hölder exponent s. No derivative order is consumed when a coefficient becomes a later birth. For cubic curvature kernels, retain only the zero source moment in the dimension-six projection. The positive source moments already give an extra factor r, and moving the curvature legs gives the original r^β remainder. Quartic and higher field terms retain their analytic grade and use the test norm directly.

At finite cutoff, the moments in RG53 are exactly the original anchor moments $\sum_a c_a z_a^\nu$. Consequently this is a new estimate for the same pointed extraction, not a new contact prescription. Reflection and translation act simultaneously on Λ, its support, its monomial moments, field paths and tensor labels. The completed family RG18 gives the same root reindexings as before. Taking absolute operator norms is done only after these exact pairings and the combined cubic-center cancellation.

After the old-root count and D6 normalization, RG53–54 therefore give the same strict factors as MT42 in $\mathfrak{T}$:

$$\|R_{j\to k}\|_{\mathfrak{T}} \le CA_j\left\{r^s a_2^{(d)}(j)t^2 + r^\beta a_3^{(d)}(j)t^3 + r^2\alpha^{d-4}g_j^{-d}t^4\right\}, \qquad d = 2,3. \tag{RG55}$$

The source Taylor remainder in the quadratic row has precisely the same $r^{2+s}r^4r^{-6} = r^s$ accounting. It is not an ungraded new descendant.

106.15.2 The original connected formation is bounded in this norm

Source localization, integration, and finite coefficient extraction are linear in a one-label test slot. Their actual frozen-coordinate formulas therefore extend to the Banach-valued coefficient space $\mathfrak{T}$. For instance,

$$\left\| \int R(B;\cdot)V(B)\, d\mu(B) \right\|_{\mathfrak{T}} \le \int \|R(B;\cdot)\|_{\mathfrak{T}} |V(B)|\, d|\mu|(B). \tag{RG56}$$

The actual one-reserve argument bounds the right side after source domain admission. The connected subtraction has the same finite partition bounds as MT9–13. Fixed current partitions multiply tests by χ_Q; multiplication is bounded on $C^{2,s}$, with a fixed constant determined by the once-chosen current geometry. Localization walls and coordinate Cauchy act on field variables, so they commute with evaluation in a test slot. No operation in this clean scalar formation requires selecting an individual microscopic anchor after its norm has been taken.

Recover newborn field kernels by RG15–17 in $\mathfrak{T}$. Apply RG53 to those kernels at their next carrying step. The geometric polynomial losses are fixed, are paid by the reserved exponent in RG26, and are restored by actual coarsening. This proves invariance of the **test-graded** physical/canonical kernel row for the completed pure-clean scalar family, subject to the already displayed fixed carrier geometry and clean source domains. It does not prove invariance of the stronger raw anchor ℓ^1 norm with a strict age factor, which is unnecessary for these scalar operations.

For the proper newborn row, retain the free companion and its degree-three discrepancy as in Section 10. The color-invariant Gaussian cancellation holds coefficientwise in the test-functional space. The actual nonlinear map, inherited interaction and density score then give the displayed $O(g_k)$ correction in that same space. The fixed-cutoff difference stays an exact flagged tail. Full lower rows and the actual dimension-four family remain in the state; the free stable-to-center zero is never used to erase them.

Combining RG55–56 with the distinguished actual clean interaction vertex gives the concrete scalar version of MT36, with canonical envelope

$$C g_k A_j \left\{ r^s \rho_{j,k}^2 q_Y^2 + r^\beta \rho_{j,k}^3 q_Y^3 + r^2 \alpha^{d-4} g_k^{4-d} \rho_{j,k}^d q_Y^4 \right\}, \tag{RG57}$$

For $d = 2$, replace the displayed cubic coefficient $\rho_{j,k}^3$ by $(g_k/\alpha)\rho_{j,k}^2$, exactly as in MT44. The full physical row has the corresponding relative factor $\rho_{j,k}^d$. The exponentially small comparison and long-carrier tails from Section 14 fit this row. All polynomial fluctuation factors are paid by one common reserve. MT34–35 make these fixed relative weights age summable.

106.15.3 What extends to finite sources, and what still requires a contact row

For a fixed finite label set I, use bounded multilinear functionals on the projective tensor product of the declared local $C^{m_I,s}$ test slots. Products with disjoint label sets are submultiplicative in this norm. The exact finite-source partition, cumulant and connected-formation formulas therefore have the same finite constants as XP. Banach-valued physical/canonical Cauchy and the higher finite field grade of Section 9 remain valid. Multiplication and the finite Leibniz map on $C^{m_I,s}$ also bound an **already coalesced** contact coefficient acting on products such as $f_i f_j$, with the derivative order declared in its own row.

This does not manufacture the contact coefficient itself. Selecting the diagonal of an arbitrary weak two-anchor coefficient is not a bounded operation merely because smooth test multiplication is bounded. For example, on a one-dimensional subchart of the source coordinates, the exact finite-anchor functional

$$T_\varepsilon(f,g)=\frac{f(0)-f(\varepsilon)}{\varepsilon}g(0) \tag{RG58}$$

has uniformly bounded bilinear dual C^1 norm. Retaining only coincident anchor pairs leaves $\varepsilon^{-1}f(0)g(0)$, whose norm diverges. Thus a sharp anchor mask can destroy the cancellations measured by the weak norm. Creation of coincident contacts, cutoff boundary currents and nonprojectable fine Haar rows must use their actual XP/WCT equations and their own quantitative incoming norms. Likewise a higher engineering-dimension regular projection needs the corresponding finite joint source/field moment calculation; it is not supplied by applying the dimension-six argument an arbitrary number of times. These are part of the outstanding finite-source/contact induction, not hidden hypotheses of the one-label scalar conclusion.

106.16 Exact remaining obligations for a full induction

The positive conclusion is scoped but quantitative: for the completed, contractible pure-clean one-label D6 family, the actual original source slot has a fixed core; its current-index representative has the required fractional regularity; RG45–51 compare the same old coefficient to its actual physical coarse kernels; RG52–57 close the graded scalar formation row without repeated test-derivative or locality loss. The finite source labels, pointed descendants, full quadratic family, vacuum and original cutoff flags are kept. This statement concerns the regular coefficient row, not by itself the complete normalized center cocycle.

The following parts of the full induction are not established here.

1. **First live entry with a straddling old read set.** A source read set meeting retained real variables and a newly prepared mixed collar still needs the full original graded-core inclusion on all its preparation walls. The R-II weaker mixed admission is not a substitute for this. The current clean construction does not change that input graph.
2. **R-origin and history-dependent carrying rows.** Healed R births have their actual shrinking physical and canonical radii, and selected histories need not have the completed translation pairing used by CD. Their joint live reserve and origin bank must pay the actual carried coefficients through every transition; the clean scalar result does not supply those rows by relabeling them.
3. **Creation and transport of new contact diagonals.** RG58 shows why the weak test norm alone is insufficient. Every partition/contact row must have its own quantitative bound under the actual XP pushforward, including cutoff boundary currents and nonprojectable fine vector densities. Keeping a symbol for that row is not the needed estimate.
4. **The full higher finite-source graded projection.** Disjoint-label formation and Banach-valued Cauchy are proved, but the larger joint source/field moment projection and all valuation-lowered current coefficients still require their explicit finite-grade bounds. Section 9 identifies the required analytic degrees; it does not prove every incoming contact bound.
5. **Alternative orders or carrier geometries.** The source-first frozen-B order and RG26a close the constructed clean filing. A coordinate change inside an already localized source or a different carrier enlargement needs its literal joint range and covering inequality, including the original R radii.

6. **The complete source-zero comparison and global closure.** RG51 compares one old coefficient on one action graph. It does not identify coefficients produced on two physical tori. The separate physical Gaussian comparison is not yet the complete nonlinear/large-field S94 estimate, and the full center feedback requires every stable/live/contact row listed above.

Accordingly the final status remains incomplete. The regular scalar kernel/formation gap has a concrete solution on its stated original clean branch. The remaining gates concern actual mixed transitions, near-insertion contact creation, and the complete source-zero hierarchy, rather than a missing bound for an unspecified periodic substitute.

107.3 A protected collar gives a derivative estimate without higher curvature jets

The scalar reconstructed-plaquette estimate S71 supplies curvature but does not itself control DX. The following local inequality identifies exactly what must be added. It is a discrete estimate, not a continuum regularity assumption.

Let $a \le \ell$, let K lie inside a collar K^+ of thickness comparable with ℓ, and choose a scalar cutoff χ equal to one on a one-stencil enlargement of K, supported in K^+, with $\|\partial_a \chi\|_\infty \le C\ell^{-1}$. In one admitted local gauge assume

$$\|U_b - I\| + \|U_b^{-1} - I\| \le CaA_0, \qquad aA_0 \le c, \tag{LM10}$$

on the complete collar. The fields may be complex. Define

$$\begin{aligned} \widehat{F}_p &= a^{-2} \max\{\|P_p - I\|, \|P_p^{-1} - I\|\}, \\ \rho_F &= \sup_{p \subset K^+} \widehat{F}_p, \\ \widehat{\mathcal{F}}^{,2}(K^+) &= a^4 \sum_{p \subset K^+} \widehat{F}_p^2, \\ \mathcal{J}^2(K^+) &= a^4 \sum_{b \subset K^+} |J_{\mathrm{phys}}(b)|^2. \end{aligned} \tag{LM11}$$

All carriers in these expressions include the finitely many displaced plaquettes occurring in the cube identities. The norms for complex matrices are Hermitian norms, not the complex bilinear continuation of a positive quadratic form.

Proposition. Under LM10,

$$a^4 \sum_K |D_\rho X_{\mu\nu}|^2 \le C \left[\mathcal{J}^2(K^+) + (\ell^{-2} + A_0^2 + a^2 \rho_F^2) \widehat{\mathcal{F}}^{,2}(K^+) \right]. \tag{LM12}$$

The constant is independent of a/ℓ. The current in this formula is the actual algebraic current $J_\nu = \sum_\mu D_\mu^* X_{\mu\nu}$. It is not an independently supplied parameter with the same name.

Proof. Use the flat differences ∂_a, ∂_a^*, and apply the exact flat discrete Gaffney identity to the compactly supported matrix-valued two-form $G = \chi X$. With the sums over ordered indices used by AI16–18, it is

$$\sum_{\rho\mu\nu} \|\partial_\rho G_{\mu\nu}\|_2^2 = 2 \sum_\nu \left\| \sum_\mu \partial_\mu^* G_{\mu\nu} \right\|_2^2 + \frac{1}{3} \sum_{\rho\mu\nu} \|\partial_\rho G_{\mu\nu} + \partial_\mu G_{\nu\rho} + \partial_\nu G_{\rho\mu}\|_2^2. \tag{LM13}$$

One can verify it by Fourier transformation, or by the commuting difference calculation underlying AI18 with $U = I$. It is an identity for complex fields with the Hermitian inner product.

On the collar, LM10 gives the operator bounds

$$\|(D - \partial_a)Z\|_2 + \|(D^* - \partial_a^*)Z\|_2 \le CA_0 \|Z\|_2. \tag{LM14}$$

Here $D^* = (T^{-1} - I)/a$ is the algebraic backward covariant difference; it need not be the Hermitian adjoint for complex fields. That distinction causes no error: LM13 uses flat Hermitian adjoints, and

LM14 compares the operators directly. Commuting the scalar cutoff through a difference costs at most $C\ell^{-1}\|X\|_{2,K^+}$. Thus LM13 implies

$$\|DX\|_{2,K} \le C\left[\|J_{\mathrm{phys}}\|_{2,K^+} + \|B\|_{2,K^+} + (\ell^{-1} + A_0)\|X\|_{2,K^+}\right], \tag{LM15}$$

where B is the exact additive cube defect AI16.

The six-face multiplicative cube identity is algebraic and remains valid for complex invertible links. Subtracting its linear part, including the inverse version needed for the holomorphic flux, shows that the additive defect contains at least two plaquette deviations. All its transports have bounded length and norm under LM10. Consequently

$$|B_{\rho\mu\nu}(x)| \le Ca \sum_{p \text{ in its enlarged cube}} \widehat{F}_p^2, \qquad \|B\|_{2,K^+}^2 \le Ca^2\rho_F^2\widehat{\mathcal{F}}^{,2}(K^+). \tag{LM16}$$

The complex holomorphic flux is bounded by $C\widehat{F}$. Squaring LM15 now proves LM12. □

107.3.1 Consequence on an actual protected reconstruction layer

Suppose the complete actual reconstruction collar satisfies

$$A_0 \le C\epsilon\ell^{-1}, \qquad \rho_F \le C\epsilon\ell^{-2}, \qquad \mathcal{J}^2(K^+) \le C\epsilon^2\ell^{-2}N_K, \qquad \epsilon \le \epsilon_*. \tag{LM17}$$

Here N_K counts ℓ-cells, so the curvature bound also gives $\widehat{\mathcal{F}}^{,2}(K^+) \le C\epsilon^2 N_K$. Then LM12 and the definition of the complementary current imply

$$\ell^2|H_{a,K}| \le C\epsilon^2 N_K. \tag{LM18}$$

For the exact cubic completion, AI24–27 and AI32b give, on the same collar,

$$\ell^2|C^\sharp_{a,K}| \le C\epsilon^3 N_K. \tag{LM19}$$

For the last divergence in AI32b one uses the exact microscopic difference bound $a^{-1}|XB| \le C\widehat{F}^3$; for the fourth-order remainder one uses $a^2\widehat{F} \le C\epsilon$. No derivative of curvature beyond LM12 is needed. Both estimates keep the physical factor ℓ^{-2}, rather than a fine-mesh factor a^{-2}.

If the actual canonical readout gives $\epsilon \le Cg(q_\infty + q_{\mathrm{ext}})$, LM18–19 become degree-two and degree-three polynomial-energy envelopes. Here $q_\infty^2 \le \sum_b |B_b|^2$, and the fixed external canonical ball contributes a constant. Carrier-size factors are retained and are paid from spare carrier decay or from the single component's record bank.

This verifies a useful implication about actual maps: **once** their common collar has the bounds LM17, a separate higher-derivative theorem is unnecessary for these two primitive moment envelopes. It does not prove LM17 for an arbitrary wall family. In particular the last bound in LM17 must concern J_{phys}. MT2–5 demonstrates why replacing it by the independent original current coordinate is false.

107.6 One joint energy reserve controls a complete finite-source coefficient

Let ω be an actual joint live history. Keep the source-free denominators, old integrations, selectors and relative measure unchanged. Suppose each incoming coefficient under consideration has been proved by LM5, LM9, LM18–19, or an already admitted analytic member bound to satisfy

$$|A_I| \le a_I(1+\mathscr{E}_\omega)^{d_I/2}. \tag{LM24}$$

Here $\mathscr{E}_\omega$ is the once-owned sum S59, and the constants a_I include the scale factors and the actual fixed external data. There is no claim that every yet-uncontrolled inherited coefficient already satisfies LM24.

The positive R-II calculation with a fixed unused fraction gives the same-kernel reserve $\int G_\omega e^{\theta\mathscr{E}_\omega} \le B_\omega$, in the notation of S86. For square-free finite source labels η, direct coefficient extraction gives

$$\left|[\eta_I]\int K_\omega e^{\sum_{J\neq\varnothing}\eta_J A_J}\right| \le B_\omega \sum_{\pi\in\Pi(I)}\prod_{J\in\pi} a_J\, C_\theta\!\left(\sum_{J\in\pi} d_J\right), \tag{LM25}$$

where one may take

$$C_\theta(D) = \sup_{t\ge 0}(1+t)^{D/2}e^{-\theta t} \le \left(1+\frac{D}{2\theta}\right)^{D/2}. \tag{LM26}$$

This is proved by expanding the finite nilpotent exponential, using $\prod_J(1+E_J)^{d_J/2} \le (1+\mathscr{E}_\omega)^{\sum_J d_J/2}$, and integrating once. There is one B_ω, for every partition. This argument preserves every mixed source coefficient; repeated source derivatives are represented by separately labeled copies before their physical values are equated. Contacts retain the supports and signs given by the exact XP multiplication rule.

For own-spacing compact primitives and their finite constant-bond EOM packets, LM24 is an established same-Wilson input. For a protected reconstruction it is established when LM17 holds on the complete joint wall domain. Thus these are concrete applications of XP39 rather than an assumed output norm.

107.7 First complete healing and the original birth radius

Consider the sector in which every incoming atom has one of the verified bounds of Sections 1–2, all its scale costs have been charged as in Section 4, and its coefficient is analytic on the actual outer pair domain used by the original source-zero operator. At first complete healing, LM25 gives its finite-source integral bound before any connected exponentiation. Applying the original localized connected construction with the finite distinguished marks yields a new R birth with a weakened carrier exponential and one once-owned rare factor. Polynomial carrier counts and the finite partition sum modify its fixed constant. No normalization by the mass of the individual live component is used.

For the purely stored compact sector, outer analyticity comes from the unchanged source-zero kernel: the compact carrying function depends on the stored real variables and is independent of the outer complex pair. The original arbitrary-carrying-function bound R-II (1.73) and its domain inclusion (1.65) therefore apply after the polynomial reserve estimate, or directly with the finite supremum LM5a and its once-paid birth cost when the local positive Wilson fraction is unavailable. This is a model-specific case in which the new R birth exists on the original shrinking domain without a new source-dependent inverse.

Denote its birth radius by α_k, and its proven supremum/carrier bound by $b_{k,I}$. On a fixed smaller **birth** domain, Cauchy's theorem gives for actual coarse pair coordinates

$$\|D^m R_{k,I}\| \le C_m \alpha_k^{-m} b_{k,I}, \qquad g_k^m \|D^m R_{k,I}\| \le C_m (g_k/\alpha_k)^m b_{k,I}. \tag{LM27}$$

If the pair has distinct radii, replace α_k^{-m} by the exact product of the corresponding radius powers for the chosen multiindex. Only a fixed number of coarse coordinates per cell is counted; a fine link-prefix count is not inserted into Cauchy's theorem. The associated finite carrier moments cost spare locality in the usual way.

These are joint analytic jets. To call them the physical quadratic and cubic field jets required by MT36, the actual own-current pullback must be admitted and differentiated on this same smaller birth domain. That is a separate map statement. Once it holds, the chain rule transfers LM27 with its proved physical pullback constants. An administrative promotion does not change α_k, supply the pullback, or enlarge the domain to a fixed clean radius.

For a later regular readout, its genuine admitted fluctuation disc has radius $\delta_{k,I}$. Schwarz/Cauchy gives a proper-difference factor $g_s/\delta_{k,I}$, not an automatically fixed clean denominator. Polynomial losses in an original shrinking R radius can be paid once by its rare birth envelope as in Section 4. The argument still requires the whole connecting disc to be inside that radius. Small rarity cannot justify evaluating an analytic function outside its domain.

107.8 Sharp selectors require a weak contact norm

The original selector functions are not generally smooth. It is false that their repeated derivatives have pointwise polynomial-energy envelopes. Already the second derivative of an interval indicator is a derivative of a delta distribution and need not be a measure of finite total variation.

There is nevertheless an exact one-reserve estimate for a **whole** kernel derivative. For a distributional derivative of a compactly supported or Haar-periodic joint kernel and a smooth test F,

$$\langle Y_1 \cdots Y_m(\chi\rho), F\rangle = (-1)^m \int \chi\rho\, Y_m \cdots Y_1 F. \tag{LM28}$$

For nonconstant vector fields the corresponding divergence adjoints are used in the stated order. If

the right-hand test derivative has a bound $a_F(1+\mathscr{E}_\omega^\circ)^p$, the same positive reserve gives

$$|\langle Y_1\cdots Y_m(\chi\rho),F\rangle|\le a_F C_\theta(2p)B_\omega. \tag{LM29}$$

This does not multiply boundary deltas, differentiate a selector after discarding its neighboring density, or divide by a conditional mass. It is directly applicable to the compact test packets of Section 1.

After pushforward, however, the test is $F=\varphi\circ q$, where q is the actual nonlinear coarse map. The right side of LM28 then contains derivatives of q and, for further marks, their contacts. Their bounds must be proved on the complete actual collar and retained variable domain. LM28 preserves the exact XP cutoff/contact bookkeeping; it does not manufacture that nonlinear derivative estimate. Summing selectors to one may cancel boundary terms in a full physical integral, but it is not a bound on an individual localized history.

107.10 A completed regular R birth has the ordinary pair domain

The intermediate mixed domain used to prepare a live integration and the domain of a finished regular R coefficient should not be identified by notation alone. The primary endpoint gives the identification that is actually needed here. R-II (1.98)–(1.100) defines the completed analytic coefficient, separates carriers still meeting the remaining large-field region into boundary coefficients, and places the regular R terms in the induction class of III. III (2.30) explicitly assigns these R terms properties (i)–(iii) after (2.27). Its property (ii) is the original I pair domain with radii

$$\alpha_{0,j} = C_0 g_j (\log x_j)^{q_0}, \qquad \alpha_{1,j} = C_1 g_j (\log x_j)^{q_1}, \qquad x_j = g_j^{-2}. \tag{LM31}$$

This endpoint does not require asserting that the intermediate IV prepared domain is itself identical to that ordinary pair domain.

For a marked birth formed by the finite carrying construction of Section 6, its integral has the same outer domain and the same prepared maps. The coefficientwise moment argument gives its bound on that domain. Thus the same endpoint applies to this verified marked sector. A still-live boundary coefficient has not reached this endpoint.

107.10.1 The current in the original completed regular source branch is physical

In I (1.15)–(1.16), each inner current is defined from its inner minimum by the Wilson current formula. I Lemma 4, (3.53), places this actual field/current pair in the original pair domain. The localized response in II (1.18)–(1.22) remains in I's admissible perturbation class (3.31) throughout the source-wall polydisc, provided $4B_0C_1e^{16\kappa_1}\epsilon_1 \le \alpha_2/2$.

Consequently, on a complete inner collar of **that original completed branch**, the curvature and current conditions LM17 are actual bounds at every source wall. LM12 then proves the derivative envelope LM18, and the compact cube algebra gives LM19. These implications use no identification of the outer independent current with the physical current. It is the inner constrained minimum that supplies the physical current. A readout with a different boundary problem, or one whose stencil leaves the completed inner collar, has not been covered.

107.10.2 An explicit smaller R radius satisfies the primary inequalities

The small constants in Lemma 4 can be chosen as functions of the birth radii; no fixed clean radius is required. To display the choice, let C_G dominate the fixed constants in its gauge and minimum bounds, and put

$$\omega_j = \min\left\{\alpha_{0,j}, \frac{\alpha_{1,j}}{4C_G M}, \alpha_*\right\}. \tag{LM32}$$

Choose an inner target curvature radius $a_{0,j} = c_0\omega_j$, an inner connection radius $a_{1,j} = C_G M\omega_j$, and fixed positive constants sufficiently small so that the following hold. Here β_{geom} is the fixed geometric small parameter called β in I, not the inverse Wilson coupling in LM1:

$$\begin{aligned} C_G M a_{0,j} &< a_{1,j}/2, \\ B_3 a_{3,j} &< a_{1,j}/2, \qquad & 2B_3 a_{3,j} &\le \beta_{\mathrm{geom}} L^{-2} a_{0,j}, \\ C_Q a_{2,j} &\le a_{3,j}/2, & 4B_0 C_1 e^{16\kappa_1}\epsilon_{1,j} &\le a_{2,j}/2. \end{aligned} \tag{LM33}$$

For example take successively $a_{3,j} = c_3\beta_{\mathrm{geom}} L^{-2} a_{0,j}$, $a_{2,j} = c_2 a_{3,j}$, and $\epsilon_{1,j} = c_1 e^{-16\kappa_1} a_{2,j}$. The first two rows are the linear size relations used in I (3.37), (3.49)–(3.52); the final row is the all-wall response condition in II. All absolute smallness conditions are met by the fixed ceiling α_*. These choices only restrict domains of existing functions; they do not change a real selector or the Wilson measure.

Restrict the analytic physical coarse input ball and physical fluctuation disc by one further fixed factor below $\epsilon_{1,j}$, including the fixed response constants. Denote that common smaller radius by ρ_j. The original low-regularity minimum bounds put that coarse input inside the enlarged outer domain of Lemma 4; its genuine inner minimum and physical current then lie strictly inside the birth domain. All source-wall parameters remain on the same fixed polydisc. In particular

$$c_* \min\{\alpha_{0,j}, \alpha_{1,j}, \alpha_*\} \le \rho_j \le C_* \min\{\alpha_{0,j}, \alpha_{1,j}, \alpha_*\}, \tag{LM34}$$

where the constants depend on the fixed geometric and wall parameters, not on j, the fine cutoff, or the ultraviolet endpoint. In the weak regime this is a shrinking radius of order g_j times a fixed power of $\log x_j$.

This proves a domain statement for the original completed regular readout, including a healed R coefficient subsequently evaluated by that same readout. It does not create new protected collars through remaining rough regions. A recent R coefficient is evaluated only on this ρ_j-sized ball until its actual later readout is admitted. The whole ordinary real canonical cutoff can be included by making the auxiliary analytic hierarchy explicit. The frozen fluctuation test in III (1.8), (3.16) has $P_j \le C_P(\log x_j)^{p_0}$. In the construction of the analytic extensions choose once $q_0 = q_1 = q > p_0$, and choose their fixed constants in the order required by I (3.37), in particular $C_1 > 2C_G M C_0$. Then LM34 implies

$$\frac{g_j P_j}{\rho_j} \le C(\log x_j)^{p_0 - q} \longrightarrow 0 \quad \text{as } x_j \longrightarrow \infty. \tag{LM34a}$$

Hence $g_j P_j < c\rho_j$ uniformly after one increase of the weak floor. The fixed extra factors from the common wall radius in LM33 are included in the constant in LM34a. They do not depend on the depth.

This is a parameter choice for the original analytic construction, not an enlargement of an arbitrary function known on a smaller disc. III (2.28) leaves the integers $q_i > 1$ and sufficiently large C_i as auxiliary choices. The absolute smallness requirements of I hold because $g(\log g^{-2})^q$ tends to zero for every fixed q. The linear radius comparisons in LM33 preserve a common logarithmic power. In the actual complex comparison for R-II (1.71), printed p.379, the perturbation of the quadratic form and

of the summed interaction is bounded by a positive power of g times a fixed logarithmic polynomial; increasing this fixed q changes that polynomial and the required weak ceiling, but not its limiting smallness. Finite inverse-radius costs are charged as in Section 4. Thus these choices are compatible with this completed regular readout. No real cutoff, action, or reference denominator changes. If an additional coordinate comparison has a proved fixed logarithmic loss, choose q larger than its exponent as well before fixing the ceiling. This observation does not establish a missing mixed readout or its complete protected collar; that separate requirement is LM47.

107.10.3 Physical coarse jets, and the locality statement they do not replace

Let $\mathcal{P}_{j,\mathbf{s}}(v,b)$ denote precisely that completed physical inner field/current pair; retain its fixed determining problem and its full readset. If $R_{j,I}$ has supremum bound $b_{j,I}$ on its original pair domain, then the actual pullback

$$G_{j,I,\mathbf{s}}(v,b) = R_{j,I}\big(\mathcal{P}_{j,\mathbf{s}}(v,b)\big) \tag{LM35}$$

is holomorphic on the physical coarse ball/disc of radius ρ_j, uniformly over its source walls, and has supremum at most $b_{j,I}$. At all walls one this is the physical readout, by the unchanged completed minimum identity. Cauchy estimates on a fixed smaller physical ball give

$$g_j^m \|D^m G_{j,I,\mathbf{s}}(0)\|_{\mathrm{op}} \le C_m (g_j/\rho_j)^m b_{j,I}. \tag{LM36}$$

Here derivatives are in actual coarse coordinates. The derivative norm is a multilinear operator norm for the declared coarse coordinate supremum norm. Its constant has no fine-prefix link count. The analytic Taylor remainder after degree three obeys

$$|G(v) - T_{\le 3}G(v)| \le C b_{j,I} (\|v\|/\rho_j)^4, \qquad \|v\| \le c\rho_j. \tag{LM37}$$

Thus the shrinking R domain does not by itself force an absolute g_j^{-m} loss in the canonical physical jets: the ratio is g_j/ρ_j. The two- and three-field bounds and the quartic tail are compatible with a rare graded R amplitude of order $b_{j,I}$.

LM36 is not automatically the exponentially localized coefficient and source-anchor moment norm in MT36. A global minimum can depend on coarse variables outside the original fine carrier. To obtain that stronger norm one must localize the **physical** pullback, verify its joint wall family, and sum the completed carriers and their moments. One cannot replace the coefficient sum by the norm of a finite jet on an ambient volume of growing size. This remaining locality issue, and the stable feedback it feeds, remain part of L2–L4.

107.13 Buffered real retention is an exact alternative, with a separate map obligation

There is another possible treatment of first entry. Declare entry at scale e using the **complete** current packet, its determining read sets, and its protected collar, before any large-field preparation changes its coordinates. If its common regular representative is then $F_{j,e,I}(v_H)$, depending only on the actual old variables in a declared finite halo H, its verified regular bound can be saved in the form

$$|F_{j,e,I}(v_H)| \le A_{j,I} W_{e,j} (1 + \mathscr{E}_H(v_H))^{D_I/2}. \tag{LM49}$$

The field used to prove LM49 is one common field for the packet. A finite halo for a still nonlocalized global minimum is not assumed: the preceding regular filing has to produce the actual finite read set.

107.13.1 The exact retention identity preserves the gained age

Let $\mu(dv)$ be the original finite-cutoff joint real Wilson integration at this stage, including its actual retained coordinates, history selectors, and constraint factors. One may keep the unexpanded positive joint density when an absolute estimate is taken. Let $q(v)$ be the actual new target and let $\pi_H v = v_H$. Define the measure on the old halo and the original new output by

$$\nu_H = (\pi_H, q)_* \mu. \tag{LM50}$$

For every bounded output test φ, finite-cutoff Fubini gives

$$\int F_{j,e,I}(v_H) \varphi(q(v))\, d\mu(v) = \int F_{j,e,I}(h) \varphi(z)\, d\nu_H(h,z). \tag{LM51}$$

No old variable has been set to an arbitrary frozen value, and no new target constraint has been removed. If the same original positive reserve obeys $\int e^{\theta \mathscr{E}_H}\, d\mu \le B$, then LM49 and the one-reserve moment inequality imply

$$\left| \int F_{j,e,I}(h) \varphi(z)\, d\nu_H \right| \le C_{D_I,\theta} A_{j,I} W_{e,j} B \|\varphi\|_\infty. \tag{LM52}$$

Thus real retention itself preserves the age already gained before entry, without invoking the complex readout inclusion LM47. It does not create additional regular decay during a later live residence; the engineering conversion and waiting costs are still LM20–23. For multiple marks use their union halo, total degree, and one reserve.

107.13.2 Retaining a variable is not yet an analytic R kernel

The joint measure LM50 generally still contains the target constraints $\delta(q(v) - z)$. If a row of q is determined entirely by retained halo variables, it is a delta constraint in the outer output with the halo fixed. Even the elementary map $q(v) = v$ gives $\nu_H(dh, dz) = \mu(dh)\delta_h(dz)$. A bounded real carrying function then gives its original function of z, which need not be analytic. Fubini and positivity alone therefore do not give the outer holomorphic kernel needed at healing.

A proposed enlarged live component must instead specify a genuine mixed output map, for example with old-depth identity rows in its halo and the actual blocked rows outside. It must preserve every crossing prefix, endpoint convention, scalar junction, and original target coordinate. Where an exact restoration map is claimed, the required identity is

$$q = r_H \circ q_H^{\mathrm{mixed}}. \tag{LM53}$$

If additional residual coordinates are needed to make this identity true, they must be retained. One can also keep the full original delta record until healing, but then its integration and eventual analytic range require a proof. Replacing constrained old fields by freely integrated Haar variables changes LM51.

MB's mixed inverse and its identity-row boundary calculation support the classical construction once the mixed hierarchy is declared. They do not identify its quantum conditional measure, Haar/coarea factor, actual reference integral, or original output restoration. In particular MB32 assumes that the whole relevant boundary stencil really has pinned identity rows; it does not turn an omitted crossing constraint into such a row. MC explicitly preserves this distinction in its final section.

107.13.3 The prospective halo cost has the right size, but its kernel must be proved

Suppose an admissible buffered version of the actual R construction has been written with $N_H \le C_I R_e^4$ extra current cells, and its extra unmarked normalization/activity cost is bounded by

$$\exp\{C_I(1+\log x_e)R_e^4\}, \qquad x_e = g_e^{-2}. \tag{LM54}$$

This is the type of cost suggested by the original R-II estimate: compact Haar integrals retain their positivity bound, positive Gaussian forms give a constant per current cell, and the displayed normalization terms cost $O(\log x_e)$ per cell. No $\exp(Cg_e^{-2}N_H)$ may be introduced by taking a Wilson action difference without its cancellation or positive part.

When $\log x_e \le CR_e$, the logarithm in LM54 is bounded by the existing local charge $C_I R_e^5$. LM21–23 then pay its future waiting charge $C_I R_e^6$, using the original strict inequality $p_0 > 6r_0$. More generally the one-time comparison of powers is $p_0 > 4r_0 + 1$; its actual parameter choice must be retained. The size and locality of the added halo and its once-owned record are part of this calculation.

The original R-II (1.79) is stated on admissible domains, with its geometric relation (1.76) between component size and actual failed-test regions. Its numerical induction uses that relation again in (1.82). A fixed buffered enlargement near a genuine failed-test seed can be charged only after its modified version of that relation, and its subsequent enlargement/merger geometry, have been verified. An arbitrary source-only component receives no new failure credit.

Consequently this alternative has a proved finite-cutoff identity and a proved saved-age estimate, LM50–52. What remains is an explicit admissible mixed partial-R map with its exact output/constraint and measure identities, plus the uniform same-Wilson estimate LM54 and its analytic healing domain. These are narrower tasks than controlling a straddling source through an unchanged complex preparation, but the existing classical identity-boundary theorem and Fubini alone do not establish them. The alternative is therefore not yet a closure of the general first-entry transition.

108.2 A finite current representation and its weak norm

On a finite product of compact groups, let X_b^A denote an invariant unit bond vector field and put $Y_b^A = gX_b^A$, with the coupling of the record on which that bond is actually stored. Retained old records keep their own coupling. At finite order m, define a test seminorm

$$\|\varphi\|_{m,g} = \max_{0\le r\le m} \sup_{b_1,A_1,\dots,b_r,A_r} \|Y_{b_r}^{A_r}\cdots Y_{b_1}^{A_1}\varphi\|_\infty. \tag{WCT1}$$

One can use a finite collection of regular group charts instead, with the corresponding fixed equivalent seminorm. The zeroth-order term is included. The supremum over coordinate lists, rather than a sum of test derivatives, is paired below with an absolute sum of current coefficients. Thus no ambient volume factor is concealed in WCT1.

A represented distribution of order at most m has the form

$$\langle T,\varphi\rangle = \sum_{r=0}^{m}\sum_{\boldsymbol{b},\boldsymbol{A}} \int Y_{b_r}^{A_r}\cdots Y_{b_1}^{A_1}\varphi \, d\nu_{r,\boldsymbol{b},\boldsymbol{A}}. \tag{WCT2}$$

The order of the vector fields is part of the label. They need not commute. The representation mass

$$\mathcal{M}_m(T;\nu)=\sum_{r=0}^{m}\sum_{\boldsymbol{b},\boldsymbol{A}}\|\nu_{r,\boldsymbol{b},\boldsymbol{A}}\|_{\mathrm{TV}} \tag{WCT3}$$

implies

$$|\langle T,\varphi\rangle|\le \mathcal{M}_m(T;\nu)\|\varphi\|_{m,g}. \tag{WCT4}$$

One may infimize WCT3 over representations to obtain a norm, but the induction keeps the displayed representation: its anchors, tensor order, carrier, and energy ownership matter. A rooted carrier weight or an engineering factor multiplies each measure before WCT3 is taken. It is not supplied for free by WCT4.

For a smooth scalar multiplier f, the ordered Leibniz rule gives

$$\|f\varphi\|_{m,g}\le 2^m\|f\|_{m,g}\|\varphi\|_{m,g}. \tag{WCT5}$$

Hence multiplication by such an f is bounded on this represented weak class. WCT5 does not define a product of two distributions. Multiple EOM insertions are first expanded into their one joint ordered representation, as in XP23–24, including the lower currents and all contacts. This is essential even when each separate distribution is small in WCT4.

108.3 Deterministic transport with explicit target-jet constants

Let q be an actual smooth target map on a neighborhood of the support of the measures in WCT2. In local charts write the output physical coordinates as Q^a and the corresponding canonical coordinates as $z^a=Q^a/g_+$. For ordered lists of ordinary input bond derivatives set

$$\Gamma_r(q)=\sup_{\boldsymbol{b},\boldsymbol{A}}\sum_a\|X^{A_r}_{b_r}\cdots X^{A_1}_{b_1}Q^a\|_\infty,\qquad 1\le r\le m. \tag{WCT6}$$

The sum includes all output coordinates actually affected by those input anchors. The corresponding invariant-coordinate version is equivalent on a fixed admitted chart. Fixed Lie-algebra dimension and chart constants are included in the constants below.

The ordered chain rule gives

$$Y^{A_r}_{b_r}\cdots Y^{A_1}_{b_1}(\varphi\circ q)=\sum_{s=1}^{r}\sum_{a_1,\dots,a_s}C^{a_1\dots a_s}_{r,s}(\partial_{z^{a_s}}\cdots\partial_{z^{a_1}}\varphi)\circ q. \tag{WCT7}$$

The coefficients are the usual finite chain-rule polynomials in the ordered target jets. They can be defined recursively by differentiating the preceding formula, so no false commutativity of the input Haar derivatives is used. Their absolute sums obey

$$\sum_{a_1,\dots,a_s}|C^{a_1\dots a_s}_{r,s}|\le\frac{g^r}{g_+^s}B_{r,s}\big(\Gamma_1(q),\dots,\Gamma_r(q)\big), \tag{WCT8}$$

where $B_{r,s}$ is a finite polynomial with positive coefficients depending only on the order. Lower terms from invariant-coordinate frames have the same form and are retained. For a mixed list of input records, the numerator g^r is the literal product of their own recorded couplings. WCT10 applies only after the corresponding relative-coupling comparison is verified.

Deterministic weak transport. The distribution q_*T has a representation of order at most m obtained by multiplying each measure in WCT2 by its coefficients in WCT7 and pushing it forward. Consequently

$$\mathcal{M}_m(q_*T) \le C_m(q;g,g_+)\mathcal{M}_m(T;\nu), \quad C_m(q;g,g_+) = C_m \max_{r\le m}\left(1+\sum_{s=1}^{r}\frac{g^r}{g_+^s}B_{r,s}(\Gamma_1,\dots,\Gamma_r)\right). \tag{WCT9}$$

This is a direct estimate of $\langle T, \varphi\circ q\rangle$. It uses no density for q_*T, no inverse marginal, and no derivative of a cutoff. At first order it is precisely $q_*(-\operatorname{div}\nu) = -\operatorname{div} q_*(Dq\nu)$. At higher order WCT7 includes all the lower currents in XP27.

When $g \le c_g g_+$ and $g_+ \le 1$,

$$\frac{g^r}{g_+^s} \le c_g^r g_+^{r-s}, \qquad s \le r. \tag{WCT10}$$

Thus lowering the current order through a nonlinear target derivative does not create an absolute inverse coupling in this convention. Geometric and engineering scale factors remain explicit in the incoming measures and in the chosen target coordinate. WCT10 does not remove them.

Successive deterministic maps compose exactly. Their target jets can either be estimated for the actual composite or propagated by WCT7. A fixed bound C_m for each step gives only C_m^n for n steps. It is not an all-depth uniform estimate. Such uniformity requires a direct bound on the actual composite target, or a graded estimate which pays the accumulated factors. This distinction is necessary when using WCT9 in the marked feedback problem.

108.5 A finite weak packet uses one energy reserve

Suppose the complete selected joint kernel has a positive majorant G_ω, with

$$\int G_\omega e^{\theta \mathscr{E}_\omega} \le B_\omega, \qquad \theta > 0. \tag{WCT13}$$

For a weak packet represented on that same joint record, suppose the absolute sum of its order-r tensor coefficients is bounded by

$$\sum_{\boldsymbol{b},\boldsymbol{A}} |A_{r,\boldsymbol{b},\boldsymbol{A}}| \le a_r (1 + \mathscr{E}_\omega)^{d_r/2}. \tag{WCT14}$$

Then its action on an output test after the admitted deterministic readout satisfies

$$|\langle T_\omega, \varphi \rangle| \le B_\omega \|\varphi\|_{m,g_+} \sum_{r=0}^{m} a_r C_r(q; g, g_+) \left(1 + \frac{d_r}{2\theta}\right)^{d_r/2}. \tag{WCT15}$$

This follows by WCT7 and the elementary one-reserve moment bound LM26. There is one B_ω, including when several source labels occupy the same live component.

For a complete finite-source coefficient, form the exact partitions and ordered contact terms before applying WCT15. Each resulting summand has one total field/energy degree, obtained by the ordered Leibniz rule. Differentiate a quadratic or cubic compact atom and its degree drops as in LM6; retain that lower row. The finite partition sum changes the constant and total degree but never supplies another copy of the rare mass. Any explicit origin-radius cost belongs to the original record and is charged only once.

If the current was defined as the derivative of this **whole** selected kernel, WCT15 is directly applicable by its distributional definition. If it arose as a physical EOM insertion before the kernel was selected, the next section explains why its selector boundary contacts still have to be included. Merely giving both objects the name "weak current'' would not make them equal.

108.6 Why arbitrary selected kernels do not preserve this weak norm

Let K be a Markov kernel, and define $K^*\varphi(x) = \int \varphi(y) K(x, dy)$. To extend K boundedly from the weak norm dual to C^m, one needs a corresponding bound

$$\|K^*\varphi\|_{m,\mathrm{in}} \le C \|\varphi\|_{m,\mathrm{out}}. \tag{WCT16}$$

Positivity and mass one give WCT16 only at order zero.

Here is an explicit obstruction at first order. Let the input be a circle with coordinate x, the output the two-point set $\{-1,1\}$, and

$$K_N(x,\{1\}) = \tfrac{1}{2}(1+\varepsilon\sin Nx), \qquad K_N(x,\{-1\}) = \tfrac{1}{2}(1-\varepsilon\sin Nx), \quad 0<\varepsilon<1. \tag{WCT17}$$

Every K_N is a smooth positive probability kernel. For the test $\varphi(y)=y$, whose output norm is one, $K_N^*\varphi(x) = \varepsilon\sin Nx$. The input current $T=\delta_0'$ has weak order-one norm at most one, whereas

$$|\langle K_N T,\varphi\rangle| = \varepsilon N. \tag{WCT18}$$

Smooth approximations to δ_0, differentiated once and concentrated on intervals of width $o(N^{-1})$, give the same obstruction with smooth current densities and uniformly bounded representing mass. This is not a Wilson counterexample. It disproves the claimed implication from a bounded positive kernel to a uniform weak-current transport bound.

The actual selected reference move XP13 has selectors depending jointly on the old variables and the reference variables. Its dual test need not be smooth in the old variables. Summing all selectors removes that particular boundary dependence before a common readout; estimating each history separately does not. The mere permission to use an original source-zero reference denominator supplies no derivative estimate. For the outer regular-pair directions such an estimate does, however, follow from the stronger original R bounds, as proved in Section 8.

More concretely, if $\nu = \rho F X\, dU$, then

$$\chi(-\operatorname{div}\nu) = -\operatorname{div}(\chi\nu) + (X\chi)\rho F\, dU. \tag{WCT19}$$

For smooth χ this is the ordinary product rule; for sharp χ it is the corresponding complete distributional identity. WCT15 bounds the first term by testing derivatives of φ. It does not, by itself, bound or remove the second term. Replacing the left side by the first term changes the selected physical source. Higher orders retain the full ordered boundary completion. Products of independently exposed surface distributions are not introduced.

For a general kernel, XP29 therefore remains the correct identity: the fine derivative hits the actual kernel, and a bound needs either WCT16, an estimate of that whole differentiated kernel against the specific smooth incoming coefficient, or exact cancellation before historywise absolute values. The full generated score in XP31 must be included whenever an adjoint is rewritten using an effective density.

108.7 What can be reduced before the sharp partition

At a fixed finite source list, primitive Wilson EOM contacts can be formed before any history cutoff is inserted. For a compactly smooth joint coefficient and the bare Wilson density,

$$\int \rho\,(\mathsf{D}^{\dagger}_{\rho,X}F)\prod_{i=1}^{n} O_i = \sum_{i=1}^{n}\int \rho\,F\,(XO_i)\prod_{h\neq i} O_h. \tag{WCT20}$$

An ordered repetition retains the exact order of the invariant derivatives and every contact. If another EOM factor appears among the O_i, its actual scalar expression is differentiated, including the Wilson score. At any fixed finite order this produces finitely many smooth compact words and score factors. LM6–9 give their polynomial-energy envelopes on the same original Wilson variables. There is no sharp-selector derivative in this preprocessing identity.

The resulting scalar contacts can be carried through the original decomposition by XP6. This settles the bare primitive contact algebra and its incoming energy bounds at that finite order. It does not automatically remove every current row generated at later field-jet projections. A new weighted adjoint uses its full current effective density, not just the bare Wilson score, and its multiplication by the remaining source polynomial creates the contacts in XP24. Those contacts must be formed before any divergence term is discarded.

108.8 The original R kernel does control its outer regular derivatives

The abstract obstruction in Section 6 is not the strongest conclusion available for the actual R operation. Write $\mathcal{T}_z$ for the original component operation R-II (1.71), with $z = (U, J)$ its admitted outer regular pair. Keep its real integration variables, retained old coordinates, history choices, and selectors in the original fixed coordinates. R-II explicitly states just before (1.73) that only the quadratic form and the analytic remainder in this operation depend on (U, J); the real selectors in that formula are frozen in this pair.

The original positive comparison and nonvanishing statement R-II (1.73)–(1.75), printed pp.379–380, give

$$Z(z) := \mathcal{T}_z 1 \neq 0, \qquad K_z := Z(z)^{-1}\mathcal{T}_z, \qquad \|K_z\|_{\mathrm{TV}} \leq M_\sigma, \quad M_\sigma = e^{3\sigma_*}, \tag{WCT21}$$

on the same original complex domain, where σ_* is the uniform bound on its displayed complex comparison error. The internal coordinates can include a discrete history label. The norm in WCT21 is then the total variation on that same joint space. The denominator is the **original source-zero body**, not a sourced marginal or the mass of an arbitrarily selected subhistory. Its scalar factor $Z(z)$ remains in the physical effective density.

The component scope in WCT21 is essential, and its comparison factor can be bounded uniformly in precisely that scope. Immediately before (1.73), R-II p.379 bounds the entire quadratic comparison for the original admissible component by a fixed construction constant times g_k, powers of R_k, and the fixed logarithmic cutoff/radius functions. The resummed comparison of the analytic remainder V is stated to have the same bound. Consequently, for the single component operation actually defined in (1.71),

$$\sigma_{*,k,X} \leq C g_k \mathcal{P}(\log g_k^{-2}), \tag{WCT21a}$$

where $\mathcal{P}$ is one fixed polynomial after the original construction parameters are chosen. The admissible component geometry has already been used in that bound; there is no additional free $|X|$ factor. R-II p.388, immediately after (1.92), explicitly uses $|\sigma| \leq 1$ and retains the resulting constant per component. Increasing the once-chosen weak floor makes the right side of WCT21a at most $1/4$, so $M_\sigma \leq e^{3/4}$ for this original single-component operation. No origin reserve is needed to obtain this particular constant.

This conclusion does not cover an arbitrary enlarged component or an unbounded product of original component kernels. Where the original formula actually factors into n such kernels, its total-variation bound is the product of their bounds, $\exp(3\sum_{i=1}^{n}\sigma_{*,i})$, at most $e^{3n/4}$ with the stated floor. The actual component count must then be retained and paid by the declared carrier/activity reserve. It is not a single geometry-independent constant. In particular the buffered modification LM53–54 has not acquired WCT21a merely by being called an R operation.

Choose a fixed smaller outer domain with a spare coordinate radius $c\rho_k$. More precisely, an outer coordinate chart used below must map this disc into the original paired domain. The physical current-unit target charts supplied by an admitted completed minimum have this property with their actual response constants. It is not asserted for arbitrary microscopic bond directions: those can have

inverse-mesh curvature costs. At a mixed straddling readout the required chart inclusion is still part of LM47.

For any bounded measurable internal function F, the original fixed-coordinate integral $K_z F$ is holomorphic and bounded by $M_\sigma \|F\|_\infty$. This follows directly from its analytic integrand, the fixed real support, and WCT21. Equivalently, K_z is holomorphic with values in the space of finite complex measures. The measure Cauchy formula, or its scalar pairing with every such F, therefore gives

$$\|D_{u_r} \cdots D_{u_1} K_z\|_{\mathrm{TV}} \le C_r M_\sigma \rho_k^{-r} \prod_{i=1}^{r} \|u_i\|, \qquad 0 \le r \le m. \tag{WCT22}$$

The finite-order constants do not depend on the dimension of the outer coordinate space. One can prove this using a polydisc in the at most r displayed directions, of radius $c\rho_k/r$, before taking the total-variation norm. Fixed analytic chart coefficients convert this into the ordered invariant derivatives. At current-scale canonical normalization the cost is

$$\|Y_{u_r} \cdots Y_{u_1} K_z\|_{\mathrm{TV}} \le C_r M_\sigma (g_k/\rho_k)^r. \tag{WCT23}$$

Lower coordinate-frame terms have the same bound on a radius at most one. Distinct physical radii retain their literal product of inverse radii. No independent-current derivative is silently replaced by a physical derivative along $J = J_{\mathrm{phys}}(U)$.

This estimate applies to smooth tests without requiring them to be holomorphic. Indeed, for $\varphi(z,\xi)$ define $\mathcal{A}\varphi(z) = \int \varphi(z,\xi) K_z(d\xi)$. Use the ordinary ordered product rule, putting derivatives of the kernel in WCT23 and derivatives of the test in its real seminorm. It follows that

$$\|\mathcal{A}\varphi\|_{m,g_k;\,\mathrm{outer}} \le C_m M_\sigma \left[\sum_{r=0}^{m} (g_k/\rho_k)^r\right] \|\varphi\|_{m,g_k;\,\mathrm{outer}\otimes L^\infty_\xi}. \tag{WCT24}$$

If the test is further composed with a nonlinear actual target, WCT7 supplies its exact additional target-jet constants. Such constants are not implied by WCT24. Tensor orders, anchors, and the lower terms from this product rule are retained. A sum over incoming anchor measures uses their existing absolute tensor norm; it does not introduce a sum over all microscopic coordinates.

For the original shrinking radii $\rho_k \gtrsim g_k(\log g_k^{-2})^q$, the canonical ratios in WCT23–24 are uniformly bounded. A finite uncanonical inverse-radius cost can instead be charged once to the actual origin reserve as in LM34a and FR19, when it belongs to that origin record. This does not pay a cost from an uncontrolled earlier regular birth.

The same Cauchy argument applies to an **unnormalized** selected measure $\mu_{z,\omega}$ whenever the original uniform complex majorant on the spare disc is $\|\mu_{z,\omega}\|_{\mathrm{TV}} \le B_\omega$. It yields

$$\|Y_{u_r} \cdots Y_{u_1} \mu_{z,\omega}\|_{\mathrm{TV}} \le C_r (g_k/\rho_k)^r B_\omega. \tag{WCT25}$$

Here B_ω is retained once. For an energy-weighted carrying coefficient its one-reserve envelope must already hold uniformly on this same domain. Dividing by a small selected mass is unnecessary.

108.8.1 What remains for internal old-variable differentiation

WCT22 differentiates only the admitted outer pair with the internal real coordinates fixed. It does not differentiate a retained old coordinate whose variation moves a selector in (1.71), changes its real chart, or changes the reference data in another normalized comparison. Those are the internal derivatives still governed by the complete XP29 identity and the boundary completion WCT19.

There is an exact useful partition identity. If a finite or locally finite partition satisfies $\sum_\omega \chi_\omega = 1$, then

$$\sum_\omega D^{\boldsymbol{r}}(\chi_\omega \nu) = D^{\boldsymbol{r}}\nu \tag{WCT26}$$

as distributions, with the displayed ordered derivatives. The individual whole-kernel derivatives have their weak bound by testing against the corresponding ordered derivatives of one smooth function. This identity does not equate $\chi_\omega D^{\boldsymbol{r}}\nu$ with $D^{\boldsymbol{r}}(\chi_\omega\nu)$; their complete boundary contact difference is retained. Nor does WCT26 cancel these differences after arbitrary history-dependent readouts: the tests would then be $\varphi \circ q_\omega$, and cancellation needs the appropriate common values and derivatives on the selector interfaces. Proving that compatibility, or directly bounding the remaining internal boundary currents, is still necessary for a historywise marked norm.

108.8.2 A relative logarithm of the same original reference body

Fix the original real comparison pair $z_0 = (\bar{U}, 0)$ before varying the outer parameters, and use the exact exponential comparison of this same original component. Thus

$$\frac{Z(z)}{Z(z_0)} = \int e^{\sigma(z,\xi)}\,\nu_0(d\xi), \qquad \nu_0 = Z(z_0)^{-1}\mathcal{T}_{z_0}, \qquad \sigma(z_0,\xi) = 0. \tag{WCT27}$$

Here ν_0 is the positive original source-zero probability kernel for that fixed comparison pair. It is not the normalization of a selected sourced island. With the single-component ceiling $\sigma_* \le 1/4$ from WCT21a, put $q_* = e^{\sigma_*} - 1 < 1$. Then

$$\left|\frac{Z(z)}{Z(z_0)} - 1\right| \le q_*.$$

The power series for $\log(1+w)$ consequently defines one holomorphic relative branch $\mathcal{L}(z)$, with $\mathcal{L}(z_0) = 0$, on the stated disc. Its magnitude is at most $-\log(1-q_*)$. On a fixed smaller disc, ordinary finite-direction Cauchy gives

$$\begin{aligned} \|D^m\mathcal{L}(z)\| &\le C_m[-\log(1-q_*)]\rho_k^{-m}, \\ \|(g_k D)^m\mathcal{L}(z)\| &\le C_m[-\log(1-q_*)](g_k/\rho_k)^m, \qquad m \ge 1. \end{aligned} \tag{WCT28}$$

The same current-scale chart and ordered-derivative qualifications as in WCT22–24 apply. The estimate controls variation of the original scalar reference body in its **outer parameters**.

Its absolute base value $\log Z(z_0)$ is retained. In particular WCT28 compares neither the original reference bodies on different physical tori nor their nonlinear vacuum increments. If an original factorization contains n component bodies, the relative logs add and their displayed derivative bounds must be summed over those actual components. A bound on each summand is not an uncounted uniform bound on the whole product's logarithm.

The external source for this section is T. Bałaban, *Large field renormalization. II. Localization, exponentiation, and bounds for the R operation*, CMP **122** (1989), (1.71), (1.73)–(1.75), printed pp.379–380, and the explicit comparison-error bound used after (1.92), printed p.388: primary paper.

109.1 An age-tagged first-source system

Remove the known dimension-six engineering factor from the two-component center. Thus a free center transfer is the identity. Let its space be $C = \mathbb{C}^2$, or any fixed finite-dimensional normed space. A birth at step j belongs to its own Banach space B_j. Its norm includes the full physical core, the separate sharp canonical grades, the declared source-test or anchor norm, and any current/contact rows which are part of this chosen system. Exact anchor data are retained algebraically. In particular, RG's clean scalar application uses its smooth-test norm; it does not infer a strict-age absolute anchor-sum bound from that norm. No different birth domains are identified.

Suppose the *exact*, first-source equations, with all lower rows already properly filed, have the form

$$\begin{aligned} b_k &= J_k c_k + \sum_{0\le j<k} K_{k,j} b_j, \\ c_{k+1} &= (I + E_k)c_k + \sum_{0\le j<k} F_{k,j} b_j. \end{aligned} \tag{FR1}$$

Here $J_k : C \to B_k$, $K_{k,j} : B_j \to B_k$, and $F_{k,j} : B_j \to C$ are bounded linear maps. The old deterministic term is carried by its old tag; it is not counted as a new birth at each step. A triangular block of genuine lower operators may be included in C, but a nonclosed quotient of the contact algebra may not.

Assume, uniformly in the ultraviolet endpoint and the finite horizon,

$$\begin{aligned} &0 < g_k \le g_*, \qquad \|E_k\| \le a g_k, \qquad \|J_k\| \le b g_k, \\ &\|F_{k,j}\| \le f g_k w_{k-j}, \qquad \|K_{k,j}\| \le d g_k w_{k-j}, \qquad 0 \le w_n \le w_* e^{-\sigma n}, \quad \sigma > 0. \end{aligned} \tag{FR2}$$

The relative coupling powers in MT34–35 may be included in w_n, after reducing σ. An absolute negative power of a birth coupling is not covered by this hypothesis.

For definiteness first take no initial stable packet. Let $C_k : C \to C$ and $B_k : C \to B_k$ denote the linear response maps from c_0 to c_k and b_k, so $C_0 = I$.

109.4 Logarithmic promotion delays do not force a loss at every step

A fixed exponential age kernel is stronger than is needed. In particular, an R birth may have to wait for a coarse readout to fit its original shrinking physical radius. The following extension keeps the delay in the summation instead of declaring it absent.

Let $x_k = g_k^{-2} \ge h$, and suppose the actual running tube obeys

$$x_k - C_0 \le x_j \le x_k + b_+(k-j) + C_1, \qquad j \le k, \tag{FR11}$$

where the constants are nonnegative and fixed. Replace the last two bounds of FR2 by the same inequalities with a two-index weight satisfying

$$w_{k,j} \le w_* \exp\{-\sigma[(k-j) - h_j]_+\}, \qquad 0 \le h_j \le A + B\log(2 + x_j). \tag{FR12}$$

Here every inverse-radius or engineering cost multiplying this weight has already been explicitly included in the norm of the actual birth. FR12 is not a license to erase such a cost. Carrier-dependent powers must also remain in the rooted locality norm.

Delayed-kernel proposition. Under FR11–12 and the unchanged E, J bounds, FR3–4 still hold after choosing the weak floor h sufficiently large. In addition,

$$\|T_k - I - E_k\| \le C g_k^2[1 + \log(2 + x_k)]. \tag{FR13}$$

The constants depend on A, B, σ, w_*, but not on the number of steps or on the individual promotion delay.

Proof. Put $H = 2b$, $D = a + 1$ as in the no-initial-packet proof, and write

$$X_k = 2 + x_k + C_1, \qquad G_k = (x_k - C_0)^{-1/2}, \qquad N_k = \lceil 2A + 2B\log X_k \rceil. \tag{FR14}$$

Choose $h > C_0$ and $X_k \ge 2Bb_+$. For $n = k - j$,

$$h_j \le A + B\log(X_k + b_+ n) \le A + B\log X_k + n/2. \tag{FR15}$$

Thus $w_{k,j} \le w_*$ for $n \le N_k$, and $w_{k,j} \le w_* e^{-\sigma(n-N_k)/2}$ for $n > N_k$. Every g_ℓ, $j \le \ell \le k$, is at most G_k by FR11. At the inductive step, the previously constructed center factors therefore give the sharper bound $\|C_j C_k^{-1}\| \le e^{2DnG_k}$.

Provided $2DG_k \le \sigma/4$, summing the initial interval and the geometric tail gives

$$\begin{aligned} \sum_{j<k} w_{k,j} g_j \|C_j C_k^{-1}\| &\le w_* G_k e^{2DG_k N_k}\left(N_k + \frac{1}{1 - e^{-\sigma/4}}\right) \\ &\le C\frac{1 + \log(2 + x_k)}{\sqrt{x_k}}. \end{aligned} \tag{FR16}$$

The supremum of the last expression over $x_k \geq h$ tends to zero as h tends to infinity. Choose h so that this supremum, denoted by ε_h, obeys $d\varepsilon_h \leq 1/2$, $fH\varepsilon_h \leq 1$, and $D/\sqrt{h} \leq 1/2$. FR8–9 then prove FR3 exactly as before, using FR16 in place of g_*M. This remains noncircular, since the inverse factors in FR16 are all from earlier steps. FR13 follows from the same sum, and FR4 follows from the already proved factors. □

The first-source recurrence does not therefore fail merely because a promotion delay grows logarithmically. A slowly varying weak coupling also becomes smaller over that interval. A raw estimate $e^{2Dg_*h_j}$ misses this fact and can introduce unnecessary ultraviolet losses. Neither proposition covers a delay of unrestricted size with no corresponding coupling or birth-bank control.

109.6 An initial packet with a logarithmic admission delay

The delayed proposition also has an initial-packet version. This matters for a primitive insertion whose initial stable lattice remainder is bounded in the chosen graded norm but is not itself a newly generated $O(g_0)$ birth. Suppose its response R_{-1} has norm at most r_*, and its two contributions at step k have operator norms bounded by $dg_k w_{k,-1} r_*$ and $f g_k w_{k,-1} r_*$, where

$$w_{k,-1} \le w_* e^{-\sigma[(k+1)-h_{-1}]_+}, \qquad 0 \le h_{-1} \le A + B\log(2+x_0). \tag{FR20}$$

The exact equations include these maps from the initial center; they are not new terms multiplied by c_k before the normalization is proved.

Set $C_* = 2w_*$,

$$H = 2(b + dr_* C_*), \qquad D = a + 1 + f r_* C_*. \tag{FR21}$$

If $H = 0$, treat the absent births directly. Choose the weak floor large enough for the delayed-kernel induction with these constants and for the following two additional bounds. With $N_0 = \lceil 2A + 2B\log(2+x_0)\rceil$, require

$$b_+ N_0 + C_1 \le x_0/2, \qquad \exp\{2D\sqrt{2/x_0}\, N_0\} \le 2, \qquad 2D/\sqrt{h} \le \sigma/2. \tag{FR22}$$

They hold uniformly for $x_0 \ge h$, because $N_0 = O(1+\log x_0)$. There is no choice of constants depending on the ultraviolet endpoint.

Only already constructed center factors are used in the next estimate. FR11 gives $x_\ell \ge x_0 - b_+\ell - C_1$, so $g_\ell \le \sqrt{2/x_0}$ for $0 \le \ell \le N_0$. For later indices use $g_\ell \le h^{-1/2}$. Thus

$$\|C_0 C_k^{-1}\| \le 2\exp\{2Dh^{-1/2}(k-N_0)_+\}. \tag{FR23}$$

Multiplying by FR20, using $N_0 \ge h_{-1}$, and applying FR22, gives

$$w_{k,-1}\|C_0 C_k^{-1}\| \le C_* e^{-(\sigma/2)[(k+1)-h_{-1}]_+}. \tag{FR24}$$

For $k \le N_0$, this follows by keeping the full initial exponential in FR20; for $k > N_0$, the extra exponent in FR23 is at most half of $\sigma[(k+1)-h_{-1}]$. This proves the bound at the inductive step without assuming the as-yet unconstructed factor T_k^{-1}.

The initial terms now cost at most $dr_* C_* g_k$ in the birth equation and $f r_* C_* g_k$ in the center equation. FR21 leaves the same half-margin for the old-birth sum FR16 and the same unit margin for its center feedback. The induction consequently proves FR3–4 with these constants. The stable contribution from generated births remains $O(g_k^2[1+\log(2+x_k)])$; the separate initial packet contributes $O(g_k e^{-(\sigma/2)[(k+1)-h_{-1}]_+})$.

This is a delayed transient, not an extra quadratic-coupling gain. Its delay is part of the original packet's admission data. This proposition does not supply a uniform r_* for a primitive whose actual physical or canonical graded norm still contains an uncontrolled bare-coupling loss.

110.2 The two constraint maps and their multipliers

Let X, Y, Z be finite-dimensional real coordinate spaces of dimensions n, m, p, respectively. Let

$$f : X \longrightarrow \mathbb{R}, \qquad q : X \longrightarrow Y, \qquad r : Y \longrightarrow Z \tag{HC1}$$

be C^3 maps. Assume there is a C^2 local family of constrained minima $x = m_q(y)$, and a corresponding multiplier $\lambda(y)$, with

$$q(m_q(y)) = y, \qquad Df(m_q(y)) = Dq(m_q(y))^{\mathsf{t}}\lambda(y). \tag{HC2}$$

For the moment these statements are made in genuine nondegenerate coordinates; Section 6 treats the inner gauge degeneracy. Define the inner minimized action

$$F(y) = f(m_q(y)). \tag{HC3}$$

Choose y_0 critical for F under the outer constraint $r(y) = z_0$, and assume the outer differential is onto after retaining its actual independent constraint rows. Put

$$\begin{gathered} x_0 = m_q(y_0), \quad Q = Dq(x_0), \quad R = Dr(y_0), \quad S = Dm_q(y_0), \\ QS = I_Y, \qquad DF(y_0) = R^{\mathsf{t}}\Lambda. \end{gathered} \tag{HC4}$$

The exact envelope identity is

$$DF(y) = \lambda(y). \tag{HC5}$$

Indeed, differentiating (HC3) and using (HC2), $DF = Dm_q^{\mathsf{t}} Dq^{\mathsf{t}}\lambda = \lambda$. Consequently the inner multiplier at the nested point is exactly $\lambda_0 = R^{\mathsf{t}}\Lambda$.

Define the two Hessian corrections

$$\begin{aligned} A &= D^2 f(x_0) - \sum_{i=1}^{m} (\lambda_0)_i D^2 q_i(x_0), \\ B &= \sum_{\alpha=1}^{p} \Lambda_\alpha D^2 r_\alpha(y_0). \end{aligned} \tag{HC6}$$

The first is the Hessian of the inner constrained Lagrangian. The second belongs to the outer constraint. Both are necessary even when the final coarse target is held fixed.

110.3 Hessian associativity

Differentiating the stationarity equation in (HC2) with respect to y gives

$$AS = Q^{\mathsf{t}} F''(y_0), \qquad S^{\mathsf{t}} A S = F''(y_0). \tag{HC7}$$

The second identity uses $QS = I$. For every $v \in \ker Q$ it also gives

$$v^{\mathsf{t}} A S = 0. \tag{HC8}$$

Thus the derivative of the inner minimizing graph is orthogonal, in the constrained Hessian, to its vertical fiber. This is a variational identity; no choice of Euclidean orthogonal lift is assumed.

For the composed constraint $h = r \circ q$, the chain rule gives

$$\sum_{\alpha} \Lambda_\alpha D^2 h_\alpha(x_0) = \sum_i (\lambda_0)_i D^2 q_i(x_0) + Q^{\mathsf{t}} B Q. \tag{HC9}$$

Hence its constrained Hessian is exactly

$$H_{\mathrm{phys}} := D^2 f(x_0) - \sum_{\alpha} \Lambda_\alpha D^2 h_\alpha(x_0) = A - Q^{\mathsf{t}} B Q. \tag{HC10}$$

On the fine tangent space $\mathcal{V} = \ker(RQ)$, every vector has the unique decomposition

$$u = Sy + v, \qquad y = Qu \in \mathcal{W} := \ker R, \qquad v \in \ker Q. \tag{HC11}$$

Uniqueness follows by applying Q. Combining (HC7)–(HC10),

$$\langle u, H_{\mathrm{phys}} u \rangle = \langle v, Av \rangle + \langle y, (F'' - B) y \rangle. \tag{HC12}$$

This is the required associativity of constrained quadratic forms. The cross term is zero by (HC8).

If A is positive on $\ker Q$ and $F'' - B$ is positive on $\ker R$, both Gaussian forms in (HC12) are nondegenerate. The theorem remains valid with a coarse gauge quadratic, as described next.

110.4 Coarse soft gauge, actual pivot coordinates, and covariance

Let K_g be the Hessian of the actual coarse soft gauge function at the selected critical representative, in the convention

$$G(0) = DG(0) = 0, \qquad G(y) = \tfrac{1}{2} y^{\mathsf{t}} K_g y + O(|y|^3). \tag{HC13}$$

The factors prescribed by fluctuation rescaling are included in this definition. Set

$$T = F'' - B + K_g, \qquad H_g = A - Q^{\mathsf{t}} B Q + Q^{\mathsf{t}} K_g Q. \tag{HC14}$$

Then (HC12) becomes

$$\langle u, H_g u \rangle = \langle v, Av \rangle + \langle y, Ty \rangle. \tag{HC15}$$

In particular the coarse soft gauge is pulled back by the **actual** derivative Q. Omitting it would generally leave a singular coarse covariance and would not identify the Gaussian used in I (2.11)–(2.13).

Let $C : \mathbb{R}^{m-p} \to \mathcal{W}$ be the actual pivot-elimination injection. It need not be an isometry. Let $L_B : Y \to \mathbb{R}^{m-p}$ be the actual selector of the independent coordinates, so that

$$RC = 0, \qquad L_B C = I. \tag{HC16}$$

The range of C must equal $\ker R$. For this nondegenerate version assume $A|_{\ker Q} > 0$ and $C^{\mathsf{t}}TC > 0$. Define Γ_g to be the covariance of the normalized Gaussian with precision H_g on $\mathcal{V}$, using induced Euclidean measure, or any other fixed linear volume element on that tangent space. Its normalization cancels in the covariance. Then

$$\boxed{Q\Gamma_g Q^{\mathsf{t}} = C(C^{\mathsf{t}}TC)^{-1}C^{\mathsf{t}}.} \tag{HC17}$$

Consequently the covariance of the actual free coordinates is

$$\boxed{L_B Q\Gamma_g Q^{\mathsf{t}} L_B^{\mathsf{t}} = (C^{\mathsf{t}}TC)^{-1}.} \tag{HC18}$$

Proof. Parameterize u by $u = SC\xi + v$, with $v \in \ker Q$. This linear bijection has a constant Jacobian, which cancels after normalizing. By (HC15), the joint Gaussian of (ξ, v) factors into precisions $C^{\mathsf{t}}TC$ and $A|_{\ker Q}$. Since $Qu = C\xi$, its covariance is (HC17). Applying L_B proves (HC18). This proves the identity without an ambient inverse for A and without requiring F'' to be positive on all of Y.

There is also an augmented representation. In the nondegenerate setting,

$$\mathbb{K}_g = \begin{pmatrix} H_g & (RQ)^{\mathsf{t}} \\ RQ & 0 \end{pmatrix} \tag{HC19}$$

is invertible, and its fine-fine inverse block is Γ_g. To prove this, the first component of $\mathbb{K}_g^{-1}(f, 0)$ belongs to $\mathcal{V}$ and solves exactly the constrained variational equation there. Thus (HC18) gives a compressed augmented inverse on the physical graph.

No assertion of uniform locality follows solely from (HC19): the actual charts, gauge section, and their auxiliary variables must be inspected before calling this a finite-range operator in the required norm.

The covariance identity is algebraic and continues to a complex physical critical branch wherever the stated constraint pivots and restricted quadratic matrices remain invertible. To see this without postulating a complex Gaussian probability measure, choose a matrix V whose columns form a basis of $\ker Q$, and put $D = (SC, V)$. The constrained inverse is $D(D^{\mathsf{t}}H_g D)^{-1}D^{\mathsf{t}}$. By (HC15) its middle precision is block diagonal with blocks $C^{\mathsf{t}}TC$ and $V^{\mathsf{t}}AV$; multiplying by Q on the left and Q^{t} on the right proves (HC17) again. All transposes in this continuation are algebraic transposes. On a constant-rank analytic patch the bases can be chosen holomorphically; the resulting constrained inverse is independent of that choice. This proves the identity on any already admitted complex branch. It supplies no cutoff-uniform lower bound on the invertible matrices, no new common complex radius, and no identification away from that physical critical branch.

110.5 Why the nonlinear outer correction has exactly this sign and factor

It is useful to derive the outer subtraction directly in the coordinates used for the actual integration. Translate y_0, z_0 to zero. Write

$$r(y) = Ry + r_2(y) + O(|y|^3), \qquad r_2(y) = \tfrac{1}{2}D^2 r(0)[y, y]. \tag{HC20}$$

Choose the actual local pivot right inverse h, with $Rh = I$. The analytic linearizing map has the expansion

$$y = \Phi(b) = b - hr_2(b) + O(|b|^3), \qquad r(\Phi(b)) = Rb. \tag{HC21}$$

Indeed this is the quadratic term of the implicit equation defining that linearizer; it is not an additional approximation to the actual map. For $b \in \ker R$,

$$\begin{aligned} F(\Phi(b)) &= F(0) + DF(0)b - DF(0)hr_2(b) + \tfrac{1}{2}F''[b,b] + O(|b|^3) \\ &= F(0) + \tfrac{1}{2}\langle b, (F'' - B)b\rangle + O(|b|^3). \end{aligned} \tag{HC22}$$

The second equality uses $DF(0) = R^{\mathsf{t}}\Lambda$, $Rb = 0$, and $Rh = I$. Therefore $DF(0)hr_2(b) = \Lambda r_2(b) = \frac{1}{2}B[b,b]$. The sign is negative because the linearizer has the correction $-hr_2$. A formula written in terms of the quadratic polynomial r_2, rather than its Hessian, consequently has the factor $-2DF(0)hr_2$ in its precision.

Because $DG(0) = 0$, substitution of $\Phi(b) = b + O(b^2)$ does not alter the quadratic part of the soft gauge. It gives K_g in (HC14), with no additional outer multiplier term.

The first inner correction has exactly the same origin. The formula $F'' = S^{\mathsf{t}}AS$, with A from (HC6), already includes the second derivative of the nonlinear inner constraint. Replacing A by $D^2 f$ would omit its contribution.

These formulas also account for nonlinear changes of the fine coordinate chart. At a critical point of $f - \lambda_0 \cdot q$, its first derivative vanishes, so the second derivative under a chart change transforms by the linear Jacobian alone. The apparent extra chart term in $D^2 f$ cancels against the corresponding term in $\lambda_0 D^2 q$. The same observation applies to the outer Lagrangian $F - \Lambda \cdot r$.

110.6 Inner gauge zero modes

The physical Wilson Hessian has gauge zero modes. One may first formulate Sections 2–5 on the appropriate inner gauge quotient. There is also a direct finite-dimensional Gaussian formulation retaining those modes.

Let $E \subset \ker Q$ be the inner gauge tangent space at the physical critical configuration. Assume

$$Ae = 0 \quad (e \in E), \tag{HC23}$$

and A is positive on $\ker Q/E$. There are two precise ways to verify the first assertion. If the inner gauge subgroup leaves both f and the whole map q invariant, the critical Lagrangian is invariant and its Hessian annihilates the orbit tangent. More generally, for any curve in the gauge orbit contained in the fixed fiber, differentiation twice gives $A[e,e] = 0$. Positivity of $A|_{\ker Q}$ then gives $A[e,v] = 0$ for every $v \in \ker Q$; (HC8) gives $A[e, Sy] = 0$ for every y. Since $X = SY + \ker Q$, together these imply $Ae = 0$. This second argument does not assume that a target-only stabilizer fixes all nearby target values. An analytic local choice of minimizing representative supplies a right inverse S, unique modulo these gauge directions; (HC7)–(HC12) are unchanged.

Choose any complement W of E inside $\ker Q$, and let $D : E_0 \to E$ be a coordinate isomorphism. Let J be a linear inner gauge row map such that JD is invertible, and let $K_i > 0$ be its soft gauge precision. Write

$$u = SC\xi + v + D\omega, \qquad v \in W. \tag{HC24}$$

The physical part of the quadratic form is $\xi^{\mathsf{t}}C^{\mathsf{t}}TC\xi + v^{\mathsf{t}}Av$, independent of ω. The gauge part is

$$\langle J(SC\xi + v) + JD\omega, K_i[J(SC\xi + v) + JD\omega]\rangle. \tag{HC25}$$

Integration over ω, by the change $w = J(SC\xi + v) + JD\omega$, gives exactly

$$(2\pi)^{\dim E/2} |\det(JD)|^{-1} (\det K_i)^{-1/2}, \tag{HC26}$$

independent of ξ, v. Dividing by this actual gauge normalization leaves the same Gaussian law of ξ. Thus (HC17)–(HC18) hold for the fine Gaussian with this inner gauge penalty as well. The gauge rows need not annihilate $SC\xi$; the shift in ω is retained by the calculation.

This proves independence of the compressed physical covariance from a legitimate **linear inner** gauge fixing. It does not allow one to discard (HC26) in a vacuum normalization. Its background dependence, coordinate Jacobian, and the actual measure convention must still be recorded. Nor does it identify nonlinear hard gauge cutoffs or truncated domains by this Gaussian calculation.

If a residual gauge row is redundant, if a stabilizer has not been removed from E_0, or if JD is not invertible, the stated normalization is invalid. The original independent gauge-coordinate construction must be used. No global gauge or holonomy variable is removed by an implicit determinant prime in these formulas.

110.7 The normalization identity that accompanies compression

For clarity consider again the nondegenerate case of Sections 2–5, with ordinary Dirac delta conventions and Q, R onto. For each y, the exact inner Gaussian fiber integral is

$$\int_X \delta(Qu - y) e^{-\langle u, H_g u\rangle/2}\, du = \mathcal{N}_{\text{in}}\, e^{-\langle y, Ty\rangle/2}, \tag{HC27}$$

where

$$\mathcal{N}_{\text{in}} = \frac{(2\pi)^{(n-m)/2}}{\sqrt{\det(QQ^{\mathrm{t}})}\sqrt{\det(A|_{\ker Q})}}. \tag{HC28}$$

Proof: translate the fiber by Sy, apply the zero cross term (HC8), and use the linear coarea formula. The restriction determinant in (HC28) uses induced Euclidean measure on $\ker Q$.

Split the current coordinates into actual pivot and free coordinates, so that $R = (R_P, R_F)$ with R_P invertible. The injection is $C\xi = (-R_P^{-1}R_F\xi, \xi)$, up to the actual coordinate ordering. Then

$$\int_X \delta(RQu) e^{-\langle u, H_g u\rangle/2}\, du = \mathcal{N}_{\text{in}} |\det R_P|^{-1} \frac{(2\pi)^{(m-p)/2}}{\sqrt{\det(C^{\mathrm{t}}TC)}}. \tag{HC29}$$

One way to see the exact normalization is to insert $\int_Y \delta(Qu - y)dy = 1$ and then integrate $\delta(Ry)$ in the actual pivot coordinates. This also proves the delta composition without a hidden metric factor.

Equation (HC29) is an exact linear Gaussian factorization. With inner gauge zero modes, the corresponding gauge normalization (HC26) and actual quotient volume convention must be included. With Haar coordinates, nonlinear coarea, or the full interaction, their remaining densities remain in the integrand. Covariance compression does not establish a comparison of those factors between tori.

110.9 What the algebra alone does not prove

The following distinctions prevent an unjustified promotion of (HC31) by itself. Sections 10–13 below resolve items 2–4 for the clean regular physical Gaussian; item 5 remains a complete-formation task.

1. **Independent current.** The physical current equals the derivative of the Wilson action and satisfies the multiplier relation in (HC2). An independently chosen off-pair current need not satisfy it. Equality on $J = J(U)$ does not identify holomorphic extensions in all (U, J) variables: an analytic term $(J - J(U))M$ vanishes on that graph without vanishing off it.
2. **Local augmented representation.** The abstract Gaussian has the same physical compression. To use V6.46–51, one must exhibit the original constraint and gauge equations as the appropriate finite-range augmented system, with the typed exponential inverse and commutator bounds. A nonlocal gauge section or the lift S is not a finite-range matrix merely because it exists. Restoring the correct auxiliary scalar rows is necessary. Sections 10–11 do this through exact equality of the raw and compatible minimized actions, instead of using the false unrestricted raw inverse.

3. **Uniform current spectrum.** The finite-cutoff positivity used above is insufficient for the required uniform positive current-scale lower and upper bounds. Section 12 supplies those bounds on the clean physical regular core by the actual compatible lift and fixed-step cochain estimate.
4. **Vacuum normalizers.** The inner fiber, gauge, and pivot factors in (HC26)–(HC29) cancel from a normalized covariance. They do not cancel from the vacuum energy. Each must be compared with its actual counterpart. Section 13 returns to the original current Gaussian integral and includes its actual pivot and Haar soft-gauge factors, rather than retaining an unidentified auxiliary determinant.
5. **Complete nonlinear/R histories.** Even a proved Gaussian torus comparison leaves the full nonlinear birth family, its complex cores and cutoffs, and all completed R histories to be compared. No second stability theorem of the form S102 has been obtained from (HC31).

The algebra in this note preserves the same action, actual nesting, both constraint-multiplier corrections, and the exact pivot-coordinate covariance. Its gain is specific: the physical covariance compression can be derived directly from the known minimizing construction, without using an unverified identity for arbitrary off-shell propagators. The quantitative clean Gaussian comparison is proved next; S94 for the complete original state remains an outstanding formation obligation.

110.10 An exact compatible-fiber replacement on the physical graph

The raw differential in (HC31) must not be inserted into MB as though it were the compatible differential. FI disproves the corresponding unrestricted raw inverse. The following replacement uses equality of the minimized **function**, not equality of those two differentials.

Fix a regular physical inner minimum U_0, and keep it fixed as the reference of CL14–15. In group-valued notation let $q_0(W)$ be the actual raw determining map and let

$$\widetilde{q}(W) = q_0(W)^{G_{U_0}(W)}, \qquad \widetilde{q}(U_0) = q_0(U_0). \tag{HC32}$$

The same actual prefixes define both maps. Work on a sufficiently small real neighborhood with a fixed larger regular collar. Then, for every nearby target y, their regular minimizing orbits have the same action value:

$$F_0(y) = F_{\mathrm{comp}}(y). \tag{HC33}$$

Here are both directions of the fiber comparison. If $\widetilde{q}(W) = y$, extend the small nodal gauge $G_{U_0}(W)$ to an allowed fine gauge by the separated node bumps of MC24. The transformed field has raw target y and exactly the same Wilson action. Conversely suppose $q_0(W) = y$. CL10 gives $\widetilde{q}(W^g) = \widetilde{q}(W)^{\gamma_{U_0}(W,g)}$. The differential of this last gauge cocycle is the full site map CL12. Its onto right inverse is uniform: use the scalar lift MB4 and its small perturbation, or the full scalar onto map MC20. The finite-dimensional analytic inverse theorem on that chosen right-inverse subspace therefore realizes the small coarse gauge needed to change $y^{G_{U_0}(W)}$ to y. The action again stays unchanged. All gauges have uniformly small physical norm, so the two minimizing fields remain in the larger common admitted collar. These two comparisons prove (HC33). Equivalently, uniqueness of the regular critical orbit identifies the two interior minimizing branches.

This argument uses the full gauge cocycle onto statement, not just its linear value at the identity. Possible stabilizers cause no onto loss: the cocycle maps fine gauge parameters onto coarse gauge parameters before either group acts on its target. The identity of minimized actions continues holomorphically on a common admitted complex target branch. In particular their first and second derivatives agree in the same target chart. One may thus use the compatible inner constraint in the

Hessian computation of Sections 2–4 while retaining the actual next raw constraint r, actual coarse gauge G, actual pivot C, and actual current variables. No current coordinate has been replaced by a gauge-inequivalent coordinate.

110.11 The multiplier Hessian is a bounded local perturbation

All lengths in this section are in current retained units. The fine spacing is $a = L^{-k}$. Write $\ell_j = La_j$ for the intermediate one-step output lengths, so $\sum_j \ell_j \le C_L$. All second variations in (HC34)–(HC35) are evaluated at the **diagonal reference input** $V_j = U_j$ of the actual compatible step. That is precisely the second variation needed in (HC36). No unrestricted off-diagonal nonlinear L^p map theorem is being asserted. On the regular core the actual physical one-step relative map Φ_j has

$$\|D^2\Phi_j[v,w]\|_{L^1(\ell_j)} \le C_L \ell_j \|v\|_{L^p(a_j)} \|w\|_{L^{p'}(a_j)}, \qquad 1 \le p \le \infty. \tag{HC34}$$

This is a literal finite-step bound. Each step has a fixed number of links and paths depending only on L. A second variation of its physical logarithmic output has coefficient $O_L(\ell_j)$: two dimensionless link variations contribute a_j^2, and the output normalization contributes ℓ_j^{-1}. The finite path and mean Taylor series are uniformly analytic on the fixed small step chart. Their absolute bilinear kernels have bounded row and column overlap. Hölder's inequality, with the physical lattice-volume weights, gives (HC34), including the endpoint transport terms of that actual step. The norm is the bond-counting L^p norm, with physical weight a_j^4 or ℓ_j^4, and the fixed invariant norm in the Lie algebra. Each bilinear kernel term is a pair of bond values at one of finitely many relative offsets. Hölder bounds its sum by $\|v\|_p \|w\|_{p'}$; the ratio of the two volume weights is the fixed number L^4. This proves the endpoint cases $p = 1, \infty$ as well. It is not a point-evaluation bound on a continuum Sobolev space.

The first derivatives of the complete prefixes and suffixes have uniform physical L^p bounds by CL17: their bounds are at most $\exp(CM \sum_j \ell_j)$. Differentiating the composition twice and using a whole prefix and suffix in each summand gives

$$\|D^2\widetilde{q}[v,w]\|_{\ell^1} \le C\|v\|_{L^p}\|w\|_{L^{p'}}, \qquad \left\|\sum_i \lambda_i D^2\widetilde{q}_i\right\|_{L^p \to L^p} \le C\|\lambda\|_{\ell^\infty}. \tag{HC35}$$

The second inequality is the bilinear dual form of the first. It is valid for the full physical prefix, and has no factor counting its depth. Intermediate physical tangent frames telescope in the chain rule. If logarithmic charts are used to write the second derivative, their intermediate second-derivative terms cancel by differentiating the inverse-chart identities. Only the initial and final chart corrections remain; the initial physical chart has coefficient $O(a)$, and the final one has coefficient $O(1)$. Both obey the same bilinear bound. Thus one must not multiply a fresh chart equivalence constant at every intermediate level. Every kernel carrier still has current diameter bounded by C_L.

At the physical inner minimum, the multiplier has $\|\lambda\|_{\ell^\infty} \le C\epsilon$. This follows directly from MC26: the five-type reference inverse applied to $(\mathcal{Q}^{\mathrm{t}}\lambda, 0, 0, 0, 0)$ has exact output $(0, 0, 0, 0, \lambda)$, and the input is the weak Wilson current, bounded by $C\epsilon$ in the admitted regular chart. The current row space has a fixed number of coordinates per cell, so its uniformly local ℓ^p bound controls ℓ^∞.

Put $\mathcal{Q} = D\widetilde{q}(U_0)$. The actual inner constrained Hessian is therefore

$$A_\lambda = H_{\log,W}(U_0) - \sum_i \lambda_i D^2\widetilde{q}_i(U_0). \tag{HC36}$$

The second term is an additional term; it is not the relative-coordinate current term already included in BI23. Linearizing MC9 at its physical solution $b = \psi = \phi = \nu = 0,\ \mu = -\lambda$ gives exactly the MB

five-type matrix with first row

$$(A_\lambda + \mathcal{Q}^{\mathsf{t}}\kappa\mathcal{Q})v + d_U\Delta_U\phi + \mathcal{Q}^{\mathsf{t}}\dot{\mu}. \tag{HC37}$$

The other four rows are the same scalar, gauge, and constraint rows as MB. By (HC35), this is an $O(\epsilon)$ perturbation in the full typed spaces. Hence the MB inverse and its exponentially summable current-cell bounds persist after decreasing the regular field ceiling once. This uses the compatible row, and never invokes the false raw inverse of FI.

The perturbation also has the required cutoff estimate. In a bilinear kernel of (HC35), the commutator has the factor $\chi(x) - \chi(x')$. Its complete carrier has bounded current diameter. For a current cutoff varying on radius R, the absolute kernel bound gains C/R. Thus the new multiplier term maps X to Y with commutator norm $C\epsilon/R$, in addition to the already checked MB34 differential, scalar, and mean commutators. This verifies a local augmented realization for the clean physical inner problem; it is not an identification of arbitrary off-pair maps.

There is a particularly useful exact readout. Let $\mathbb{K}_\lambda$ denote (HC37) with the other four rows, let I_B put a current vector into its constraint row, and let P_μ read its multiplier row. Differentiating the actual inner minimum with respect to its target gives

$$F'' = -\kappa I - P_\mu \mathbb{K}_\lambda^{-1} I_B. \tag{HC38}$$

Indeed the physical multiplier at target y is $-\kappa y - \mu(y)$. The auxiliary scalar force vanishes along the nonlinear physical minimizing family by MC20–22; its derivative therefore vanishes as well. Equation (HC38) is an identity for the Hessian of (HC33), with no replacement of the Wilson objective. It gives a current-coordinate Schur readout of a uniformly bounded typed inverse, without any fine-coordinate determinant trace.

110.12 A uniform current spectral bound

Let b be the current field before pivot elimination, with the actual next constraint $Rb = 0$. The actual one-step coarse soft gauge has quadratic form $K_g[b, b]$. The following fixed-L cochain estimate is uniform in the number of current cells:

$$\|b\|_2^2 \le C_L\{\|d_V b\|_2^2 + \|Rb\|_2^2 + K_g[b, b]\}. \tag{HC39}$$

For clarity, its flat proof keeps the actual raw paths. Write Jb for the rooted within-block path integrals appearing in the linear axial gauge. On a gradient, $Jd\varphi$ is the endpoint value minus the root value, independently of the path. Set the root values of $\varphi = Jb$ to zero and subtract $d\varphi$. This kills the gauge row and leaves the raw coarse constraint unchanged, because $Rd\varphi = d_L N\varphi = 0$. Every internal bond of the resulting field differs from its rooted path difference by a sum of plaquette curls inside its block. Every crossing bond differs from the actual averaged coarse path by such a sum inside the two adjacent blocks. These finite path loops have fillings of bounded size depending only on L. Squaring, summing, and using bounded block overlap proves the flat estimate, including its actual positive path-gauge weights. On a regular nonflat current background, use a common small analysis chart on each such two-block carrier. The raw row, the path gauge, and the curl differ by $O_L(\epsilon)$ bounded current matrices. Their errors are absorbed in the left side. This proves (HC39) without assuming a global gauge with small potential or removing holonomies. The quadratic gauge is transported together with its base representative. In particular the analysis uses the variation of the relative rooted transport $P_{V'}P_V^{-1}$; it does not apply the absolute gauge-fixing function afresh to a background that has been moved out of its original axial representative.

In the fine compatible projected gauge, the MB energy estimate gives

$$\|v\|_{H^1}^2 \le C\{\|d_U v\|_2^2 + \|\mathcal{Q}v\|_2^2\}. \tag{HC40}$$

Here the projected gauge is $\Pi_U \delta_U v = 0$, where Π_U is the orthogonal projection onto $\Delta_U \ker S_U$. The following argument records both its coercivity and its zero-mode content. Replace the Wilson Hessian in the MB saddle system by the positive covariant curl Hessian. This is a uniformly small $H^1 \to H^{-1}$ perturbation on the regular core, by the direct unconstrained Wilson second-variation bound; that bilinear estimate does not use (HC40). Hence the full typed saddle inverse still exists. Eliminating its scalar rows gives, on $\ker \mathcal{Q}$, the symmetric nonnegative form $\|d_U v\|_2^2 + \|\Pi_U \delta_U v\|_2^2$. Its inverse maps the dual energy space to the energy space with a uniform bound. Applying the Hilbert-space Riesz map therefore makes it a positive self-adjoint operator with inverse norm bounded by that same constant. Its spectrum is bounded below uniformly, so the form controls $\|v\|_{H^1}^2$ on $\ker \mathcal{Q}$. This is an energy-space statement, not a bound on a microscopic Euclidean condition number.

For general target $b = \mathcal{Q}v$, subtract the bounded row lift in the projected gauge constructed below. Its difference lies in $\ker \mathcal{Q}$ and still obeys the projected gauge, so the preceding coercivity gives (HC40). The lift construction uses only the scalar onto estimate and row inf-sup, not (HC40).

At the flat reference the absence of omitted modes is also explicit. A zero-curl periodic field is $d\varphi + h$, with harmonic one-form h constant. The boxed mean retains h, and $Qd\varphi = d_c S\varphi$. Thus $Qv = 0$, summed around coarse cycles, first forces $h = 0$, then forces $S\varphi$ to be constant. The projected gauge makes φ the constrained scalar minimum for those constant averages; it is constant, so $v = 0$. The scalar constant itself is removed from the auxiliary inversion by $S\phi = 0$. No flat toron or nonzero field gauge direction is omitted. The positive-form inverse argument above extends this control to the admitted nonflat regular references without an irreducibility premise.

The complete compatible row also satisfies the cochain comparison

$$\|d_V \mathcal{Q}v\|_2 \le C\|d_U v\|_2 + C\epsilon\|v\|_2. \tag{HC41}$$

At zero background this is the exact box-and-line cochain identity, whose two-form average is an L^2 contraction up to fixed normalization constants. In each common local analysis chart, CL17 and its whole-prefix sum give $\mathcal{Q} = Q_{\mathrm{bl}} + O(\epsilon)$ in L^2 row norm. The output mesh is one, so differentiating this error costs a fixed constant, not a^{-1}. Replacing ordinary by covariant curls introduces only $O(\epsilon)\|v\|_2$. Summation over the bounded overlap of complete carriers proves (HC41).

The regular Wilson Hessian estimate and (HC35) give the real energy comparison

$$|A_\lambda[v,v] - \|d_U v\|_2^2| \le C\epsilon\|v\|_{H^1}^2. \tag{HC42}$$

This energy version follows directly from the BI23 second-variation proof: the potential coefficients are small in $W^{1,p}$, $p > 4$, and the two varied fields are in $H^1 \subset L^4$. The flux and coordinate-current terms are bounded by the same local Hölder pairings. The multiplier contribution uses (HC35) with $p = 2$. All comparisons can be made in the fixed regular covariant energy norms, using the local CE charts.

There is a uniformly bounded lift of current data into this fine compatible gauge. Begin with MB12. Correct its actual row by $E_0(\mathcal{Q}E_0)^{-1}$, using the small full-space row error. Then subtract a gauge variation with $S_U\varphi = 0$, solving the scalar projected gauge equation by MB4–6. The exact chain $\mathcal{Q}d_U = d_V S_U$ preserves the target in that correction. The resulting lift has H^1 norm at most $C\|b\|_2$. The same construction proves positivity of A_λ on the inner gauge-fixed constraint kernel for a smaller fixed ceiling. Its Schur minimum is the actual $F''[b,b]$, by (HC33) and constrained-Hessian associativity.

The outer multiplier satisfies $\|\Lambda\|_\infty \le C_L\epsilon$, either from its actual next-scale multiplier estimate or from $\lambda = R^{\mathsf{t}}\Lambda$ and the bounded actual pivot right inverse. Its fixed-step correction obeys $|B[b,b]| \le C_L\epsilon\|b\|_2^2$.

Apply (HC39)–(HC42) to the inner minimizing lift of $b \in \ker R$. First (HC40) absorbs the fine $\epsilon\|v\|_{H^1}^2$ error. Next (HC39) and (HC41) show that $\|b\|_2^2 \le C(\|d_U v\|_2^2 + K_g[b,b])$, after absorption of the remaining $\epsilon^2\|v\|_2^2$ term. Thus, for one fixed sufficiently small regular ceiling,

$$c\|b\|_2^2 \le (F'' - B + K_g)[b,b] \le C\|b\|_2^2, \qquad Rb = 0. \tag{HC43}$$

For the upper bound, use the bounded fine lift just constructed as a trial field for the inner Schur form and the bounded fixed-step gauge and outer correction. The lower bound holds for every admissible inner lift, so in particular for its minimizer.

The original current pivot map has fixed-L support and bounded norm on this regular core. Its actual selector gives $\|\xi\|_2 \le \|C\xi\|_2 \le C_L\|\xi\|_2$. Consequently, with precisely the original current precision,

$$\mathcal{D} = C^{\mathsf{t}}(F'' - B + K_g)C, \qquad c_0 I \le \mathcal{D} \le C_0 I, \qquad C^{(k)} = \mathcal{D}^{-1}. \tag{HC44}$$

These are current-scale bounds, uniform in the hidden fine spacing and depth. They do not assert a uniform Euclidean condition number for $\mathbb{K}_\lambda$.

110.13 The clean physical Gaussian has an exponential torus defect

Clean Gaussian torus-comparison theorem. On the admitted clean regular physical core, at the same Wilson coupling and with the same actual averages, gauge insertion, and current measure, the one-step Gaussian normalization obeys

$$\left|\frac{\log Z_{G,T}}{\mathrm{Vol}(T)} - \frac{\log Z_{G,T'}}{\mathrm{Vol}(T')}\right| \le C\ell_k^{-4} e^{-cT_*/\ell_k}$$

for a torus and an integer periodic cover with copied physical target data. The constants are independent of the fine cutoff, the number of hidden blocking levels, and the cover. Exact local measure and soft-gauge constants have the same density on both sides. The proof below uses HC33–44 and establishes the complete Gaussian assertion; it does not include the nonlinear normalizer N_k.

Consider a periodic target on a torus and its periodic copy on an integer cover, at the same Wilson coefficient. The actual raw path maps commute with this copying. Repeat the fine critical configuration and its multipliers. Their local Euler equations and target equations remain exactly satisfied on the cover. Uniqueness of the admitted regular minimizing orbit identifies this with the covered physical minimum, up to its prescribed gauge. Thus the base backgrounds and multipliers used in (HC36)–(HC38) can be chosen as literal copies. This is uniqueness only in the regular branch specified by the original variational construction and its common bounds. Copying preserves those local bounds and hence stays in that branch. No uniqueness assertion is made for the full compact Wilson action or for other critical or topological branches outside that class. This statement concerns the base critical field. Its responses to an individual nonperiodic current perturbation need not be copies.

All entries of $\mathbb{K}_\lambda$ on a common lifted carrier are then literally the same: the actual prefix paths, scalar rows, Wilson plaquettes, and multiplier-Hessian terms have been retained. The commutator argument (V6.46)–(V6.47), applied to (HC38), gives

$$|F''_T(x,y) - F''_{T'}(x,y)| \le Ce^{-cD} \tag{HC45}$$

for current roots in a ball whose complete surrounding common collar has current radius proportional to D. It also gives exponentially summable current entries for F''. The local outer correction, soft gauge, and pivot coefficients agree exactly on that collar. Therefore the same comparison and entry

bounds hold for $\mathcal{D}$. Combine these facts with (HC44). The weighted resolvent and principal-restriction proof (V6.49)–(V6.51) gives

$$\begin{aligned} |C_T^{(k)}(x,y) - C_{T'}^{(k)}(x,y)| &\le Ce^{-c'D}, \\ |(\log \mathcal{D}_T)(x,x) - (\log \mathcal{D}_{T'})(x,x)| &\le Ce^{-c'D}. \end{aligned} \tag{HC46}$$

The first line is for roots with the same protected placement as above. No fine-coordinate trace occurs. There are at most $4 \dim G$ free current coordinates per current cell. Taking D proportional to the shorter torus circumference in current units consequently proves the per-current-cell Gaussian log-determinant defect

$$\left| \frac{\log \det \mathcal{D}_T}{N_T} - \frac{\log \det \mathcal{D}_{T'}}{N_{T'}} \right| \le Ce^{-cT_*/\ell_k}. \tag{HC47}$$

The same conclusion holds for the **original current Gaussian normalization**. In its original current variables its integral is

$$Z_G = \int \delta(Rb) \exp[-\tfrac{1}{2}(F'' - B + K_g)[b,b]]\, db = |\det R_P|^{-1} (2\pi)^{N_{\text{free}}/2} (\det \mathcal{D})^{-1/2}. \tag{HC48}$$

The actual pivot factor is a product of fixed-step local factors and therefore agrees per cell under copying. The number N_{free} is the actual current bond count minus the actual outer constraint count; the soft gauge variables have not been deleted. Fixed Haar linearization densities and fluctuation scaling powers, wherever the original convention places them, have their exact current bond and constraint counts and agree per physical volume at equal coupling. If fine scalar or inner gauge inverses are introduced to compute this integral, their auxiliary determinants must reproduce (HC48), as in (HC26)–(HC29); they are not additional unexplained vacuum factors.

Thus the clean physical Gaussian normalization has a proved source-zero wrapping bound. This statement does not identify the full nonlinear normalizer N_k, which I defines by the entire remainder integral at the identity. In particular (HC48) is not a redefinition of that normalizer. Its own-scale sum is bounded by TD11–12.

The original soft gauge insertion can be checked separately and exactly, including its cutoff. I (0.11), (0.14)–(0.16) uses the averaged rooted contour $P_V(y,x)$, a one-group-variable weight w_k (the exponential gauge weight with its stated restriction), and one gauge parameter $g(x)$ for each nonroot site of a current block. Set $g(y) = 1$ at every next-scale root. Endpoint covariance and Haar invariance give

$$\begin{aligned} &\int_{g(y)=1} \prod_y \prod_{x \in B(y),\, x \ne y} w_k(P_{V^g}(y,x)) \prod_x dg(x) \\ &\qquad = \prod_y \prod_{x \in B(y),\, x \ne y} \int_G w_k(h)\, dh = z_k^{N_{\text{nr}}}. \end{aligned} \tag{HC49}$$

Indeed $P_{V^g}(y,x) = P_V(y,x) g(x)^{-1}$, and each change $h = P_V(y,x) g(x)^{-1}$ preserves normalized Haar measure. The identity holds even though the averaged path is nonlinear in the links. The original insertion divides by exactly these copies of z_k. At quadratic order its gauge derivative on a root-fixed parameter is the endpoint value, with unit triangular Jacobian; there is no unrecorded background-dependent Faddeev determinant. Here $N_{\text{nr}} = (L^4 - 1) N_{\text{next}}$, with the original normalization of group coordinates. These are current-scale sites, not hidden microscopic sites. Thus this factor also agrees exactly per physical volume under a cover at equal coupling. The root-fixed gauge group has no unfixed global gauge parameter. No determinant prime or division by a cutoff-dependent global gauge volume appears.

The clean Gaussian comparison has a common complex physical neighborhood as well. Use the fixed real reference and the first-order complex perturbation norm of MC2, retaining its separately imposed own-scale matrix-curvature bound. The typed coefficient estimates, (HC35), and the resolvent identity make the change of the current precision bounded by $C\epsilon_c$ in current ℓ^2 operator norm. Choose this one physical complex radius so that $C\epsilon_c < c_0/2$. Every full or principal current precision is then accretive relative to its real reference, with real part at least $c_0/2$. The inverse, weighted resolvent estimates, and logarithm branch used in (HC45)–(HC48) persist on that radius. The branch is fixed by continuation from the positive real determinant. This is a current-operator perturbation estimate, not a count of microscopic eigenvalues. The complex curvature condition is the already proved physical-branch CC/MC condition; it has not been inferred from $W^{1,p}$ alone or extended to arbitrary independent currents.

110.14 Fixed-coordinate Gaussian sensitivity and the remaining formation

The preceding argument establishes the primitive physical Gaussian comparison on the clean regular core. It avoids an unnecessary off-pair identity requirement: a direct comparison of physical Gaussian formation values only uses the physical minimized action. The original paired analytic extension may still supply bounds, but it need not equal the compatible construction away from the graph.

There is also a useful exact comparison which avoids moving the clean sharp tests. On any common finite set of actual current coordinates, let the normalized Gaussian precisions be D_1, D_2, with the uniform bounds of (HC44), and set $E = D_2 - D_1$. In these same coordinates their density ratio is

$$\frac{d\mu_2}{d\mu_1}(b) = \exp\left\{\tfrac{1}{2}\log\frac{\det D_2}{\det D_1} - \tfrac{1}{2}b^{\mathsf{t}}Eb\right\}. \tag{HC50}$$

If n is the number of those current coordinates and $\|E\| \le \eta < c_0/4$, then

$$\left|\log\frac{d\mu_2}{d\mu_1}(b)\right| \le Cn\eta + \tfrac{1}{2}\eta|b|^2. \tag{HC51}$$

The determinant term follows by integrating $\mathrm{Tr}[(D_1+tE)^{-1}E]$. No regularity of a common measurable selector is used in these formulas. A fixed selector can be kept inside both integrals. The changed measure is then exactly a constant plus quadratic carrying potential, controlled by a single reserved Gaussian tilt. Principal covariance marginals retain the same positive spectral bounds; entrywise errors on a carrier of n coordinates cost at most the explicit polynomial n when converted to operator norm. Such current-carrier powers can be paid from spare locality, with no hidden fine-coordinate count.

For the clean step, I (2.9) uses probes of the current relative field B'. Its one-step nonlinear straightening $B' = b - hD(b)$ depends on the copied current base and fixed-step paths, so these clean selectors are the **same functions** of b on copied nonwrapping data. The soft-gauge selector in (HC49) has the same property. Using (HC50), rather than introducing different covariance square roots, avoids an artificial movement of those boundaries. A genuinely changed nonlocal reference or an old stored selector is a different case, not covered by this observation.

For the original complete source-zero state one still needs an actual one-step estimate of the form TD2 or TD5. Its input difference must include the inherited activities and coupling rows, the original cutoffs and retained variables, and every completed rough history. The Gaussian comparison above supplies an exponentially small primitive forcing term, not that complete estimate. A stable analytic cluster map is Lipschitz on a common admitted activity ball by the fundamental theorem of calculus

and its bounded derivative; this fact is applicable only after the two actual completed integrands, their domains, and their normalization are put on that same ball.

In particular the following items are still not supplied by HC:

- A uniform comparison of the original sharp nonlinear curvature and retained-history selectors that cannot be kept fixed by the preceding common-coordinate comparison. TD14–15 applies to specified nondegenerate current linear probes; it is not an estimate for every old or fine nonlinear selector.
- A bounded one-step sensitivity of all completed R formation and healing operations, with their original denominators and all origin/area reserves retained once.
- The accompanying nonlinear vacuum increment estimate in the original N_k convention. A Gaussian determinant bound alone does not control this scalar.

Once these actual estimates are obtained, TD-L or TD-H propagates the defect without requiring a contracting one-step norm; TD11–12 then provides the ultraviolet summation. They have not been assumed in the proof of the clean Gaussian result, and S94 remains open.

111.3 One common complex domain for this matrix construction

The determinant and inverse construction above admits a uniform holomorphic extension. This addresses its Gaussian matrix domain; it does not establish the nonlinear curvature-domain inclusion for the entire Yang–Mills iteration.

Let K^0 be any of the real matrices satisfying (V6.3), and let $K = K^0 + E$ be a complex matrix with the same stencil. Use the algebraic transpose, not the conjugate transpose, in $K^{\mathsf{t}}K$. Since

$$\|K^{\mathsf{t}}K - (K^0)^{\mathsf{t}}K^0\| \le 2M\|E\| + \|E\|^2, \tag{V6.12}$$

the condition

$$2M\|E\| + \|E\|^2 \le m^2/2 \tag{V6.13}$$

implies

$$\|I - M^{-2}K^{\mathsf{t}}K\| \le \rho_{\mathbb{C}} := 1 - m^2/(2M^2) < 1. \tag{V6.14}$$

For example $\|E\| \le m^2/(8M)$ is sufficient. This radius does not contain the matrix dimension or torus size. The two series (V6.5), (V6.7) therefore converge uniformly on every smaller such neighborhood, with $\rho_{\mathbb{C}}$ in place of ρ. For the inverse-tail bound the prefactor M^{-1} is replaced by $(M + \|E\|)/M^2$. They are holomorphic there.

For the determinant define

$$\mathcal{D}(K) = D \log M - \frac{1}{2}\sum_{k\ge 1}\frac{1}{k}\operatorname{Tr}(I - M^{-2}K^{\mathsf{t}}K)^k. \tag{V6.15}$$

At real K^0, this is $\log|\det K^0|$. Moreover $\exp(2\mathcal{D}(K)) = (\det K)^2$. On a connected neighborhood it is the analytic branch determined by that positive real normalization. Thus orientation and square root choices are fixed explicitly. No dimension-dependent choice of a new branch is made on a second torus.

Exact copied-stencil equality (V6.10) and its tail estimate hold throughout a common complex tube whenever (V6.13) is uniform. For finitely many holomorphic field/source variables, Cauchy's formula on a smaller common polydisc also gives the same exponential torus error for their derivatives. This derivative statement has the usual explicit Cauchy radii and factorials; it is not an all-order bound with those costs omitted.

111.4 Leaving the gauge and constraint variables uneliminated

Here is the exact Gaussian algebra that makes the preceding matrix theorem relevant to scalar/coarse inverses and projectors. It also records which normalization must accompany any change of the localization of those inverses.

Let $C : \mathbb{R}^N \to \mathbb{R}^r$ be onto, and let H be a real symmetric matrix strictly positive on $\ker C$. No positivity on the full ambient space is needed. Define the saddle matrix

$$\mathbb{S} = \begin{pmatrix} H & C^{\mathsf{t}} \\ C & 0 \end{pmatrix}. \tag{V6.16}$$

Choose an orthonormal matrix Z whose columns span $\ker C$, and put $H_0 = Z^{\mathsf{t}} H Z$. Complete Z by the orthonormal columns $Y = C^{\mathsf{t}}(CC^{\mathsf{t}})^{-1/2}$. Then $CY = (CC^{\mathsf{t}})^{1/2}$. In this orthonormal basis, block elimination, or expansion in the last block row, gives

$$\det \mathbb{S} = (-1)^r \det(CC^{\mathsf{t}}) \det H_0. \tag{V6.17}$$

This also proves invertibility of $\mathbb{S}$.

With the ordinary Dirac delta on $\mathbb{R}^r$, coarea gives

$$\begin{aligned} I(H, C) &:= \int_{\mathbb{R}^N} \delta(Cx) e^{-\langle x, Hx\rangle/2}\, dx \\ &= \frac{1}{\sqrt{\det(CC^{\mathsf{t}})}} \int_{\ker C} e^{-\langle x, Hx\rangle/2}\, d\sigma(x) \\ &= \frac{(2\pi)^{(N-r)/2}}{\sqrt{|\det \mathbb{S}|}}. \end{aligned} \tag{V6.18}$$

If the desired measure is induced Euclidean measure on $\ker C$, rather than $\delta(Cx)dx$, the additional factor $\sqrt{\det(CC^{\mathsf{t}})}$ must be retained. If a normalized gauge delta or an independently normalized soft gauge factor is used, its precise factor must likewise be included. Formula (V6.18) is not permission to identify these different conventions.

The covariance on the constrained physical coordinates is

$$Z H_0^{-1} Z^{\mathsf{t}} = \left(\mathbb{S}^{-1}\right)_{xx}. \tag{V6.19}$$

Indeed the first block of the solution of $\mathbb{S}(x, \lambda) = (f, 0)$ belongs to $\ker C$ and satisfies its restricted variational equation. Thus the covariance can be reconstructed from the local augmented matrix without separately localizing an already eliminated projector or coarse Schur inverse.

111.4.1 Quantitative saddle invertibility

Suppose, in addition,

$$CC^{\mathsf{t}} \geq \gamma^2 I, \qquad H_0 \geq hI, \qquad \|H\| \leq H_*, \quad \|C\| \leq C_*. \tag{V6.20}$$

These conditions imply a dimension-free bound on $\|\mathbb{S}^{-1}\|$, depending only on γ, h, H_*. To see it, put $R_C = C^{\mathsf{t}}(CC^{\mathsf{t}})^{-1}$, so $\|R_C\| \leq \gamma^{-1}$. For $\mathbb{S}(x, \lambda) = (f, b)$, write $x = R_C b + z$, with $z \in \ker C$. The restricted equation gives

$$\|z\| \leq h^{-1}(\|f\| + H_* \gamma^{-1} \|b\|), \qquad \|\lambda\| \leq \gamma^{-1}(\|f\| + H_* \|x\|). \tag{V6.21}$$

These are the required inverse estimates; an upper bound for $\|\mathbb{S}\|$ follows from H_*, C_*. If H, C have fixed finite stencils, the matrix theorem applies.

This proof retains all multiplier variables. It does not assume that the separately eliminated scalar or coarse inverse is local. Conversely, if a proposed set of gauge rows has redundant rows, (V6.20) is false. One must exhibit the independent local rows or use the correct full augmented system; dropping a global gauge or holonomy mode by hand can change the normalization count and is not allowed.

111.4.2 Matrix theorem versus the existing typed estimates

The boxed and mixed inverse notes prove bounds in typed Sobolev norms, including fine-lattice differential orders. One cannot substitute those bounds into (V6.3) without checking the coordinate normalization and the matrix upper bound M. An unpreconditioned fine Laplacian has norm of order a^{-2}. With a fixed current-scale spectral gap, (V6.4) then has $1-\rho$ of order a^4, and the polynomial degree needed for a useful tail can exceed the circumference in fine units. That would not prove (V6.11) with a cutoff-independent current-scale rate.

For a single blocking step with fixed fine-to-current ratio L, all stencil sizes and local coordinate dimensions are fixed, and this obstruction is absent. For an arbitrary multiscale auxiliary matrix it is a real additional check. A local current-scale representation, or a proved local preconditioner with its own carrier and determinant factors, is needed. A global Sobolev preconditioner cannot simply be declared local.

111.5 Wilson-specific primitive naturality at nonflat backgrounds

The matrix theorem is general. Some of its geometric inputs can be verified directly for the Wilson action without the unavailable definitions in the old background-propagator paper.

Use exactly the action S1–S2 and, on a fixed branch of the complexified group, its holomorphic plaquette expression

$$s_{\mathbb{C}}(U_p) = 1 - \frac{\operatorname{Tr} U_p + \operatorname{Tr} U_p^{-1}}{2N}. \tag{V6.22}$$

On the compact real group this equals $1 - N^{-1} \operatorname{Re} \operatorname{Tr} U_p$. Write a link in a relative chart as $e^{A_b} U_b$, with the usual inverse for the oppositely oriented link. Every derivative of one plaquette action uses only its four oriented links. In particular

$$H_U^W(b, b') = 0 \quad \text{if no plaquette contains both } b, b'. \tag{V6.23}$$

If the full oriented plaquette and its background link values are copied to another torus, the two analytic functions of the relative variables are identical. Their Hessians and every higher derivative therefore agree exactly. This holds for nonflat U; no truncation at $U = 1$ has occurred.

On any fixed relative-chart tube the derivatives of each plaquette term have uniform bounds, because they are finite products of matrix exponentials, inverses, and traces with fixed length. Each link meets only a bounded number of plaquettes. The matrix Schur bound then gives

$$\|H_U^W - H_{\widetilde{U}}^W\| \le C \max_b \|U_b - \widetilde{U}_b\| \tag{V6.24}$$

when the two backgrounds are compared in the same local frame and the same chart. The constant is independent of the number of cells. An overall β_a factor is included on both sides if it has not already been removed by the prescribed fluctuation rescaling.

The same entrywise argument applies to a **specified** finite path average or gauge row: include every link in each reference-transport path, its chosen logarithm branch, and its endpoints in the carrier. Products, their derivatives, and the covariant difference matrix then coincide on copied data. Their row and column bounds are uniform if path lengths and overlaps are uniform at the current scale. Any current-dependent Lagrange-multiplier contribution to the action Hessian must be copied as well.

For a finite local Haar integral, copying the full variable list and the integrand also copies the measure. In relative Lie-algebra coordinates the Haar density is the same local function. Coarea Jacobians are the same finite Jacobians, with the same independent row selection and coordinate normalization. Thus there is no new gauge-volume factor merely from the embedding of this specified finite integral into a larger torus.

These are verified primitive statements for the same Wilson action. They do **not** prove the following three facts that would be needed for the complete application:

- that the original effective covariance is a specified block of this particular local augmented inverse, with all its determinant factors identified;
- that the uniform bounds (V6.3), in current-scale coordinates, hold on every original regular and large-field transition domain;
- that every inherited effective activity is itself the same analytic function of the copied complete carrier. Its local external-field support does not by itself establish that last equality.

111.5.1 A model-specific fixed-L Gaussian layer

There is a useful case in which the uniform spectral inputs can also be proved directly. This subsection concerns **one** homogeneous layer of fixed blocking factor L, in a small relative-link chart. It is not an unelimination of an arbitrary number of old scales.

Put the fine spacing equal to one in this subsection; the current retained spacing is the fixed integer L. Partition sites into disjoint L-blocks $B(y)$, one root at each retained site y. Take the reference torus large enough that two adjacent blocks have a common injective lift. Choose a spanning tree in each block by the same coordinate rule. The trees have uniformly bounded lengths and stay inside their own blocks. Let $P_{\mathrm{tr}}A$ be the list of values of a real Lie-algebra one-form on the oriented tree edges.

The flat box-and-line average can be written as

$$(Q_0A)(y,\mu) = L^{-4} \sum_{x\in B(y)} \sum_{i=0}^{L-1} A_\mu(x + ie_\mu), \tag{V6.34}$$

On the zero-tree subspace each straight path can be closed by the tree paths from its two endpoints to their respective block roots. The added line integrals vanish. Those tree paths are used to prove the estimate below; they are not added to the definition of Q_0. This is the box-and-line differential appearing in BI1; it is not the NODE-compatible mixed relative differential distinguished in MB6. Only its homogeneous single-layer version is used here. With any fixed positive normalized path weights, the same proof works. Equation (V6.34) uses the bond-integral convention. If a physical-component convention divides its line sum by L, conjugate that row by this fixed factor and retain its delta/coarea factor. Since L is fixed, all bounds below persist; the two conventions must not be interchanged inside an exact normalization formula.

Set

$$C_0A = (P_{\mathrm{tr}}A, Q_0A). \tag{V6.35}$$

The rows of C_0 have fixed range and overlap depending on L only. They are onto with an explicit bounded right inverse. To see this, given retained bond values $b(y,\mu)$, put $R_b b = b(y,\mu)$ on **all** fine links crossing the μ-face from $B(y)$ to $B(y+Le_\mu)$, and zero elsewhere. A straight path in (V6.34) crosses that face exactly once and crosses no other block face. Hence

$$P_{\mathrm{tr}} R_b = 0, \qquad Q_0 R_b = I, \qquad \|R_b\| \le C_L. \tag{V6.36}$$

Here and below the finite number of link and row types per block is absorbed into C_L. Given tree values t, let $E_t t$ be their extension by zero to all other links. Then

$$R_0(t,b) = E_t t + R_b(b - Q_0 E_t t), \qquad C_0 R_0 = I, \qquad \|R_0\| \le C_L. \tag{V6.37}$$

This proves $C_0 C_0^{\mathsf{t}} \ge \gamma_L^2 I$ uniformly in the number of blocks. In particular there are no unremoved constant constraint rows.

Next, on the simultaneous zero-tree and zero-average subspace,

$$\|A\|_2^2 \le C_L \|dA\|_2^2 \qquad (C_0 A = 0). \tag{V6.38}$$

Here is a local proof. For a link inside a block, close it by tree paths. The resulting loop lies inside the cubical block. It bounds a plaquette two-chain there. Since the block is finite with size depending on L only, the number and multiplicity of plaquettes in a chosen filling are bounded by a constant C_L. The tree integrals vanish, so the original link value is bounded by the plaquette curls in that block.

For a fine link crossing between two adjacent blocks, compare it with a fixed reference crossing link by tree paths. Their difference is a loop inside the union of those two blocks. That union is cubically contractible; choose its plaquette filling by a fixed finite rule. Its line integral is consequently bounded by the curls in the two blocks. Each path in (V6.34) has integral equal to the reference crossing value plus such a controlled error. Since $Q_0 A = 0$, the reference value itself is bounded by those curls. Every crossing link has the same bound. Squaring, summing over links, and using the bounded overlap of the one- and two-block neighborhoods proves (V6.38). This proof also rules out harmonic one-forms: a harmonic or closed zero-mode in $\ker C_0$ must vanish.

At the identity background the Wilson quadratic form, after its prescribed positive group factor and coupling rescaling, is $\|dA\|_2^2$. Equations (V6.37)–(V6.38) therefore verify the three bounds (V6.20) for this actual one-layer constrained Wilson Gaussian. All constants depend on G, L, not on the torus size.

This conclusion persists in a small nonflat chart. Let C_U be the differential of the same finite-path average together with the relative tree-gauge rows, and let $H_{U,\lambda}$ be the second variation of the Wilson Lagrangian at a constrained critical point, including the multiplier term. Finite-path differentiation and (V6.24) give

$$\|C_U - C_0\| + \|H_{U,\lambda} - H_{I,0}\| \le C_L \left(\max_b \|U_b - I\| + \|\lambda\|_\infty \right). \tag{V6.39}$$

The multiplier norm refers to the actual current-scale row convention and prescribed fluctuation rescaling; it is not a fine-scale force divided by an unrecorded power of the mesh. The bounded overlap of the rows is what makes (V6.39) independent of volume.

For completeness, positivity on the perturbed kernel does not require identifying that kernel with the old one. If $C_U z = 0$, let $z_0 = z - R_0 C_0 z$. Then $C_0 z_0 = 0$ and

$$\|z - z_0\| \le \|R_0\| \, \|C_U - C_0\| \, \|z\|. \tag{V6.40}$$

The flat coercivity (V6.38), the uniform upper bound on $H_{I,0}$, and Cauchy–Schwarz show that its quadratic form on z is at least $h_L\|z\|^2/2$ when the right side of (V6.39) is sufficiently small. The additional Hessian perturbation costs at most its operator norm times $\|z\|^2$; reducing the chart radius once more gives

$$C_U C_U^{\mathsf{t}} \geq \tfrac{1}{2}\gamma_L^2 I, \qquad \langle z, H_{U,\lambda} z\rangle \geq \tfrac{1}{4} h_L \|z\|^2 \quad (C_U z = 0). \tag{V6.41}$$

Thus the full saddle matrix has the uniform singular bounds required in (V6.3). Its exact Gaussian covariance and normalization obey (V6.18)–(V6.19) and the nonflat torus estimate (V6.11). A further fixed complex shrink gives the common tube (V6.13).

For application of Section 2, assign every fine coordinate and multiplier in an L-block to its retained root. There are $(5L^4 + 3)\dim G$ saddle coordinates per retained cell and only a fixed current-cell interaction range. The torus side in (V6.9) is then the number of retained cells, not the fine-lattice side. This is the explicit current-scale representation needed for the root count ℓ^{-4}.

The number of integrated physical directions is also explicit. With N_f fine vertices and N_c retained vertices there are $4N_f \dim G$ fine bond variables, $(N_f - N_c)\dim G$ tree-gauge rows, and $4N_c \dim G$ coarse rows. Therefore

$$N - r = 3(N_f - N_c)\dim G = 3(L^4 - 1)N_c \dim G. \tag{V6.42}$$

There is no extra cutoff-dependent global constant in this dimension count. The retained coarse gauge and holonomy variables have not been removed. Gauge normalization, however, needs a separate check. For the actual flat box-and-line differential, $Q_0 d = d_c S_0$, with S_0 the block site average. Its residual infinitesimal gauge space is $S_0\omega = 0$, not the space of root-fixed gauge parameters.

For one block with $m = L^4$ vertices, let B_{tr} be its tree incidence matrix. The square matrix

$$J_0 = \begin{pmatrix} S_0 \\ B_{\mathrm{tr}} \end{pmatrix} \quad \text{satisfies} \quad |\det J_0| = 1. \tag{V6.43}$$

Indeed each maximal minor of the tree incidence matrix is ± 1, with the signs in its cofactor null vector all equal, because that null vector is constant. Expanding in the first row $S_0 = (1/m, \ldots, 1/m)$ gives magnitude one. Consequently, with ordinary Lebesgue delta conventions,

$$\int d\omega\, \delta(S_0\omega)\delta(B_{\mathrm{tr}}\omega - u) = 1. \tag{V6.44}$$

If instead one integrates using induced Euclidean measure on $\ker S_0$, the Jacobian is $\sqrt{m}$; the two conventions are not interchangeable without their coarea factor. For Lie-algebra-valued variables the same identity holds in each component.

For a specified nonflat residual gauge map, the corresponding square derivative $J_U = (S_U;\, P_{\mathrm{tr}} D_U)$ must be retained. If its rows are the prescribed finite local rows and are a sufficiently small perturbation of (V6.43), its determinant is nonzero, local, and identical on copied data. Inserting the gauge delta then carries $|\det J_U|$. The source-zero construction's actual gauge map and its normalization must still be identified with this specified convention. The fact that (V6.43) equals one at the flat point does not make a nonflat determinant equal to one.

In particular the saddle calculation proves the exact Gaussian integral with its explicitly specified constraints. It does not declare that an arbitrary tree-gauge representation already has the original RG normalization. The Haar density and nonlinear coarea density, when present, remain part of the full local integrand and are not discarded by this quadratic calculation.

The coupling power is equally explicit. If the quadratic action is $\beta H/2$ before fluctuation rescaling, then

$$I(\beta H, C) = \beta^{-(N-r)/2} I(H, C). \tag{V6.45}$$

This follows from the restricted Gaussian integral, or from (V6.17). Equations (V6.42) and (V6.45) make the power of β exactly extensive in the retained cell count. It cancels per physical volume when comparing compatible tori at the same coupling. It is not discarded as a uniformly bounded error.

This is a Wilson-specific verification for a raw fixed-L layer, not just an assumed spectral gap. Its limits are equally specific. It does not cover a small-curvature field with arbitrary global holonomy by assuming it is in one small link chart. It does not identify the effective coarse Schur Hessian obtained after an arbitrary number of previous integrations. It does not prove the nonlinear density/large-field bounds needed for a second source-zero stability construction. Consequently it closes the specified fixed-L constrained Gaussian subproblem, not its unverified identification with every original gauge-normalized factor and not O4.

111.6 Full finite Gaussian integration is continuous under the comparison

One need not stop at a finite Taylor jet of a local observable when comparing two Gaussian integrals. Here are two precise finite-dimensional estimates.

Let C_0, C_1 be positive semidefinite covariance matrices on the same s coordinates, and let $F \in C_b^2(\mathbb{R}^s)$. With $C_t = (1-t)C_0 + tC_1$, Gaussian integration by parts, first for nonsingular covariances and then by a vanishing common regularization, gives

$$\mathbb{E}_{C_1} F - \mathbb{E}_{C_0} F = \frac{1}{2} \int_0^1 \sum_{i,j} (C_1 - C_0)_{ij} \mathbb{E}_{C_t}[\partial_i \partial_j F], dt. \tag{V6.25}$$

Hence

$$|\mathbb{E}_{C_1} F - \mathbb{E}_{C_0} F| \le \frac{1}{2} \sum_{i,j} |(C_1 - C_0)_{ij}| \|\partial_i \partial_j F\|_\infty. \tag{V6.26}$$

The whole function F is retained, including all its nonlinear orders. The estimate assumes the same function F on the identified coordinates.

For a merely bounded measurable F, including a characteristic cutoff, suppose instead that the two covariances are nondegenerate and

$$C_0 \ge cI, \qquad \|C_0^{-1/2}(C_1 - C_0)C_0^{-1/2}\| \le \eta \le \tfrac{1}{2}. \tag{V6.27}$$

The Gaussian relative entropy formula, or a direct diagonalized density computation, gives

$$D(N(0, C_1) \| N(0, C_0)) = \frac{1}{2} \sum_{i=1}^{s} \{\lambda_i - \log(1 + \lambda_i)\} \le \frac{1}{2} \sum_i \lambda_i^2, \tag{V6.28}$$

where $|\lambda_i| \le \eta$. The elementary total-variation/entropy inequality then yields

$$|\mathbb{E}_{C_1} F - \mathbb{E}_{C_0} F| \le \|F\|_\infty \sqrt{s}\, \eta. \tag{V6.29}$$

For completeness, the entropy inequality used here is $\mathrm{TV}(P, Q) \le \sqrt{D(P\|Q)/2}$, with $\mathrm{TV} = \frac{1}{2}\|P - Q\|_1$, and $|\int F(dP - dQ)| \le 2\|F\|_\infty$ TV. For a short proof, let $A = \{dP/dQ \ge 1\}$. The log-sum inequality lowers relative entropy to that of the Bernoulli probabilities $p = P(A), q = Q(A)$. As a function of p, that entropy has second derivative $1/[p(1-p)] \ge 4$, and vanishes with its first derivative at $p = q$. It

is therefore at least $2(p-q)^2 = 2\,\mathrm{TV}(P,Q)^2$. Endpoint cases follow by limits. No cutoff-dependent constant is present.

This version keeps hard cutoff functions intact. However, an actual constrained covariance may be singular in the displayed coordinates. One must then exhibit a common independent coordinate system with the lower bound in (V6.27), or use (V6.25) when its smoothness assumptions hold. A lower bound on an unrelated Gaussian covariance is insufficient.

Neither (V6.26) nor (V6.29) provides a lower bound for the nonlinear integral. Therefore passing to the logarithm of a local integral, dividing by it, or using it as a normalized R mass needs its own admitted nonzero-body/lower-bound estimate. These identities do not normalize a still-live field island.

111.7 The wrapping sum, if the complete coefficients are identified

The estimates above have the scale needed by S94. Write the current physical spacing as ℓ_j and the reference side as ℓ_*. Then $n_j = \ell_*/\ell_j$, and (V6.11) gives a Gaussian determinant density error at most

$$C\ell_j^{-4} e^{-c\ell_*/\ell_j}. \tag{V6.30}$$

The factor ℓ_j^{-4} is explicitly the number of roots per physical volume. The Gaussian dimension factor in (V6.18), if constant per cell, cancels exactly between the two volumes before this estimate. Any additional determinant ratio with local bounded matrices has the same tail estimate; the finite number of such factors is included in C.

For $\ell_j = L^{-j}$,

$$\sum_{j\ge 0} L^{4j} e^{-c\ell_* L^j} < \infty. \tag{V6.31}$$

One explicit proof splits the exponential in half: $x^4 e^{-c\ell_* x/2}$ is bounded for $x \ge 1$, and $\sum_j e^{-c\ell_* L^j/2}$ converges. The bound is uniform in the terminal ultraviolet cutoff and in the number of copies q.

The argument tolerates fixed polynomial losses in g_j^{-1} along a logarithmic weak-coupling trajectory. More precisely, for fixed nonnegative u, v,

$$\sum_{j\ge 0} L^{4j}(1+j)^u \exp\left[-\frac{c\ell_* L^j}{(1+j)^v}\right] < \infty. \tag{V6.32}$$

Indeed $(1+j)^v \le L^{j/2}$ for all sufficiently large j; the remaining exponential then beats $L^{4j}(1+j)^u$. This is useful if the matrix condition number or a local derivative bound costs a power of the inverse coupling. It does not tolerate an untracked power of the fine-to- current scale ratio.

For the complete vacuum one would need to apply this reasoning **before** summing its root count to all regular and R births, including all historical arguments and normalization factors. The following is a sufficient formulation, not a claim proved for those births in this note:

$$|b_{j,T_*}(\text{root}) - b_{j,qT_*}(\text{corresponding root})| \le C(1+j)^u \exp\left[-\frac{c\ell_*/\ell_j}{(1+j)^v}\right]. \tag{V6.33}$$

Exact equality on nonwrapping complete carriers plus an exponential carrier tail implies (V6.33). A proved approximate comparison on the same complex integration domains would also suffice. Then (V6.31)–(V6.32), together with the explicit dimension/measure counts, would imply (V6.1).

The point of the matrix theorem is that it proves (V6.33) for a well-specified class of full Gaussian determinant factors and gives an exact replacement expansion. It has not proved that all the nonlinear birth factors in the original construction belong to that class.

111.7.1 A scale-correct compressed-resolvent alternative

The Euclidean condition-number obstruction in Section 4.2 can be avoided for a **current-scale covariance**, under a more specific, verifiable set of operator hypotheses. The following comparison lemma does not assume a bounded Euclidean condition number for the full fine matrix.

Suppose two fine augmented operators K_T and $K_{T'}$ have the same finite local stencils and coefficients on a common lifted ball of current-cell radius $4D$. Assume:

- their inverses have uniformly exponentially summable block bounds from their typed forcing space to their typed solution space;
- smooth current-cell cutoff multiplication is uniformly bounded in those spaces, and the commutators $[K, \chi]$ are bounded between those spaces with no fine-mesh loss;
- the current covariance has the **exact** representation $A_T = P_T K_T^{-1} I_T$, where P_T, I_T have uniformly bounded, fixed-range current-cell stencils that agree on the common region;
- on the real backgrounds under discussion, A_T is a positive symmetric matrix on at most d_0 coordinates per current cell, with $c_0 I \le A_T \le C_0 I$, uniformly on both tori.

Then the inverse comparison gives, for current coordinates x, y in the concentric ball of radius D,

$$|A_T(x,y) - A_{T'}(x,y)| \le Ce^{-cD}. \tag{V6.46}$$

To prove it, apply a local source at y, write $u' = K_{T'}^{-1} I_{T'} e_y$, and choose $\chi = 1$ on the radius-$2D$ ball with support inside the common radius-$3D$ ball. The cutoff solution, transported and extended by zero, obeys the exact identity

$$K_T(\chi u') = I_T e_y + [K, \chi]u'. \tag{V6.47}$$

The commutator is supported in the remote annulus, allowing for the fixed stencil thickness. Applying K_T^{-1} and then P_T near x, exponential block bounds give (V6.46). Summing the polynomial number of current cells in the annulus only reduces the exponential rate. All multipliers and commutators here act in the stated typed spaces; no fine-coordinate trace is taken.

The same hypotheses give an exponential off-diagonal entry bound for A_T. This, the positive spectral bounds, and (V6.46) imply the diagonal logarithm comparison

$$|(\log A_T)(x,x) - (\log A_{T'})(x,x)| \le Ce^{-c'D} \tag{V6.48}$$

for roots in a fixed smaller concentric ball. Here is a proof that records the needed functional-calculus estimates.

Restrict both current matrices to the same ball Ω of radius D around the root. The two principal restrictions inherit the lower bound $c_0 I$. Their entrywise difference is (V6.46), so their operator-norm difference is at most $C(1+D)^4 e^{-cD}$. For either principal restriction or full matrix, conjugation by a small Lipschitz exponential weight changes A in operator norm by at most $C\alpha$, using its exponentially summable entries. Choose α with $C\alpha \le c_0/2$. The resolvents then satisfy

$$|(A + tI)^{-1}(u,v)| \le \frac{C}{c_0 + t} e^{-\alpha d(u,v)}, \qquad t \ge 0. \tag{V6.49}$$

This follows by the Neumann perturbation of the weighted resolvent, and holds equally for the principal restriction.

Writing P for restriction to Ω and $Q = 1 - P$, the exact resolvent identity is

$$P(A + tI)^{-1}P - (PAP + tP)^{-1} = -(PAP + tP)^{-1} PAQ(A + tI)^{-1}P. \tag{V6.50}$$

At the root its right side is bounded by $Ce^{-cD}(c_0+t)^{-2}$: combine (V6.49) with the exponential entry bound for A, and sum the two intermediate current-cell coordinates. Any connecting product reaches Ω^c, so the spare exponential pays the distance D. The difference between the two restricted resolvents has the same integrable $(c_0+t)^{-2}$ bound times their operator difference. Integrate these identities in

$$\log A = (\log C_0)I - \int_0^\infty \{(A+tI)^{-1} - (C_0+t)^{-1}I\}\,dt. \tag{V6.51}$$

This proves (V6.48). Its constants involve the **current** coordinate count, not the number of hidden fine variables.

For periodic copied backgrounds, choose D a fixed small fraction of the reference torus circumference in current cells. Sum (V6.48) over roots and use $\operatorname{Tr}\log A = \log\det A$. The determinant-density error has exactly the form (V6.30), even though the full fine matrix has an unbounded Euclidean condition number.

This lemma isolates the correct way to use an AL/MB type inverse estimate for a determinant comparison. The companion Hessian-associativity note now supplies its clean physical application through HC32–49, as summarized next. The proof does more than invoke finite-cutoff covariance equality: it identifies the compatible minimized action, adds the actual constraint-multiplier Hessian, proves the current spectral bounds, and retains the original current Gaussian normalization. The complete nonlinear/R birth comparison remains a further step.

111.7.2 The application now proved for the clean physical Gaussian

The original raw constraint cannot be put directly into the MB inverse: FI rules out that unrestricted raw negative-Sobolev estimate. HC32–33 instead proves equality of the minimized action as a function of its target for the raw and compatible means. One fiber comparison uses the actual nodal target recovery; the reverse uses the full scalar gauge cocycle onto. Both preserve the original Wilson action. Hence their target Hessians are identical.

Linearization at the physical multiplier adds the genuine term $-\lambda D^2\widetilde{q}$ to the MB first row. HC34–37 proves its uniform bound by composing finite-step bilinear estimates and summing the intermediate lengths, with no depth factor. Its kernels have bounded current range and the same typed cutoff commutator gain. HC38 then reads the exact target Hessian from the multiplier row of this local augmented inverse. The actual current raw outer constraint and original soft gauge remain unchanged.

HC39–44 proves the needed positive current spectral bounds by the actual fixed-step path-gauge/cochain inequality, the compatible fine cochain estimate, and a bounded compatible fine lift. This proof never assigns a bounded Euclidean condition number to the full microscopic matrix. Copied target minima and multipliers solve the same local critical equations on a cover. The common-carrier commutator comparison therefore gives HC45–47 for the target Hessian, the actual current covariance, and its logarithmic determinant.

Finally HC48 returns to the original current Gaussian integral. Its actual pivot determinant, current coordinate count, Haar-origin factors, and fluctuation scaling powers are retained exactly. HC49 checks the original soft-gauge insertion by its one-variable Haar identity, including the original cutoff; its normalization is one fixed factor for every nonroot current site. No fine gauge determinant or global determinant prime is left over from the auxiliary inverse. Thus the complete clean physical Gaussian normalizer satisfies

$$\left|\frac{\log Z_{G,T}}{\operatorname{Vol}(T)} - \frac{\log Z_{G,T'}}{\operatorname{Vol}(T')}\right| \le C\ell_k^{-4}e^{-cT_*/\ell_k}.$$

Its ultraviolet sum is finite by V6.31–32 or TD11–12.

This is not yet a comparison of the entire one-step nonlinear normalizer, the completed R histories, or the full source-zero state. HC50–51 gives an exact density-ratio comparison in common current coordinates, which preserves the clean sharp selectors. For all remaining selectors and reference-dependent R operations, one still has to prove the actual complete-state estimate TD2 or TD5, keeping the original vacuum convention. Those estimates are not inferred from the Gaussian theorem.

112.1 The scale and the defect must refer to the same object

Let $\ell_k = L^k a$, with $L > 1$, and compare a torus of fixed physical side T_* with an integer periodic cover, using the same bare Wilson coupling and the same Haar normalization. Put

$$d_k = T_*/\ell_k, \qquad d_{k-n} = L^n d_k. \tag{TD1}$$

Only levels with $d_k \geq d_0 > 0$ are considered here. A bounded number of terminal levels can be treated separately if a bound at their actual physical scale is already available.

The relevant defect δ_k is a rooted norm of the difference of the **complete source-zero state** after identifying the common local input data. It cannot be just the difference of two Gaussian covariances if

the next operation also uses old regular activities, coupling rows, retained variables, large-field records, or gauge and scalar normalizers. Vacuum increments may either be retained as an additional row or bounded from that same state in each step. A choice which silently resets a torus-dependent scalar at every scale changes the question.

For a primitive Wilson entry whose complete path lies inside the common region, the two evaluations agree exactly. Thus a natural initialization has $\delta_0 = 0$; any nonlocal initialization defect must instead be included explicitly. No equality of the subsequently generated states is being asserted here.

112.2 A bounded step amplification is harmless

Proposition TD-L. Suppose the actual state comparison obeys

$$\delta_{k+1} \le C\delta_k + Ae^{-cd_{k+1}}, \qquad \delta_0 = 0, \tag{TD2}$$

where A, C, c are positive and independent of the ultraviolet endpoint, the number of steps, and the torus cover. Then

$$\delta_k \le AM(C, L, c, d_0)e^{-cd_k}, \qquad M = \sum_{n\ge 0} C^n e^{-cd_0(L^n-1)} < \infty. \tag{TD3}$$

In particular no contraction condition $C < 1$ is needed.

Proof. Iteration of TD2 gives

$$\delta_k \le A\sum_{n=0}^{k-1} C^n e^{-cd_k L^n}. \tag{TD4}$$

Factor out e^{-cd_k} and use $d_k \ge d_0$. The ratio of successive terms in the series defining M is $C\exp[-cd_0(L-1)L^n]$, which tends to zero. This proves the claim. □

A fixed engineering amplification L^{pn} is included by replacing C with CL^p. Thus a finite relevant scaling power does not invalidate this comparison by itself. A factor growing with the number of hidden fine coordinates is a different issue and is not covered by TD2.

112.3 A fixed Hölder response can also suffice

Sharp real selectors can make a Lipschitz comparison of two complete integrands harder than an analytic activity comparison. The geometry allows a weaker modulus.

Proposition TD-H. Suppose $0 < \theta < 1$, $\theta L \ge 1$, and the actual state comparison satisfies

$$\delta_{k+1} \le C\delta_k^\theta + Ae^{-cd_{k+1}}, \qquad \delta_0 = 0. \tag{TD5}$$

Then $\delta_k \le Me^{-cd_k}$ uniformly in the ultraviolet endpoint. For example any constant

$$M \ge \max\{2A, (2C)^{1/(1-\theta)}\} \tag{TD6}$$

has this property.

Proof. If the assertion holds at step k, TD1 and TD5 imply

$$\delta_{k+1} \le CM^\theta e^{-c\theta Ld_{k+1}} + Ae^{-cd_{k+1}} \le (CM^\theta + A)e^{-cd_{k+1}} \le Me^{-cd_{k+1}}. \tag{TD7}$$

The initialization is zero, so induction proves the result. This argument also covers the borderline $\theta L = 1$; it does not rely on a divergent sum of independently estimated forcing terms. □

For $\theta L > 1$, a direct iterated estimate is also informative: concavity gives

$$\delta_k \le \sum_{n=0}^{k-1} C^{(1-\theta^n)/(1-\theta)} A^{\theta^n} e^{-cd_k(\theta L)^n}. \tag{TD8}$$

The powers of C and A in this formula are uniformly bounded, and its geometric-distance sum converges. TD7 is both shorter and valid at equality. If $\theta L < 1$, an old nonzero defect propagated only by $u \mapsto Cu^\theta$ loses the displayed exponential rate; this weaker modulus is not sufficient for this argument.

112.3.1 Fixed scale powers

Suppose $d_k \ge 1$, and replace TD5 by

$$\delta_{k+1} \le C d_{k+1}^q \delta_k^\theta + A d_{k+1}^p e^{-cd_{k+1}}, \tag{TD9}$$

where $p, q \ge 0$. Choose $r \ge \max\{p, q/(1-\theta)\}$. Then the same induction gives

$$\delta_k \le M d_k^r e^{-cd_k}, \tag{TD10}$$

provided $M \ge \max\{2A, (2CL^{r\theta})^{1/(1-\theta)}\}$. Indeed the first power of d_{k+1} is $q + r\theta \le r$, and $\theta L \ge 1$ controls its exponential. A fixed polynomial in a coupling may be treated this way only after the actual running tube has related that coupling to d_k. An unrestricted ultraviolet-dependent constant is not a fixed power of the current physical scale.

112.4 The ultraviolet root count

Suppose the complete original vacuum increment at scale ℓ_k has a per-current-cell intertorus discrepancy bounded by $Cd_k^r e^{-cd_k}$. The discrepancy per physical volume is at most

$$C\ell_k^{-4} d_k^r e^{-cd_k} = CT_*^{-4} d_k^{r+4} e^{-cd_k}. \tag{TD11}$$

When summed over geometric scales with $d_k \ge d_0$, this is bounded uniformly in the fine cutoff. For example, fix the coarsest permitted scale, write $d_k = d_{\text{coarse}} L^n$ in reverse order, and compare with

$$CT_*^{-4} \sum_{n \ge 0} (d_{\text{coarse}} L^n)^{r+4} e^{-cd_{\text{coarse}} L^n} < \infty. \tag{TD12}$$

The bound is uniform when the coarsest dimensionless distance ranges over a fixed compact subinterval of $(0, \infty)$. Consequently a proved TD2 or TD5 for the actual complete state, together with the corresponding actual vacuum-increment bound, is sufficient for S94. This conclusion compares the original normalization on the two tori. It does not introduce a second vacuum convention whose equivalence would itself need a stability theorem.

112.5 A selector estimate that is available, and its exact limit

Here is a finite-current-coordinate estimate relevant to TD5. Let X be a centered real Gaussian vector and $\lambda_1, \dots, \lambda_m$ fixed linear probes. Assume each $\lambda_i X$ has variance at least $v_0 > 0$. For thresholds $r_i \ge 0$, put

$$\chi(X) = \prod_{i=1}^m \mathbf{1}_{\{|\lambda_i X| \le r_i\}}. \tag{TD13}$$

If perturbed probes obey $|\widetilde{\lambda}_i(X) - \lambda_i X| \le \eta$ on an event $\mathcal{A}$, then

$$\mathbb{P}(\chi(X) \neq \widetilde{\chi}(X), \mathcal{A}) \le \frac{4m\eta}{\sqrt{2\pi v_0}}. \tag{TD14}$$

Indeed a change of any one indicator requires $\big||\lambda_i X| - r_i\big| \le \eta$. The corresponding set on the real line has length at most 4η, and the one-dimensional Gaussian density is bounded by $(2\pi v_0)^{-1/2}$. A union bound proves TD14 without independence of the probes. For a weight W with $\mathbb{E}|W|^2 \le B^2$, Cauchy–Schwarz gives

$$\mathbb{E}\big[|W|\,|\chi - \widetilde{\chi}|\,\mathbf{1}_{\mathcal{A}}\big] \le B\left(\frac{4m\eta}{\sqrt{2\pi v_0}}\right)^{1/2}. \tag{TD15}$$

This is a genuine $\theta = 1/2$ comparison for this specified selector class. At $L \ge 2$, TD-H permits precisely that exponent. The complement of $\mathcal{A}$, if present, has to be charged to its actual tail estimate. An existing one-joint-integral positive reserve may supply the weighted moment only if enough of that same reserve remains; it cannot be counted again for every probe.

TD14 does **not** apply automatically to a nonlinear gauge-invariant curvature norm, a selector on a retained old variable, or a maximum over an unbounded number of fine-coordinate probes. For those cases one must prove an appropriate uniform sublevel bound, count the probes at their own retained scale, and retain the complete history and origin banks. Analyticity by itself gives none of these quantitative anti-concentration statements. In particular the sharp real indicators are not differentiated as holomorphic functions.

115.3 A local background coefficient need not have a finite current-variable readset

RG37 defines

$$P_k^{D_0}(U) = U_k(D_0, M'(U)), \tag{ME2}$$

with the literal boundary trace included in the carrier. On the physical graph, its value agrees with the restriction of the global minimum $U_k(V)$. This is an exact and useful identity. It does not say that $U_k(V)|_{D_0}$ is determined by finitely many current variables V in a uniformly bounded neighborhood of D_0. The actual traces on ∂D_0 can carry information from far away. The following computation shows that this issue is present in the Wilson model itself.

115.3.1 A transverse commuting sector of the Wilson Hessian

Work first at the flat configuration on a compatible four-dimensional torus. Choose a nonzero Cartan generator T, and consider transverse variations

$$A_2(x) = a(x_1)T, \qquad A_\mu = 0 \quad (\mu \neq 2). \tag{ME3}$$

The quadratic Wilson action on this sector is a positive constant times

$$\frac{1}{2}\sum_n (a_{n+1} - a_n)^2. \tag{ME4}$$

The harmless color and transverse-volume constants do not change its minimizer. The literal one-step average of I (0.12), linearized at the identity and expressed in physical connection units, is the block mean

$$\frac{1}{L}\sum_{r=0}^{L-1} a_{mL+r} = v_m. \tag{ME5}$$

Here the current mesh has been set to one. To check ME5 directly, the two rooted endpoint contour contributions for a direction-2 row cancel: they have identical direction-1 coordinates and identical direction-2 offsets. The straight line contributes $La(x_1)$. Averaging over the starting block gives ME5 after division by the coarse length L. For a direction-1 row the remaining direction-2 root contribution is odd in the transverse offset and its block average vanishes; the other components vanish as well. Thus this is the original averaging differential on an invariant transverse sector, not a new scalar averaging model inserted in its place.

Gauge degeneracies do not affect this calculation. The curvature in ME3 is transverse, the coarse means fix its constant mode, and the linearized constrained minimizer can be chosen in this invariant sector. Uniqueness of its physical curvature then gives the same curvature response as the full gauge-fixed linearized problem.

Let the one-dimensional periodic lattice have N blocks of length L, with N even. The Euler equation for ME4–ME5 implies that the discrete flux increases linearly within each block. Set

$$f_m = a_{mL} - a_{mL-1}, \qquad g_m = f_{m+1}.$$

For a block multiplier λ_m, one has

$$\begin{aligned} a_{mL+r} &= a_{mL-1} + (r+1)f_m + \frac{r(r+1)}{2}\lambda_m, \\ f_{m+1} &= f_m + L\lambda_m, \\ v_m &= a_{mL-1} + \frac{L+1}{2}f_m + \frac{L^2-1}{6}\lambda_m. \end{aligned} \tag{ME6}$$

Subtract the last identity at adjacent blocks and eliminate λ_m. The result is the exact tridiagonal equation

$$A_L g_{m-1} + B_L g_m + A_L g_{m+1} = v_{m+1} - v_m, \qquad A_L = \frac{L^2-1}{6L}, \quad B_L = \frac{2L^2+1}{3L}. \tag{ME7}$$

In particular, $A_L > 0$ for the odd blocking factors used here and $B_L > 2A_L$. Define

$$\Delta_L = \sqrt{B_L^2 - 4A_L^2} = \sqrt{\frac{L^2+2}{3}}, \qquad \rho_L = \frac{B_L - \Delta_L}{2A_L} \in (0,1). \tag{ME8}$$

The inverse on the infinite block lattice has kernel

$$G_m = \Delta_L^{-1}(-\rho_L)^{|m|}. \tag{ME9}$$

Substituting ME9 into the three-term recurrence proves the identity at every $m \neq 0$; at zero the normalization follows from $B_L - 2A_L\rho_L = \Delta_L$. On the periodic lattice it is its absolutely convergent periodization. Equivalently, for $0 \le m \le N$,

$$G_m^{(N)} = \frac{(-\rho_L)^m + (-\rho_L)^{N-m}}{\Delta_L(1-(-\rho_L)^N)}. \tag{ME10}$$

For the target $v_m = \mathbf{1}_{\{m=0\}}$,

$$g_m = G^{(N)}_{m+1} - G^{(N)}_m. \tag{ME11}$$

When N is even, consecutive $G^{(N)}_m$ have opposite signs, so ME11 is nonzero at every block. By translation, the response of a fixed local crossing curvature to a distant coarse block mean is also nonzero. The exponential size is useful; its exact nonvanishing is what rules out strict finite locality.

For a gauge-invariant example, take the Wilson plaquette energy at that crossing, with normalized trace $\operatorname{tr} 1 = 1$. Its Taylor expansion on the analytic small-field minimum begins with

$$1 - \operatorname{Re}\operatorname{tr} U(\partial p) = c_T \, (a_{n+1} - a_n)^2 + O(\|a\|^3_\infty), \qquad c_T > 0. \tag{ME12}$$

The second derivative with respect to a remote block mean is $2c_T$ times the square of the nonzero response in ME11. A function of a bounded set of current variables could not have this derivative outside that set. Taking larger compatible even tori disproves any uniform finite-readset assertion for this local observable of the original background minimum.

This is not a counterexample to exponential locality, to RG37, or to the unknown LM47 inclusion. It is a counterexample to a particular shortcut: exact locality in the background field does not imply the finite-current-variable hypothesis in LM49.

115.3.2 Consequence for the proposed retention argument

A retained copy of $U_k(V)|_{\partial D_0}$ can be added as a coordinate, but its defining constraint is then

$$h_{\partial D_0} = U_k(V)|_{\partial D_0}. \tag{ME13}$$

ME13 remains nonlocal in V. Adding that copy has not converted the current measure into a kernel with a finite local readset. Nor can its delta constraint be dropped: doing so changes the measure.

Therefore the buffered strategy requires one of the following actual constructions before the elementary mixed map below can apply to a general generated coefficient:

- an exact current-target localization expansion with controlled tails;
- an original preparation whose retained variables include every needed dependency with a proved quantitative kernel bound;
- or the direct same-record domain inclusion ME1.

The first option is carried out in a restricted Wilson sector in Section 6. The others remain to be proved in the full mixed setting.

115.5 Quantitative local coordinates and exact normalization

115.5.1 The original pivot coefficient

Let $n_G = \dim G$. At the flat field, vary only the pivot $v_{p(c)} = \exp(tX)$. In the linearization of (0.12), the central path contribution outside the exponential cancels its negative copy inside the logarithm. The surviving coefficient is the average of translated crossing paths. There are L choices of the longitudinal starting coordinate and one central transverse choice out of L^4 starting sites. Hence, in dimensionless group tangent coordinates,

$$D_{p(c)} q_c\big|_1 = L^{-3}\,\mathrm{Id}\,. \tag{ME21}$$

For physical coarse connection coordinates an additional coarse length factor is present; ME21 is the group/Haar-coordinate version used in this section. Confusing the two conventions would give an incorrect normalization.

There are only finitely many variables in each one-step readset at fixed L. The original means are analytic on their admitted small-loop branch. Thus there is a positive fixed chart size ε_L for which, in the corresponding transported frames,

$$\|L^3 A_c(v) - \mathrm{Id}\,\| \le \tfrac{1}{2}, \qquad A_c(v) = D_{p(c)} q_c(v), \tag{ME22}$$

simultaneously for every row whose complete readset lies in that chart. The constants are independent of the torus volume and of the number of retained blocks. At a regular nonflat real reference, fixed local gauge transports give the same conclusion on a sufficiently small chart; their real adjoint actions preserve the chosen invariant norm. The chart hypothesis concerns the complete readset. A large or unprepared seam is not covered by naming a pivot.

By ME15, the derivative with respect to all kept pivots is exactly block diagonal. In particular, on the real chart,

$$(\tfrac{1}{2}L^{-3})^{n_G} \le |\det A_c(v)| \le (\tfrac{3}{2}L^{-3})^{n_G}. \tag{ME23}$$

The inverse function theorem can be applied separately to each outside row. Its complex neighborhood and each fixed-order jet bound are uniform in the number of rows. With $w = v_{E\setminus(H^+\cup\{p(c):c\in O_H\})}$, this gives

$$v = \Psi_H(h, z, w), \qquad (h, z, w) = (v_{H^+}, q_{O_H}(v), w), \tag{ME24}$$

on a fixed smaller local chart, with

$$\|D^m \Psi_H\|_{\ell^\infty\to\ell^\infty} \le C_{L,m} \quad \text{for each fixed } m. \tag{ME25}$$

The multilinear operator norm is meant for $m > 1$. The bound comes from a uniformly finite row problem with its own pivot, not from an inverse of an extensive matrix. The same type of bound holds for M_H and r_H. No all-depth mixed-history bound is being inferred from this one-step assertion.

115.5.2 Haar coarea with every old factor retained

Use normalized Haar forms and their left-trivialized tangent coordinates. Let μ be the exact finite-cutoff measure at the step under consideration, including its original Wilson density, all history selectors, all old delta constraints, exterior data, and original source-zero normalization. It may be an unnormalized history contribution. Nothing below divides by its mass.

In the chart ME24, the ordinary product-Haar Jacobian for the changed variables is

$$J_{O_H}(v) = \prod_{c\in O_H} |\det A_c(v)|^{-1}. \tag{ME26}$$

If logarithmic coordinates are used instead, all Haar coordinate densities must be included; they are already included in ME26's Haar-form statement. Every old density and constraint is composed with Ψ_H. Old gauge deltas are not declared independent of these variables, removed, or normalized a second time.

Define the mixed output measure by the actual pushforward

$$\nu_H = (M_H)_*\mu. \tag{ME27}$$

For every integrable marked coefficient $F(v_{H^+})$ and bounded test φ,

$$\int F(v_{H^+})\varphi(q(v))\, d\mu(v) = \int F(h)\varphi(r_H(h, z))\, d\nu_H(h, z). \tag{ME28}$$

This is a factorization of the same measure. In particular,

$$(r_H)_*\nu_H = q_*\mu, \qquad \nu_H(1) = \mu(1). \tag{ME29}$$

At a prescribed original target y, ME28 is understood with the remaining original constraint

$$\delta(q_{I_H}(h) - y_{I_H}) \tag{ME30}$$

and with $z = y_{O_H}$. ME30 is indispensable. Solving it using the retained inside pivots supplies the omitted inside Jacobian factors and recovers precisely the original full-pivot expression. At a common configuration,

$$\frac{J_{O_H}(v)}{J_C(v)} = \prod_{c \in I_H} |\det A_c(v)|. \tag{ME31}$$

Thus the coordinate cost of changing which rows are retained is bounded above and below by fixed L-dependent constants per omitted row, with its exact L^{-3n_G} factor. This comparison is local in the changed rows. It is not a conditional partition-function estimate. Rescaling variables by the original coupling carries along the corresponding Haar/coordinate powers on both sides of the exact identity; no source-dependent or island-dependent inverse is created.

The original reference move of XP is a separate identity. Its variables C, source-zero denominator, and coupling partition must be retained exactly if that move is used before or after ME28. The physical map q in this section is not substituted for the matched-reference coordinate by notation.

115.5.3 Why this is not yet an analytic R kernel

Even when Ψ_H is holomorphic, a retained sharp selector becomes

$$\chi(\Psi_H(h, z, w)). \tag{ME32}$$

The expression ME32 has no holomorphic extension merely because its argument does. In a one-dimensional real chart, the smooth positive Wilson weight $\exp[-\beta(1 - \cos t)]$ multiplied by $\mathbf{1}_{|t|<a}$, pushed through the identity coordinate, still has that discontinuous selector. Multiplication by a smooth positive density and a nonvanishing analytic Jacobian does not remove its support boundary.

This example only refutes an algebraic inference; it is not a claim that the complete original R-II kernel lacks analyticity. The original construction handles such restrictions by its specific preparations, real variables, and literal implications among tests. To use the buffered map inside that construction, one must prove that all retained selectors remain on real integration variables, or prove their permitted removal before a complex continuation. The coarea formula by itself proves neither alternative.

Likewise ME23 does not control the change of classical action when an integration order is altered. Dropping positive action terms while retaining an unmatched reference action can create a factor of order $\exp(Cg^{-2}|\mathcal{R}|)$. Such a factor is not hidden in ME31 and is not claimed to be paid by a rare bank. The required Wilson/Gaussian cancellation has to be established on the literal mixed constraint manifold.

115.6 Source jets, contact distributions, and origin ownership

All identities in Sections 3–4 extend coefficientwise to the exact finite-source algebra. A product with label set I is formed in the original variables before the single pushforward. Every partition of I, old source anchor, original lower row, and old birth label is unchanged. There is one original history density and one original origin factor, even if several marks meet the same retained component.

For a whole finite-order contact distribution $\mathcal{C}$, the correct statement is

$$\langle (M_H)_* \mathcal{C}, \psi \rangle = \langle \mathcal{C}, \psi \circ M_H \rangle, \qquad (r_H)_* (M_H)_* \mathcal{C} = q_* \mathcal{C}. \tag{ME33}$$

It follows directly by composition of tests. Delta constraints, selector derivatives and their density partners must remain in the same distribution. Separate products of singular boundary factors are not introduced.

For clarity, let $\|\cdot\|_{-m}$ denote the dual norm of bounded C^m tests with the WCT convention: the supremum of each mixed coordinate partial derivative is bounded, and current coefficients are summed absolutely over their coordinate labels. The row bound ME25 is not, by itself, the column estimate needed for this convention. The one-step geometry supplies that estimate independently. A fixed input coordinate belongs to at most C_L complete one-step readsets, and a fixed tuple of input coordinates belongs to no more of them. Each row has a fixed-order jet bound $C_{L,m}$. The retained identity output adds at most one further first-order entry. Consequently every fixed-order derivative column of M_H has absolute sum bounded by $C_{L,m}$, independently of the volume or size of the retained halo. The WCT chain rule therefore implies

$$\|(M_H)_*\mathcal{C}\|_{-m} \le C_{L,m}\|\mathcal{C}\|_{-m}. \tag{ME34}$$

For a local current or vector density, the ordinary full tensor and divergence-adjoint rules are still necessary. ME34 is a one-step test-pullback bound; it does not prove an incoming norm that is unknown, control the source-zero score, or remove a derivative in a retained fluctuation direction.

There is also an important limit on iterating it. A retained input can affect both its own identity output and a kept coarse row whose readset crosses the buffer boundary. Thus the complete mixed derivative column need not have norm at most one. One cannot say that retention merely pauses a contraction and then multiply only the outside contraction factors. A coherent compatible prefix, including all cross-level endpoint/gauge records, or the original live-duration/energy estimate is needed for an all-history bound.

If two source buffers overlap, their marked carriers may be joined for bookkeeping. Their original failed-seed record is the union of the already present origins. Neither the number of buffers nor the number of source labels increases the number of independent rare components. Every old failed test, S merger, and reserve expenditure remains attached to its original history. ME28–34 are consistent with that ownership rule because none introduces a new weight.

115.7 A quantitative repair in a small commuting Wilson sector

This section proves an exact target-local shell expansion for the same restricted Wilson sector used in Section 2. It is included to make the proposed alternative constructive. It is not an extension to arbitrary nonabelian histories.

Take an SU(2) Cartan generator normalized so that the reduced plaquette action is $1-\cos d_n$, where $d_n = a_n - a_{n-1}$. A corresponding Cartan subgroup in SU(N) only changes a positive overall action constant. In the small commuting branch the original logarithmic means are exactly ME5, not just their linearizations. All path variables commute, and the endpoint cancellations used in Section 2 are exact in logarithms.

The constrained critical equation is

$$\sin d_{n+1} - \sin d_n = \lambda_m \quad (mL \le n < (m+1)L). \tag{ME35}$$

Put $\tau_m = \sin d_{mL}$ and $g_m = \tau_{m+1}$. Within a block,

$$\sin d_{mL+r} = (1 - r/L)\tau_m + (r/L)\tau_{m+1}, \qquad 0 \le r \le L. \tag{ME36}$$

Use the branch of arcsin near zero. Counting how often each fine difference enters the difference of two adjacent block means gives the exact nonlinear equation $\mathcal{H}(g)_m = v_{m+1} - v_m$, where

$$\begin{aligned} \mathcal{H}(g)_m = \frac{1}{L}\sum_{r=1}^{L} r \,\arcsin\big((1 - r/L)g_{m-1} + (r/L)g_m\big) \\ + \frac{1}{L}\sum_{r=1}^{L-1}(L - r)\,\arcsin\big((1 - r/L)g_m + (r/L)g_{m+1}\big). \end{aligned} \tag{ME37}$$

Its derivative at zero is exactly the tridiagonal matrix in ME7. Write $\mathcal{H}(g) = T_L g + N_L(g)$. Since $\arcsin'(z) = 1 + O(z^2)$, on a sufficiently small complex ℓ^∞ ball,

$$\|DN_L(g)\|_{\infty\to\infty} \le CL\|g\|_\infty^2. \tag{ME38}$$

Choose $\eta > 0$ strictly below the decay exponent in ME9. The absolute row sum of the inverse kernel with the weight $e^{\eta\,\mathrm{dist}(m,n)}$ satisfies

$$\sup_m \sum_n e^{\eta\,\mathrm{dist}(m,n)} |(T_L^{-1})_{mn}| \le C_{L,\eta}. \tag{ME39}$$

The bound is uniform in the periodic volume: periodization can be estimated by the infinite-lattice sum. For fixed L its constant is finite. In fact it is $O(L^{-1})$ for a fixed sufficiently small η, uniformly over $L \ge 3$; that strengthening is not needed here. The row of ME38 has range one, so its weighted version differs by only a fixed factor.

For sufficiently small $\|v\|_\infty \le \epsilon_L$, the map

$$g \longmapsto T_L^{-1}(Dv - N_L(g)), \qquad (Dv)_m = v_{m+1} - v_m, \tag{ME40}$$

is a contraction on one fixed small complex ball, with contraction constant below one half. This proves the existence and uniqueness of its analytic solution there. The same weighted inverse proves

$$\left|\frac{\partial g_m}{\partial v_n}\right| \le C_L e^{-\eta\,\mathrm{dist}(m,n)}. \tag{ME41}$$

Here a possible fixed shift by one block from D is absorbed in C_L. To verify that the solution is the actual periodic constrained Wilson critical field, reconstruct its differences by ME36 and arcsin. Summing ME37 over m gives exactly the sum of all reconstructed fine differences. The right side sums to zero, so the reconstruction is periodic. Its adjacent block means differ by the prescribed amounts; an additive constant then makes all means equal to v. On the real small branch the Hessian has positive weights $\cos d_n$, and the mean constraints remove the constant mode. The critical field is consequently the unique local constrained minimum on that convex branch.

Let $\Phi(v)$ be the Wilson energy of one fixed crossing plaquette of this minimum. It equals

$$\Phi(v) = 1 - \sqrt{1 - g_m(v)^2} \tag{ME42}$$

on the same branch. For radius R, keep the target coordinates within block distance R of that crossing and set the other target coordinates to zero. Call this target $v^{[R]}$, and put $\Phi_R(v) = \Phi(v^{[R]})$. Each Φ_R is a function of finitely many actual coarse target variables. All interpolations between consecutive truncations remain in the same small target ball. ME41 and the fact that $\Phi'(g) = O(g)$ give

$$|\Phi_R - \Phi_{R-1}| \le C_L \epsilon_L^2 e^{-\eta R}. \tag{ME43}$$

On a finite torus the expansion terminates when its ball covers the torus; on the infinite lattice it converges absolutely. In either case,

$$\Phi = \Phi_0 + \sum_{R \ge 1} (\Phi_R - \Phi_{R-1}) \tag{ME44}$$

is exact. Banach-valued Cauchy on one smaller target ball gives, for each fixed r,

$$\|D^r(\Phi_R - \Phi_{R-1})\| \le C_{L,r} \epsilon_L^{2-r} e^{-\eta R}. \tag{ME45}$$

ME43–45 supply the quantitative input that a finite readset argument actually needs in this sector: each summand has a finite target readset and the enlarged readsets are paid for by a genuine exponential tail. Applied termwise, the mixed map in Section 3 retains the correct real variables without changing the source.

The restriction matters. The truncation to a fixed zero target is admitted on this common small global chart. It has not been shown to preserve arbitrary curved nonabelian references, mixed-depth constraints, old shrinking R radii, nontrivial global holonomy sectors, or the original prepared source-wall domains. The full LM47 problem cannot be reduced to this sector by a gauge choice.

115.9 Common complex curvature is a separate domain requirement

Real constraint records and a weak norm do not alone give a common complex curvature bound. On a fine four-dimensional lattice of spacing a, take a pure imaginary connection bump supported on a fixed number of neighboring fine bonds, with physical connection amplitude

$$h_a = \delta a^{1-4/p}, \qquad p > 4. \tag{ME46}$$

Its discrete $W^{1,p}$ norm is $O(\delta)$, whereas its physical curvature supremum is of order

$$\delta a^{-4/p}. \tag{ME47}$$

The dimensionless link angle is $ah_a \to 0$; therefore this is not a failure of the local group logarithm. The obstruction is the fine derivative. The example can also preserve a fixed finite collection of current coarse constraints. Choose more pairwise separated commuting bumps than the number of independent coarse rows reading their support, and choose a nonzero vector in the kernel of that finite linear mean matrix. Normalize its largest coefficient to one. The number of bumps is fixed, their supports are disjoint, and at least one curvature spike remains. In the commuting small chart the logarithmic mean is linear, so those constraints vanish exactly. Thus a fixed set of coarse constraints does not upgrade the weak connection norm to a curvature supremum.

This is not a counterexample to an elliptic bound for the true constrained minimum. It explains why that true-minimum equation, its complete collar, and its multiplier bounds have to be used. For a completed regular branch the established LM17/RG39–44 argument supplies the needed physical current and curvature control. At a mixed preparation seam, using the MC inverse on a common complex domain still requires verifying its separate curvature hypothesis on that domain, or deriving

it from the actual minimum equation there. It cannot be inferred from a real selector or from weak analytic closeness alone.

115.10 Exact disposition of the remaining obligations

Assertion	Result of this chapter	Remaining issue
Background-local means current-variable-local	False in general; ME6–12 gives a Wilson linearization counterexample	A genuine target localization expansion is required
Exact finite-halo mixed target	Constructed for the original one-step mean by ME15–20	Arbitrary older mixed prefixes and reference records need their own factorization
Local coordinate singularity at an outside pivot	Removed on the complete regular chart by ME21–25	Unprepared/large-field seams are not included
Preservation of the original normalizer	Exact by ME27–31, with the original inside target deltas retained	No uniform conditional mass or Wilson/Gaussian ratio has been bounded
Finite sources and contacts under this map	Exact by ME28 and ME33; one-step weak norm bound ME34	Incoming contact norms and all-history score/transport bounds are separate
Finite-current-variable localization with tails	ME37–45 in the commuting sector; ME49–59 for noncommuting targets and primitive roots; joint reference-block locality ME60–70 on one global small weak chart	Arbitrary mixed/reference charts, actual mixed admission and the generated-source weak analytic interface remain
Analyticity after buffering	Not implied by holomorphic coarea coordinates; ME32 identifies the selector obstruction	Original selector-compatible preparation/removal must be proved
Common complex curvature	The weak-norm shortcut fails by ME46–47	Verify the actual mixed-domain curvature/current bounds
Stable marked feedback MT36	Not established for all histories	Requires ME1 or a fully bounded same-measure alternative, and the contact induction

The proof of four-dimensional interacting Yang–Mills existence remains incomplete. The unresolved mixed-entry obligation is now more sharply specified: it is not the existence of a formal mixed target or a finite-dimensional inverse. It is the uniform analytic and graded estimate for that target with the original old selectors, all determining records, all current-source derivatives, and the same Wilson normalization through every allowed R transition.

116.2 Fixed data, spaces, and lattice operators

Rescale the protected retained length to one and write $a \leq 1$ for the hidden fine spacing. Fix $p > 8$. Let Ω be a fixed finite union of current cells containing the complete source carrier, every prefix needed by the compatible target rows which read it, and a collar. All fine differences below include their

actual neighboring stencils. Choose a scalar χ equal to one on a one-stencil enlargement of the source carrier, supported strictly inside Ω, with uniformly bounded first physical differences. The complete support of χ and its differentiated stencils stays inside the box. A box of fixed current diameter is sufficient.

Use a fixed admitted reference U, and a common local analysis gauge in which its physical potential is small in $W^{1,p}$. Write $V = e^{ab}U$, with b in a small $W^{1,p}$ ball. The coarse multiplier μ is retained in the same typed space as MC and V7N. The reference multiplier $\bar{\mu}$, row mass κ, and compatible logarithmic target map $q_U(b)$ are the actual ones. Let $E(b)$ be the logarithmic-to-physical tangent factor in that KKT system.

All norms of forms include their physical volume weights. The bilinear invariant pairing is used in holomorphic identities; Hermitian absolute norms are used for estimates. On real fields the bilinear continuation of a squared norm is the original squared norm. For antisymmetric q-forms the pairing is $a^4/q!$ times the sum over all ordered distinct indices.

Let d_U be the literal based forward covariant exterior difference, and δ_U its algebraic backward transpose in the physical form pairing. Their degree-dependent normalizations are chosen consistently with AI6 and AI16. In particular, for the exact holomorphic flux $F(V)$ of AI4,

$$\delta_V F(V) = J_{\mathrm{phys}}(V), \qquad d_V F(V) = B(V). \tag{WH1}$$

Here $B(V)$ is the additive cube defect, not zero. These are identities of the actual lattice operators. They do not use the continuum Bianchi identity or assume $d_U^2 = 0$. Explicitly, $(d_U A)_{\mu\nu} = D_{U,\mu}A_\nu - D_{U,\nu}A_\mu$, $(\delta_U F)_\nu = \sum_\mu D^*_{U,\mu}F_{\mu\nu}$, and $(d_U F)_{\rho\mu\nu}$ is the cyclic sum in AI16. With $D^* = (T^{-1} - 1)/a$, the positive flat Hodge operator is $\sum_\mu D^*_\mu D_\mu$, namely $-\Delta_a$ in AI17's scalar convention.

116.3 Flux and cube defect on the weak ball

There is a fine-mesh-uniform holomorphic map from the weak connection ball to local L^p fluxes. In a common potential chart, put $V_b = e^{av_b}$. Expanding its four-link plaquette and subtracting the linear curl gives

$$\begin{aligned} a^{-2}(P_p(V) - 1) &= (d_a v)_p + G_{+,a}(v)_p, \\ a^{-2}(P_p(V)^{-1} - 1) &= -(d_a v)_p + G_{-,a}(v)_p, \\ \|G_{\pm,a}(v)\|_{L^p} &\le C\|v\|_{L^\infty}\|v\|_{L^p}. \end{aligned} \tag{WH2}$$

The constant can be enlarged on one fixed small potential ball. Every word beyond the linear curl has at least two link potentials and a coefficient a^{n-2} at word order $n \ge 2$. The translations are isometries in the physical L^p sum. The analytic word series and all its Banach derivatives converge uniformly because $a\|v\|_\infty$ is small. The relative-to-absolute potential change $e^{ab}U = e^{av}$ has the same uniform analytic bound by the local BCH series and the $W^{1,p}$ algebra property.

Define $\mathcal{F}$ as the pair of the two plaquette deviations on the left of WH2. Consequently

$$\|F(V)\|_{L^p} + \|\mathcal{F}(V)\|_{L^p} \le C\epsilon, \qquad \epsilon = \|U\|_{W^{1,p},\mathrm{pot}} + \|b\|_{W^{1,p}}. \tag{WH3}$$

The reference notation denotes its potential norm in the declared analysis gauge. It is not the norm of its group elements.

The exact six-face product identity and its inverse version give the holomorphic analogue of LM16. Each monomial in the additive cube defect has at least two plaquette deviations. Its finite transports are bounded on the weak potential ball. Thus

$$\|B(V)\|_{L^{p/2}} \le Ca\|\mathcal{F}(V)\|^2_{L^p}. \tag{WH4}$$

The cubic deviation words satisfy the same bound because a dimensionless plaquette deviation is uniformly small in supremum: the elementary link product bound already gives $\|P(V)-1\|_\infty \le Ca\|v\|_\infty$. No bound on the physical curvature supremum is inferred from that fact. On a fixed current box, $p/2 > 4 > 2$, so WH4 in particular controls the L^2 cube defect. A local formulation can use the same estimate in each complete unit collar.

Equations WH2–4 are holomorphic Banach-space estimates. The notation $\|\mathcal{F}\|$ is a majorant; the actual functions being differentiated are the two holomorphic plaquette deviations. Absolute values are not analytically continued.

116.4 The actual compatible current as an independent analytic readout

For an arbitrary weak pair (b,μ) in this chart, define

$$\mathcal{J}_U(b,\mu) = -E(b)^{-\mathsf{t}} Dq_U(b)^{\mathsf{t}}\{\kappa q_U(b) + \bar{\mu} + \mu\}. \tag{WH5}$$

Every transpose and endpoint logarithm factor in WH5 is the actual one in V7N3. In particular the raw averaging differential is not substituted for the compatible differential. CL17 and the once-only endpoint chart factors give

$$\|\mathcal{J}_U(b,\mu)\|_{L^2(\Omega)} \le C\big(\|\bar{\mu}\|_{\mathrm{row}} + \|b\|_{W^{1,p}} + \|\mu\|_{\mathrm{row}}\big). \tag{WH6}$$

The complete prefix is retained in the read set. The row norm in WH6 includes the finitely many current row cells meeting the complete collar, or the corresponding uniformly local norm with that cell count retained. The same argument gives local higher L^p bounds when their row norm is used. The map WH5 is holomorphic on a fixed weak ball, with uniform fixed-order Banach derivatives.

At a true solution of the physical KKT equations, MC20 eliminates ϕ. The first equation is then exactly

$$J_{\mathrm{phys}}(V) = \mathcal{J}_U(b,\mu). \tag{WH7}$$

For nonzero target forcing the row $q_U(b)$ in WH5 is that forcing. No zero-target simplification is made. Away from a physical solution, WH7 is not asserted. The independent variable μ in this source extension is not relabeled as a proved physical current.

116.5 A precise auxiliary Dirichlet inverse

Extend two-forms on the interior fine cells of Ω by zero. Let P_Ω be restriction to these cells. Define the operator

$$\mathcal{L}_{U,\Omega} = P_\Omega(d_U\delta_U + \delta_U d_U + 1)E_\Omega, \tag{WH8}$$

where E_Ω is zero extension. This is a componentwise zero-exterior Dirichlet realization of the full operator. It is not an unspecified relative or absolute Hodge boundary convention. The reference is defined on every boundary stencil. In particular no boundary trace or harmonic projection is silently omitted.

At $U = 1$, the flat discrete div–curl identity identifies WH8 with the componentwise massive Dirichlet Laplacian, up to the fixed form normalizations. Its bilinear weak inverse has the uniform bound

$$\|\mathcal{L}_{1,\Omega}^{-1} f\|_{H_a^1} \le C\|f\|_{H_a^{-1}}. \tag{WH9}$$

The mass is one in the current units. Zero extension and the flat identity prove the estimate without a fine-grid Poincare constant.

For the admitted reference, $d_U - d_1$ and $\delta_U - \delta_1$ are bounded zeroth-order maps of size $C\|U\|_{\infty,\mathrm{pot}}$. In the weak bilinear form of WH8 their perturbation has norm $C\epsilon$ from H^1_a to H^{-1}_a. Choose the reference ceiling so that this is smaller than half the inverse margin in WH9. A Neumann series, not a positivity claim for complex coefficients, gives

$$\|\mathcal{L}^{-1}_{U,\Omega} f\|_{H^1_a} \le C\|f\|_{H^{-1}_a}. \tag{WH10}$$

It is holomorphic in the reference on that ball. The constants are independent of the fine mesh; the mass also avoids a loss from a large box diameter. For the primitive theorem only fixed protected boxes are needed.

116.6 Exact reconstruction of the differentiated flux on the physical graph

Define the two retained right-hand sides

$$\begin{aligned} r_\delta(b,\mu) &= \chi\{\mathcal{J}_U(b,\mu) + (\delta_U - \delta_V)F(V)\} + [\delta_U,\chi]F(V),\\ r_d(b) &= \chi\{B(V) + (d_U - d_V)F(V)\} + [d_U,\chi]F(V). \end{aligned} \tag{WH11}$$

Each commutator is the literal lattice commutator, including its shifted cutoff value. Define the source reconstruction

$$\mathfrak{F}_U(b,\mu) = \mathcal{L}^{-1}_{U,\Omega} P_\Omega\{d_U r_\delta + \delta_U r_d + \chi F(V)\}. \tag{WH12}$$

The last term is required by the mass in WH8.

On a physical KKT solution, WH1 and WH7 give the exact equalities

$$r_\delta = \delta_U(\chi F(V)), \qquad r_d = d_U(\chi F(V)). \tag{WH13}$$

The two-form $\chi F(V)$ has the required zero-exterior trace. Its support and full differentiated stencils are interior. Therefore

$$\mathfrak{F}_U(b,\mu) = \chi F(V). \tag{WH14}$$

This follows by applying WH8 to the right side and using uniqueness of its inverse. It does not require $d_U^2 = 0$. On the enlarged source carrier, where $\chi = 1$,

$$D_V \mathfrak{F}_U(b,\mu) = D_V F(V) \tag{WH15}$$

for every directional forward or backward derivative required by the chosen source.

Off the physical graph, WH12 remains a well-defined analytic readout. The differences $d_U - d_V$ and $\delta_U - \delta_V$ are zeroth-order multipliers bounded by $C\|b\|_\infty$. The cutoff commutators are bounded by $C\|\partial_a \chi\|_\infty$. Using WH3–6 and WH10 gives

$$\begin{aligned} \|\mathfrak{F}_U(b,\mu)\|_{H^1_a} + \|D_V \mathfrak{F}_U(b,\mu)\|_{L^2} \le C\{&\|\mathcal{J}_U(b,\mu)\|_{L^2} + (1+\|b\|_\infty)\|F(V)\|_{L^2}\\ &+ a\|\mathcal{F}(V)\|^2_{L^4}\}. \end{aligned} \tag{WH16}$$

All terms are bounded on the weak ball. The inclusion $L^p \subset L^4$ uses the fixed local physical volume, not a count of fine points. The covariant derivative on the left is the background derivative plus a bounded zeroth-order term, so the estimate introduces no a^{-1} loss.

116.6.1 Covariance and the role of the box

Under a simultaneous gauge transformation of U, V, all based forms, target rows, and multipliers transform by their actual adjoint endpoint rules. WH5 is the physical fine-link current readout. Thus WH11 transforms as a one-form and three-form pair. The scalar cutoff and zero boundary values are gauge invariant, and WH8 is conjugated by the same two-form action. Consequently WH12 transforms as a two-form. This is joint covariance; the analysis gauge used for estimates is not a change of the source.

The artificial boundary belongs only to this source readout. The physical KKT system, its action boundary and all its constraints are untouched. Equality WH14 is proved using a compactly supported two-form inside the auxiliary box, rather than by claiming that two different Yang–Mills boundary value problems have equal minima. A local source packet uses one declared box and one actual physical field throughout. No root-dependent physical fields are introduced into its reflected pairing.

116.7 The actual quadratic primitive

For a direction ν, let the three directional current pieces of the reconstructed flux be

$$\mathfrak{J}_{\nu;\mu} = D^*_{V,\mu}\mathfrak{F}_{\mu\nu}, \qquad \mathfrak{Q}_{\nu;\mu} = \mathfrak{J}_{\nu;\mu} - \tfrac{1}{3}\sum_{\rho\neq\nu}\mathfrak{J}_{\nu;\rho}. \tag{WH17}$$

Keep the exact source stencil and its existing finite hypercubic symmetrization. Define

$$\widehat{H}_U(f;b,\mu) = a^4\sum_{x,\nu}\sum_{\mu\neq\nu} f(x)\langle\mathfrak{Q}_{\nu;\mu},\mathfrak{Q}_{\nu;\mu}\rangle. \tag{WH18}$$

Equations WH14–15 identify this exactly with the original $H_a(f)$ in AI34–35 whenever (b,μ) is the actual physical solution. For arbitrary weak inputs, WH16 proves

$$|\widehat{H}_U(f;b,\mu)| \le C\|f\|_\infty(\epsilon + \|\bar{\mu}\|_{\mathrm{row}} + \|\mu\|_{\mathrm{row}})^2. \tag{WH19}$$

This is a uniform holomorphic quadratic envelope on the weak ball. At the flat reference, the constant and linear Taylor coefficients vanish. Fixed-order Banach Cauchy estimates follow on a smaller ball, including their actual engineering and canonical powers.

The source has not been set equal to a total-current square. The three separate directional terms in WH17 remain. In particular the non-EOM hypercubic direction detected by AI35 has not been discarded. The construction reconstructs the actual derivative tensor before making that quadratic combination.

116.8 The exact cubic completion needs no differentiated weak curvature

Use the original smeared identity AI32a with its actual flux, transports and cube defect:

$$C^\sharp_a(f) = C^{\mathrm{split}}_a(f) - R_{4,a}(f) - \frac{a^4}{6}\sum f\,\langle B,B\rangle - \frac{a^4}{2}\sum(D_\rho f)\langle T_\rho F,B\rangle. \tag{WH20}$$

Every sum has the original ordered indices and finite shifts. Formula WH20 is just the smeared form of the specified bare observable. The test derivative has not been replaced by a derivative of a weak field.

The product bounds used in AI23–27, now on the complex weak chart, give

$$\begin{aligned} |C_a^{\mathrm{split}}(f)| &\le C\|f\|_\infty \|\mathcal{F}\|_{L^3}^3, \\ |R_{4,a}(f)| + \left|a^4 \sum f\langle B,B\rangle\right| &\le Ca^2\|f\|_\infty \|\mathcal{F}\|_{L^4}^4, \\ \left|a^4 \sum (D_\rho f)\langle T_\rho F, B\rangle\right| &\le Ca\|D_a f\|_\infty \|\mathcal{F}\|_{L^3}^3. \end{aligned} \tag{WH21}$$

These are uniform local physical norms. In particular, for $a \le 1$ and a small fixed ϵ,

$$|C_a^\sharp(f)| \le C\|f\|_{C^1}\epsilon^3. \tag{WH22}$$

The constant, linear and quadratic jets vanish at the flat reference. The complete expression is holomorphic on the weak ball, with uniform fixed-order derivatives. It is not a statement that the quantum quartic mixing in AI32 is negligible: all quartic completion terms in WH20 remain part of this observable.

Restoring a retained length ℓ, the integrated dimension-six factor in WH19 and WH22 is ℓ^{-2}. The physical test norms are rescaled with the same convention as LM18–19. If the entire input strength, including the reference potential and its reference multiplier, is $O(gP)$, the primitive quadratic and cubic bounds are Cg^2P^2 and $C(g^3P^3 + a^2g^4P^4)$, respectively, in their dimensionless own-scale normalization. The fixed small-ball condition $gP \le \eta$ absorbs the last term into the stated cubic envelope. Dividing by the declared g^2 and g^3 then gives the corresponding canonical grades.

This conclusion cannot be obtained by scaling only the fluctuation while leaving a reference of fixed physical strength. In that case the bound retains $\epsilon_{\mathrm{ref}} + gP$, together with $\|\bar{\mu}\|$, and a normalized quadratic source can contain $\epsilon_{\mathrm{ref}}^2/g^2$. The physical and canonical cores must therefore still be tracked separately, as in RG and LM; the auxiliary source lift has not removed that issue.

116.9 What the lift supplies to current-target localization

On the fixed protected reference chart, the KKT solution for a small current target is analytic in the weak variables used above. Compose it with WH18 or WH20. Since their values agree at every physical target in that ball, the resulting target function is exactly the actual original primitive evaluated on the Wilson minimum. All fixed target derivatives agree as well.

The complete weak input read set of WH18 is the auxiliary box plus the full target prefixes needed in WH5. It has fixed current size in this theorem. WH20 has its literal microscopic carrier inside that set. Thus these are legitimate analytic roots for the connected-carrier expansion in the mixed-entry chapter. Their uniform quadratic and cubic envelopes, and their fixed canonical jets, are available before the inverse paths are summed. This avoids treating a curvature supremum as a consequence of an arbitrary weak potential norm.

The conclusion has the following limits.

1. It proves a weak analytic lift for these two primitives, the specified quadratic descendant $Q_a(\Delta_a f)$, the displayed test-derivative terms, and a fixed finite polynomial in the already defined integrated functionals. It does not assert that an arbitrary newly coalesced contact density has this lift. Nor does it prove that every previously generated marked coefficient has such a lift with its previously gained regular age.
2. Equality on the physical graph does not identify an old paired coefficient away from that graph. Any replacement of its source extension must preserve the exact endpoint and prove its own graded source-wall bounds.

3. The reference and all target prefixes must lie in the admitted common chart. An unprepared mixed seam or rough identity row has not been admitted by this auxiliary Hodge equation.
4. This source-only change introduces no new probability measure. It consequently supplies no bound on an internal R score, reference denominator, or changed selector support.

The all-history LM47 and MT36 obligations therefore remain. The new point is narrower and quantitative: the actual primitive H and C-sharp roots can be used in a connected current-target expansion without postulating a fine-curvature bound on every off-shell weak field. Their physical current equation and exact lattice cube identity supply a uniform analytic extension with all stated normalizations retained.

117.2 A boundary-current estimate without a surface measure

Let the finite configuration space have a smooth reference volume $d\lambda$. For a fixed ordered list of smooth vector fields $\boldsymbol{Y} = (Y_1, \dots, Y_r)$, define the whole adjoint distribution by

$$\langle \mathscr{D}_{\boldsymbol{Y}} \nu, \varphi \rangle = \int Y_r \cdots Y_1 \varphi \, d\nu. \tag{C7.1}$$

The order is part of the data. In particular no commutation of Haar derivatives is used. Suppose $\nu = \rho F \, d\lambda$ and its indicated whole directional adjoint are ordinary finite measures, and write

$$\mathscr{D}_{\boldsymbol{Y}} \nu = \rho S \, d\lambda. \tag{C7.2}$$

Smooth ρ, F on a compact chart is one sufficient condition. It is stronger than necessary: fixed measurable factors independent of all differentiated coordinates may remain in both measures. The scalar S contains the complete actual score and every lower term in the ordered adjoint. It need not be replaced by a bare Wilson score if ρ is an effective density.

For an arbitrary measurable selector $0 \le \chi \le 1$, its complete boundary commutator is the distribution

$$\mathcal{B}_{\chi,\boldsymbol{Y}} = \chi \rho S \, d\lambda - \mathscr{D}_{\boldsymbol{Y}}(\chi \nu). \tag{C7.3}$$

This definition uses multiplication of ordinary measures by a bounded function. It does not define a product of exposed surface distributions. Since C7.2 holds on the full reference domain,

$$\mathcal{B}_{\chi,\boldsymbol{Y}} = -\left[(1-\chi)\rho S \, d\lambda - \mathscr{D}_{\boldsymbol{Y}}((1-\chi)\nu) \right]. \tag{C7.4}$$

Thus the represented weak mass generated by this literal ordered vector-field list is bounded by

$$\mathcal{M}_r(\mathcal{B}_{\chi,\boldsymbol{Y}}) \le \int (1-\chi)|\rho| \, (|S| + |F|) \, d\lambda. \tag{C7.5}$$

For the actual canonical Haar fields gX_b or current fluctuation coordinates ∂_{B_b}, this is the WCT representation. If a different smooth field is expanded into that representation, its actual coefficient and finite-jet bounds have to be included. A notation for an arbitrary smooth field does not give those bounds for free.

The same statement holds for Banach-valued source coefficients, replacing absolute values by their declared test-functional norm. A fixed finite sum of tensor/anchor components is summed in C7.5, before taking any additional weak quotient.

Proof. Subtract C7.3 from the identically zero expression $\rho S\, d\lambda - \mathscr{D}_{\boldsymbol{Y}}\nu$. The right side of C7.4 is represented by one order-zero measure and one order-r measure. Their total variations give C7.5. This proof works for sharp selectors and degenerate level sets without a limiting coarea argument. □

The estimate is useful because $1-\chi$ can be a genuine original failure sector with a positive action reserve. A bound on the mass in the *good* sector would generally lose that rarity. Nor does C7.5 estimate an individual formal term such as $(Y_1\chi)(Y_2\chi)\rho F$; that object was never required.

117.2.1 Several tests and one ownership record

If $\chi = \prod_{i=1}^{N} \chi_i$ with zero-one original tests in a fixed order, set

$$\eta_i = \left(\prod_{h<i} \chi_h\right)(1-\chi_i), \qquad \sum_{i=1}^{N} \eta_i = 1-\chi. \tag{C7.6}$$

These are disjoint first-failure sectors. Formula C7.4 is the sum of the corresponding expressions with η_i, and C7.5 is their sum of actual failure-sector integrals. A joint finite-source packet uses its one total degree in each integral. It does not receive a separate reserve for every derivative, label, or displayed contact.

Where the original history already records the first failed test, use that record. An auxiliary ordering in C7.6 is only a partition of the same failure set; it does not create a new failed-test credit. Actual component/record counts still occur in the original large-field bound.

For a fixed list of canonical coordinate anchors there is a useful local form of this identity. Split the original selector into $\chi = \chi_{\mathrm{far}}\chi_{\mathrm{near}}$, where every probe in χ_{far} is independent of every differentiated coordinate. Haar or Euclidean Fubini then gives

$$\mathscr{D}_{\boldsymbol{Y}}(\chi_{\mathrm{far}}\nu) = \chi_{\mathrm{far}}\mathscr{D}_{\boldsymbol{Y}}\nu \tag{C7.6a}$$

as ordinary measures whenever the displayed directional adjoint of ν is one. Apply C7.4 to χ_{near} and the reference measure $\chi_{\mathrm{far}}\nu$. Only failures of the dependent probes are now counted. The other original selectors remain exactly in the integral. One does not replace a boundary at the specified anchors by the event that some probe anywhere on the torus has failed.

This reduction requires actual independence, not small influence. An old reconstruction-dependent selector can have a nonlocal current readset. In that case its full dependence must be retained, as ME's Wilson example demonstrates.

117.2.2 Local extension on a spare admitted domain

A coefficient defined only on an analytic fluctuation chart needs a minor but essential modification. Let ψ be a fixed smooth cutoff supported in a larger admitted chart and equal to one on a neighborhood of the full good selector support, so $\chi\psi = \chi$. Extend $\widetilde{\nu} = \psi\rho F\, d\lambda$ by zero. It is a globally smooth measure in those actual unconstrained fluctuation coordinates. An invariant measurable factor χ_{far} may then be reinserted as in C7.6a; smoothness in the differentiated directions is sufficient. On the good support, $\mathscr{D}_{\boldsymbol{Y}}\widetilde{\nu} = \mathscr{D}_{\boldsymbol{Y}}\nu$, so its C7.3 is the same boundary current. Its C7.4 representation lies in the failure collar inside $\operatorname{supp}\psi$.

All derivatives of ψ are kept in its scalar adjoint packet. On the support of χ_{far}, their support is outside the success set of χ_{near}. The extension collar is therefore charged to these same original failed tests. It is not a new event with a newly postulated rare bank. One may use a product of coordinate cutoffs with a fixed transition width in canonical coordinates; a fixed ordered derivative then hits only its displayed coordinates and has no ambient dimension count. If a block-cutoff convention is used instead, its actual finite block size belongs in the constants. The construction changes no selected physical integral: it extends a density only outside its good support to represent the already defined boundary distribution.

A cutoff function depending on a rough nonlinear curvature probe does not thereby have a bounded derivative. C7.5 differentiates the smooth reference extension, not that probe. What is required is its actual failure-sector energy bound and a genuinely admitted extension collar.

117.4 Strict derivative-column decay for a regular averaging prefix

117.4.1 Physical volume becomes a dimensionless column contraction

Let the actual compatible one-step maps be those of CL3–4, and consider their literal composition CL14. Work on the admitted regular domain where all intermediate reference and relative fields have the stated small physical supremum bound M. The whole retained spacing is rescaled to one; adjacent spacings satisfy $a_{i+1} = La_i$.

Use dimensionless relative link coordinates u_i, so their physical tangents are $v_i = u_i/a_i$. The physical L^1 estimate CL17 gives

$$a_{i+1}^4 \sum_c |v_{i+1,c}| \le (1 + C_L a_{i+1} M) a_i^4 \sum_b |v_{i,b}|.$$

Consequently the absolute first-derivative columns in dimensionless coordinates obey

$$\sup_b \sum_c \|(D\Phi_i)_{cb}\| \le L^{-3}(1 + C_L a_{i+1} M) \le \lambda < 1. \tag{C7.13}$$

Here Φ_i is the actual one-step compatible map in the chosen relative charts. The finite chart factors change the error constant, not the factor L^{-3}. In local common analysis gauges the flat part is a positive normalized box-and-line average; the absolute column error is the CL17 error. Thus a fixed weak ceiling permits, for example, $\lambda = 2L^{-3}$. Color norms and the fixed group dimension are included in C_L.

This is a statement about first derivatives at the actual admitted points. It does not require a cutoff-independent complex disc obtained by varying an arbitrary microscopic bond while retaining the physical curvature bound.

117.4.2 Higher one-step jets have finite absolute kernels

Every one-step map uses a fixed number of links, paths, logarithms, and analytic group means depending only on L. In a common strictly admitted relative group chart, its ordinary dimensionless coordinate jets therefore satisfy, for each fixed $r \ge 2$,

$$\sup_{b_1,\dots,b_r} \sum_c \|D_{b_r} \cdots D_{b_1} \Phi_i^c\| \le H_r, \tag{C7.14}$$

uniformly in the step index and total volume. This follows either by finite-dimensional Cauchy or by differentiating the finite path/mean formulas. A fixed input tuple can affect only a fixed number of output coordinates. The logarithmic mean has its invertible normalized implicit derivative on this chart. The constants may depend on L, r, the group and the spare chart margin.

Equivalently, the absolute multilinear kernel of that jet maps $(\ell^1)^r$ to ℓ^1 with norm at most H_r. No uncontrolled factor is iterated in the next argument: the strict first-column contraction is separated from the higher-jet constants.

117.4.3 The fixed-order composition theorem

Let $Q_n = \Phi_{n-1} \circ \cdots \circ \Phi_0$. For fixed input coordinate anchors define

$$\Gamma_r(n) = \sup_{b_1,\dots,b_r} \sum_c \|D_{b_r} \cdots D_{b_1} Q_n^c\|. \tag{C7.15}$$

Then for every fixed r there is A_r, independent of the depth and ultraviolet cutoff, such that

$$\Gamma_r(n) \le A_r \lambda^n. \tag{C7.16}$$

Proof. For ordinary coordinate derivatives Q_0 is the identity, $\Gamma_1(0) = 1$, and $\Gamma_r(0) = 0$ for $r \ge 2$. The chain rule gives

$$\Gamma_r(n+1) \le \lambda \Gamma_r(n) + \sum_{\substack{\pi \in \Pi([r]) \\ |\pi| \ge 2}} H_{|\pi|} \prod_{B \in \pi} \Gamma_{|B|}(n). \tag{C7.17}$$

For $r = 1$, C7.16 is immediate. Induct on r. Every block in the last sum has size below r, so its value is at most $C_r \lambda^{2n}$. Iterating gives

$$\Gamma_r(n) \le C_r \sum_{t=0}^{n-1} \lambda^{n-1-t} \lambda^{2t} \le \frac{C_r}{\lambda(1-\lambda)} \lambda^n. \tag{C7.18}$$

This proves the claim. Ordered input Haar derivatives are finite linear combinations of ordinary derivatives of orders at most r, with fixed relative-chart coefficients. Convert at the original input once. Converting output coordinate derivatives to invariant derivatives also occurs only at the terminal chart. Their lower terms obey the same bound; no chart constant is multiplied at every depth. □

The complete determining carriers remain those of the actual composed maps. Their physical diameter is bounded by the sum of the original step lengths. C7.16 does not identify this composition with a different macro-average or discard its prefix paths.

117.5 Consequences for whole currents and retained observations

117.5.1 Terminal weak-current transport

Let the current be represented as in WCT2 with positive orders $1 \le r \le m$, and let the original input and terminal couplings be g_{min}, g_{mout}. The ordered chain rule and C7.16 give

$$\mathcal{M}_m((Q_n)_* T) \le C_m \lambda^n \max_{1 \le s \le r \le m} \frac{g_{min}^r}{g_{mout}^s} \mathcal{M}_m(T). \tag{C7.19}$$

The zeroth-order input row is excluded from the decay assertion; its mass has the ordinary pushforward bound. If the actual running tube gives $g_{min} \le c_g g_{mout} \le c_g$, the coupling ratios in C7.19 are bounded

at fixed order. Every term of a positive-order chain rule differentiates the output test at least once; no scalar row is manufactured by differentiating a constant test.

If the declared current packet has exactly the dimension-six engineering conversion L^{2n}, and no further uncounted scale factor, C7.19 gives the engineering-normalized decay

$$C_m(L^2\lambda)^n \le C_m(2/L)^n. \tag{C7.20}$$

This corollary concerns that explicitly normalized whole-current representation. A current tensor with a different physical convention must retain its own conversion. It is not an assertion about the scalar part of a selected EOM source.

The compatible and actual raw terminal observations satisfy the exact gauge relation CL15. Therefore, for any smooth terminal test invariant under the coarse gauge group,

$$\varphi(\mathcal{M}^n(V)) = \varphi(Q_n(V)). \tag{C7.21}$$

Their ordered input derivatives agree on the same admitted open domain. Thus C7.19 is an actual raw-observation weak bound on this physical test class. No equality of the unrestricted raw and compatible differentials is asserted.

117.5.2 A literal compatible stack is bounded; a general mixed record is different

For a stack of observations $(Q_0, Q_1, \dots, Q_N)$ of the same admitted compatible prefix, its derivative columns obey

$$\sum_{n=0}^{N} \Gamma_r(n) \le \frac{A_r}{1-\lambda}, \tag{C7.22}$$

with the finite identity-chart constants included at $n = 0$. This is a quantitative way to retain ancestral target records without paying their number blindly.

For raw stacked observations, the gauge used to relate Q_n to $\mathcal{M}^n$ depends on the field and on n. A terminal gauge-invariant test satisfies C7.21. A joint function linking colors or transporters across depths need not be invariant under the separate restorations. Such a function must keep the common endpoint/gauge records or have its own joint identity proved. C7.22 does not erase those records.

Similarly, retaining identity coordinates in a mixed map does not by itself pause a contraction for free. A retained input can also affect crossing coarse outputs, so its total column may contain the identity plus additional components. A mixed R/reference construction must be identified with the literal prefix stack, or receive a separate bound using its actual retained geometry and origin reserve. The map in C7.13 is not claimed for rough compact coordinates outside its chart.

117.6 Scalar lifting of a fixed finite contact family

There is an exact way to avoid exposing boundary distributions in the first place, useful independently of C7.5. Fix the entire finite source list at the original Wilson cutoff. Expand its joint polynomial in the nilpotent source algebra. Perform any primitive Wilson EOM identities with their complete ordered contacts before inserting a history partition. LM6–9 bounds every resulting compact scalar coefficient with its actual lowered degree and common energy.

Then carry that finite scalar source polynomial coefficientwise through the original source-independent operations. XP6–13 applies to each coarea map, exact constraint, matched reference insertion, and

retained joint integration. Source differentiation never differentiates a selector. If a later field-jet splitting is used, keep its exact scalar projection and complement. Do not rewrite a generated scalar current monomial as a weighted divergence merely in order to discard it.

For a still-live component, retain these scalar numerators jointly and defer field-jet projection until the original completed analytic output exists. The actual R arbitrary-carrying-function inequality then estimates any already admitted scalar envelope using one energy reserve. On a completed output, WCT22–28 controls the genuine outer field derivatives and relative original normalization. This order creates no internal derivative of a sharp cutoff by formal manipulation.

This is an exact representation theorem at every finite history depth. It does not by itself prove uniform bounds on all scalar numerators. In particular it does not turn a physical (U_k(V)|_{D}) readout into a strictly finite current-variable readset. Its complete dependencies and localized tails must be retained in the actual entry analysis.

117.6.1 Why the old two-dimensional center cannot simply be reused

The physical dimension-six center in DN/MT was two-dimensional after the full EOM, contact and descendant structure was accounted for. Unreduced scalar current terms can add further jet directions. Some have zero separated physical Gram norm or contribute only through contacts. Keeping them is legitimate, but deleting them or assigning the old two-by-two inverse to the enlarged system is not.

There are two precise possible continuations. One can build an invariant enlarged finite jet system, keeping its null/contact block and normalizing only the physical quotient after proving the required triangular structure. Alternatively one can prove that an exact observable-level contact elimination performed at the original joint source level covers every later join. Neither conclusion follows from scalar lifting alone. The full higher finite-source projection remains an obligation.

117.6.2 A stronger diagonal norm is not automatically a source algebra

Adding the diagonals already present within each label block to its norm does not solve the multiplication problem. Let

$$\Lambda_\varepsilon(f) = \frac{f(0) - f(\varepsilon)}{\varepsilon}. \tag{C7.23}$$

Its dual C^1 norm is bounded, and a single label has no nontrivial contact diagonal to add. But the coincident-anchor restriction of $\Lambda_\varepsilon \otimes \Lambda_\varepsilon$ is

$$\varepsilon^{-2}[f(0)g(0) + f(\varepsilon)g(\varepsilon)], \tag{C7.24}$$

whose norm diverges. Therefore a partition-completed norm must prove bounded *new cross-block* joins, not merely include old diagonals by definition. Bare joint contact formation before the weak norm avoids this particular false implication, but does not prove the larger renormalized jet system.

117.9 Exact remaining interface

The proved additions in this chapter are the whole boundary/failure representation C7.4–5, its actual compact-Wilson and completed-clean applications C7.9 and C7.12, the exact bare contact-creation grades and compact estimates C7.9a–g, and the actual admitted regular-prefix jet/current estimates C7.16–22. The scalar lifting construction preserves the finite Wilson correlations and primitive Ward identities without exposing selector derivatives. C7.29–34 supplies the exact paired-measure alternative, including its normalization commutators and the hypotheses needed for a quantitative application.

These do not establish the following remaining assertions:

1. A straddling old marked source and its finite jets are admitted on the common real/complex mixed entry and failure collar required by LM47. Scalar lifting does not identify or bound that readout.
2. The still-live mixed transition has either the controlled internal adjoint packet C7.2 on an actual constrained coordinate realization, or the normalization-preserving paired scalar/current representation C7.29–34 with quantitative failure envelopes for both partners. Positivity of its original body proves neither alternative. Its delta constraints and reference denominators remain.
3. All history-dependent mixed targets and gauge-linked old records have the coherent joint jet bound needed to apply C7.19 or C7.22. The regular forward-prefix theorem alone is insufficient.
4. The higher finite-source scalar/current/contact jet basis and its physical quotient have the graded transition bounds needed to instantiate FR. An enlarged list of coefficients is not a proof of those bounds or of a nondegenerate physical normalization.

Accordingly the all-history MT36 estimate and the Yang–Mills existence proof remain incomplete. Section 7 also makes explicit why the current prefix theorem is insufficient for the full higher finite-source problem even on an entirely clean history. The boundary problem has been narrowed: on controlled original Wilson or completed clean reference densities, a new geometric anti-concentration theorem for each sharp cutoff is not needed. The unresolved part is the actual mixed density or its normalization-preserving paired representation, admission and joint-readout structure, together with the higher graded source system.

118.2 A target-forced physical problem on one fixed reference

Use current retained units, so the current spacing is one and the fine spacing is $a_f = L^{-k}$. Let T be a compatible current torus, and T' an integer periodic cover. Let $\bar{U}_T$ be an admitted clean regular physical minimum of the unscaled Wilson action $\mathscr{W}_{a_f}$, with its actual raw target $\bar{V}_T$. Copy both to the cover, including their fixed prescribed gauge representatives and actual determining prefixes. HC's regular-branch uniqueness identifies the copied minimum on T'. Its current multiplier $\bar{\lambda}$ is also copied. No uniqueness of other compact-action or topological branches is assumed.

The multiplier and all background coefficients obey the same fixed small regular ceiling. Fix these backgrounds as references in the compatible map; they are not reselected separately for different target perturbations. Write

$$q_T(b) = \log\Big(\mathcal{B}_{\bar{U}_T}(e^{a_f b}\bar{U}_T)\bar{V}_T^{-1}\Big), \qquad q_T(0) = 0, \qquad Q_T = Dq_T(0). \tag{V7N2}$$

This is MC7 in a uniform regular hierarchy, with the actual complete prefix and relative endpoint gauges. Its physical read set has a fixed current diameter R_0, independent of a_f and k. The output logarithm is the current target chart throughout this chapter.

For a small current target θ, impose $q_T(b) = \theta$. Let $\bar{\mu} = -\bar{\lambda}$, with the sign convention $D\mathscr{W}_{a_f}(\bar{U}_T) = Q_T^{\mathrm{t}}\bar{\lambda}$.

The shifted five equations are

$$\begin{aligned} E(b)^{\mathsf{t}}\{J(W)+d_W\Delta_{\bar U_T}\phi\}+Dq_T(b)^{\mathsf{t}}\{\kappa q_T(b)+\bar\mu+\mu\}&=0,\\ \Delta_{\bar U_T}\psi+S_T^{\mathsf{t}}\nu&=0,\\ -\delta_{\bar U_T}b+\psi+\Delta_{\bar U_T}\phi&=0,\\ S_T\phi&=0,\\ q_T(b)=\theta,\qquad W=e^{a_f b}\bar U_T.& \end{aligned} \tag{V7N3}$$

The unknown is $x=(b,\psi,\phi,\nu,\mu)$. Its zero value solves the zero-target problem. These equations retain the original Wilson objective, actual constraint and current normalization. Only the last right-hand side has changed from the homogeneous target problem. The scalar and gauge rows are not deleted.

118.2.1 The shifted derivative and nonlinear domain

Linearization at zero is exactly the HC multiplier-corrected matrix $\mathbb{K}_{\bar\lambda,T}$. In particular its field Hessian is

$$H_{\log,W}(\bar U_T)-\sum_i \bar\lambda_i D^2 q_{T,i}(0), \tag{V7N4}$$

not the uncorrected Wilson Hessian. The current term already present in the logarithmic Wilson Hessian and the displayed constraint term are distinct.

In the fixed covariant typed spaces of MC3 the equations have the form

$$\mathbb{K}_{\bar\lambda,T}x=I_B\theta-\mathcal{N}_T(x),\qquad \mathcal{N}_T(0)=D\mathcal{N}_T(0)=0. \tag{V7N5}$$

HC gives a uniformly bounded exponentially local inverse. MC's weak nonlinear estimates give, on a fixed small ball,

$$\|\mathcal{N}_T(x)\|_{Y,\mathrm{uloc}}\le C\|x\|^2_{X,\mathrm{uloc}},\qquad \|D\mathcal{N}_T(x)\|_{X\to Y}\le C\|x\|_{X,\mathrm{uloc}}. \tag{V7N6}$$

The additional shifted term is $(Dq_T(b)-Dq_T(0)-D^2q_T(0)b)^{\mathsf{t}}\bar\lambda$. CL17 bounds the actual compatible adjoint in physical L^p uniformly on the fixed small complex potential ball, at every $1\le p\le\infty$. The once-only endpoint logarithmic chart factor is analytic there. Operator-valued Cauchy applied to this adjoint, with $b\in W^{1,p}\hookrightarrow L^\infty$, bounds the displayed second-order remainder by $C\|\bar\lambda\|\|b\|^2_{W^{1,p}}$, already in local L^p and therefore in the field negative type. This is an estimate for the compatible adjoint with its actual endpoint factors; it is not obtained by dualizing a $W^{1,p}$-to-row bound at the wrong exponent. Nor is it an unrestricted raw-prefix adjoint estimate.

All nonlinear vertices in V7N6 are filed in weak form before taking negative norms. They have the actual plaquette, mean-prefix, and incidence carriers, of diameter at most a fixed R_0. This finite current range applies to the equations, not to their inverse. The fixed local analytic radius and constants are independent of the number of fine links.

A once-smaller target ball $\|\theta\|_{\ell^\infty}\le\eta_0$ makes the contraction constant less than $1/2$. It yields a unique solution satisfying $\|x_T(\theta)\|_{X,\mathrm{uloc}}\le C\|\theta\|_\infty$. The same conclusion holds on a fixed complex target ball. The current mesh has only a fixed number of target coordinates per cell, so its uniformly local ℓ^p and supremum bounds are equivalent with fixed constants.

118.2.2 Physical endpoint and curvature

At the unweakened physical endpoint, the fine gauge cocycle still satisfies the exact chain relation with the target $e^{\theta}\bar{V}_T$. Take a physical gauge tangent $d_W\zeta$ with $\mathcal{S}_{\bar{U},W}\zeta = 0$, and insert its logarithmic coordinate representative $\delta b = E(b)^{-1}d_W\zeta$ into the first equation. The full compatible chain makes $Dq_T(b)\delta b = 0$, including the final logarithm derivative, even though $q_T = \theta$ is nonzero. Wilson gauge invariance removes the current term. The remaining equality is $\langle \Delta_W\zeta, \Delta_{\bar{U}_T}\phi\rangle = 0$. Applying MC20 with the retained scalar row annihilates the auxiliary gradient force and gives $\phi = 0$. The remaining equations are the true Wilson KKT equations with this target, in the fixed compatible gauge. HC32–33 restores the original raw target without changing the minimized Wilson action.

The protected Noether and elliptic argument of MC13–18 applies on all current cells in this uniform clean problem. Its coarse source is still bounded by the current-unit multiplier, with a fixed target incidence constant. It supplies the separate curvature and current bounds for the complex physical solution on a smaller target ball. Thus the common complex domain used here is not inferred from a $W^{1,p}$ potential bound alone. No arbitrary independent-current wall is admitted by this argument.

The scalar fields ψ, ν need not vanish for nonzero θ. They implement the relative gauge slice and will be retained in the comparison below. Nor is a comparison of arbitrary raw-gauge potentials asserted: the field comparison is in the common compatible reference frame, and gauge-invariant values have their actual physical meaning.

118.5 A common-selector relative integral estimate with no small-mass division

The previous bound can be integrated in a useful restricted sector without moving its sharp real selectors. Here the common finite coordinate set consists of actual current free pivot variables in a short carrier. Let C_1, C_2 be their covariance marginals from the two original clean Gaussians, with the HC spectral bounds. Let $D_i = C_i^{-1}$, let n be their common dimension, and suppose HC's entry comparison on this protected carrier has size ε_D. Then

$$\|D_1 - D_2\| \le Cn\varepsilon_D, \qquad \left|\log \frac{d\mu_2}{d\mu_1}(B)\right| \le Cn^2\varepsilon_D + Cn\varepsilon_D|B|^2. \tag{V7N19}$$

The first estimate follows from covariance inversion and the common spectral bounds. The second is the actual Gaussian density ratio, including its determinant constant, as in HC50–51. This is a current coordinate count. No fine determinant or hidden fine-coordinate volume occurs.

Let $\chi \ge 0$ be any common measurable selector supported on $\|B\|_\infty \le P$. Keep its exact shape, including any intersections of the original fixed current cutoffs. For real potentials W_1, W_2 on that support suppose $\sup|W_1 - W_2| \le \varepsilon_W$. Define the positive integrals

$$I_i = \int \chi(B) e^{W_i(B)}\, d\mu_i(B). \tag{V7N20}$$

If $I_1 > 0$, the pointwise density comparison gives

$$\left|\log \frac{I_2}{I_1}\right| \le \varepsilon_W + Cn^2(1+P^2)\varepsilon_D. \tag{V7N21}$$

Indeed the ratio of the two positive integrands on their identical support lies between the exponentials of the positive and negative right-hand side. Integrating preserves both inequalities. The proof does not require a lower bound on the selected probability, and does not estimate an absolute numerator error and then divide by a tiny mass.

For the actual classical fluctuation vertices just proved, $\varepsilon_W \le Cg_k n_Y P^3 e^{-cD}$, and HC supplies $\varepsilon_D \le Ce^{-cD}$. Thus V7N21 proves an exponentially small relative normalizer defect for this restricted current integral, with explicit polynomial carrier and cutoff costs. All literal common Haar, gauge, and Jacobian factors may be placed in χe^{W_i} and cancel in the ratio comparison. The original Gaussian determinant constant was already retained in V7N19.

The scope is necessary. V7N20 is an actual finite-coordinate integral only when its integrand reads those coordinates and the original marginalization or component factorization justifies that operation. It is not a permission to replace a globally coupled action by its restriction, to integrate retained delta constraints as free variables, or to declare an arbitrary selected history a normalized positive kernel. At source zero the real positive comparison is valid; a bound on complex normalized logarithms of a different complete integrand requires its own nonvanishing or cluster argument.

118.6 Restricted extension to an actually admitted mixed hierarchy

The same argument has a mixed version, but its hypothesis is the actual MC domain. It cannot be applied merely because a large-field history has a mixed scale label.

Take a separated determining hierarchy satisfying MB27 and MC1–2. Retain its actual dynamic action graph, prescribed exterior data, bare pins, every primary pillar, every inward junction scalar row, and every complete prefix carrier. Suppose its reference $\bar{U}$ is an actual regular constrained critical configuration of that literal Wilson action and raw mixed target, in the MC small branch. Its actual multiplier is small in the adapted norm by MC26. On the cover copy this complete record, including the action boundary class and the reference coefficients. This copying statement is a hypothesis about the actual two records; a newly selected or prepared reference body does not satisfy it by name.

Use the physical target convention of MC7,

$$q_{m,\bar{U}}(b)_c = r_c^{-1} \log[\mathcal{B}_{m,\bar{U}}(e^{ab}\bar{U})_c M_m(\bar{U})_c^{-1}], \qquad q_{m,\bar{U}}(b)_c = \theta_c. \tag{V7N22}$$

The dimensionless forcing norm is $\eta = \sup_c r_c|\theta_c|$. The shifted unknowns have exactly the factors in MC3. In particular the row multiplier has scaled size $r_c^3\mu_c$, and its physical pairing with the target uses the weight r_c^4. The row mass remains $\kappa_c = \kappa_{0,c} r_c^{-2}$.

Here are the additional uniformity checks needed for this extension. On an adaptive tile of constant primary length ℓ, rescale by ℓ as in MB30. Every actual row touching its protected carrier has r_c/ℓ in a fixed adjacent-scale range. The multiplier-Hessian contribution is the literal bilinear form

$$-\sum_c r_c^4\, \bar{\lambda}_c D^2 q_{m,\bar{U},c}(0)[v, w]. \tag{V7N23}$$

After this rescaling it is the HC35 full-prefix form, followed at most by the fixed-complexity mixed top map. Its norm is bounded by $C \sup_c r_c^3|\bar{\lambda}_c|$ in the MB typed operator norm. At a bare identity row the second derivative is zero. Bounded overlap of the complete carriers gives the global bound; no factor counts the number of determining levels. Thus it is a small perturbation of the actual MB mixed inverse. Its cutoff commutator has the same complete-carrier difference factor as HC37.

The nonlinear adjoint remainder is bounded after the same constant tile rescaling using CL17/CL46 and operator-valued Cauchy, as in Section 2.1. Adjacent rescalings differ only by fixed L-dependent factors; no derivative of a discontinuous scale field is taken. This gives the shifted nonlinear contraction in the actual adapted MC spaces, for a fixed small η.

Measure locality in the adaptive cell distance d_{ad} used by MB's summable inverse. Each complete vertex meets a bounded number of adjacent adaptive cells. Hence a weight $e^{\gamma d_{\mathrm{ad}}(\cdot,Y)}$ costs a fixed factor on every vertex, and the proof of V7N8–12 applies in those adapted types. A smooth adaptive cutoff is obtained from the existing MB partition on complete tiles. Its first and second derivatives in each tile's own rescaled coordinates have fixed bounds. Cut off all five components and retain all target, junction, scalar, gauge, and field residual rows. The fixed exterior variables are never introduced as unknowns.

If the protected adaptive collar of Y has distance D_{ad} from its transplant boundary, this proves

$$\sup_{c\in Y} r_c^3|\mu_T(\theta; c) - \mu_{T'}(\theta; c)| \le C\eta e^{-cD_{\mathrm{ad}}}. \tag{V7N24}$$

At the physical endpoint, MC20 still removes ϕ for nonzero target forcing. MC24's actual nodal gauge restores the raw target $e^{r_c\theta_c}M_m(\bar{U})_c$, including at every inward row and pinned boundary. The resulting action is the physical critical value on the admitted regular branch. If the reference is the original regular minimum, local constrained coercivity identifies this continuation with that minimizing branch. No statement about competing rough critical branches is needed or supplied.

In the physical row pairing the exact derivative is

$$DF_{m,T}(\theta)[h] = -\sum_c r_c^4 \langle \kappa_c\theta_c + \bar{\mu}_c + \mu_T(\theta; c), h_c\rangle. \tag{V7N25}$$

The factor $r_c^4 r_c^{-3} r_c^{-1} = 1$ cancels row by row. Consequently, with n_Y the actual number of retained row coordinates in Y,

$$\begin{aligned} &|[F_{m,T}(\theta) - F_{m,T}(0)] - [F_{m,T'}(\theta) - F_{m,T'}(0)]| \\ &\qquad \le C n_Y \eta^2 e^{-cD_{\mathrm{ad}}}. \end{aligned} \tag{V7N26}$$

The cubic analytic remainder obeys the corresponding bound $C\eta_0^{-1} n_Y \eta^3 e^{-cD_{\mathrm{ad}}}$.

There are two substantial limits to this statement. First, MC1–2 includes a separate own-scale curvature ceiling even on the literal bare collar. An arbitrary rough component or arbitrary retained old field has not been shown to satisfy it. Its preparation, constraints, selectors and reference normalization must remain in place until that admission is proved. Second, D_{ad} is not automatically T_*/r_c for an old row. A path through coarser adaptive cells can be shorter. The universal geometric implication only compares it to physical distance divided by the largest cell length encountered. Therefore V7N26 does not by itself provide an old-age exponential $e^{-cL^{k-j}D_k}$ needed to compensate an $L^{4(k-j)}$ fine-cell count. Those scale-sensitive counts and metric lower bounds must be proved for the original body. A regular mixed locality theorem alone cannot replace them.

118.7 Boundary-data locality in one fixed actual reference

The compact-support estimate also controls two arbitrary small target arrays that agree near an observation. This strengthening does not require that either array vanish far away.

Keep the fixed physical reference, shifted multiplier, gauge rows and typed coefficients of Section 2. On T and T', let θ_1, θ_2 lie in one once-smaller target ball of radius $\eta \le \eta_1$. Suppose they agree on a common injectively lifted region U containing a complete prefix collar. The background reference and all of its determining/gauge records are still the literal copies fixed before either target was varied. Let K be a protected observation carrier with distance at least D from the outside of that common region.

Choose a cutoff χ equal to one on a slightly smaller common region and zero outside the available common lift. Cut off all five components of $x_2 = x_{T'}(\theta_2)$, as in Section 3, and transplant them to T. Call the result $x_{2,\mathrm{ext}}$. The cutoff has uniformly bounded typed multiplication norms, so $\|x_{2,\mathrm{ext}}\|_{X,\mathrm{uloc}} \le C\eta$. Shrink η_1 once so this vector and its line segment to $x_1 = x_T(\theta_1)$ remain in the fixed weak analytic ball of V7N6. These auxiliary spliced vectors need not be physical minima; the argument uses the weak analytic equations on them, while the two physical endpoints retain the separate curvature bounds.

Set $z = x_1 - x_{2,\mathrm{ext}}$ and define the actual local linearized remainder

$$\mathcal{M} = \int_0^1 D\mathcal{N}_T(x_{2,\mathrm{ext}} + tz)\, dt. \tag{V7N27}$$

Every row is included in this derivative: the field equation with its shifted multiplier, the scalar and gauge equations, the nodal constraint, and the nonlinear target equation. Its local typed majorant is $C\eta$, with the same bounded complete carriers as $\mathcal{N}_T$. The inverse

$$(\mathbb{K}_{\bar\lambda,T} + \mathcal{M})^{-1} = [I + \mathbb{K}_{\bar\lambda,T}^{-1}\mathcal{M}]^{-1}\mathbb{K}_{\bar\lambda,T}^{-1} \tag{V7N28}$$

therefore has a uniformly summable exponential majorant, after decreasing η_1. Indeed a sufficiently small fixed exponential weight costs at most a fixed factor on each finite-carrier term of $\mathcal{M}$; the weighted Neumann ratio is then less than one half. This proves locality of the full linearization along the segment, rather than assuming it from the unshifted flat operator.

The exact difference equation is

$$(\mathbb{K}_{\bar{\lambda},T} + \mathcal{M})z = I_B\theta_1 - [\mathbb{K}_{\bar{\lambda},T}x_{2,\mathrm{ext}} + \mathcal{N}_T(x_{2,\mathrm{ext}})] =: \mathcal{R}. \tag{V7N29}$$

On every complete carrier where the cutoff is one and $\theta_1 = \theta_2$, the right side is exactly zero: the true equations, reference coefficients and target rows coincide there. Elsewhere it is bounded by $C\eta$ in the uniformly local negative and positive types. This includes the target difference outside U and every cutoff residual; no target delta or scalar residual has been discarded.

Applying the exponential majorant of V7N28 to this separated forcing gives

$$\|x_T(\theta_1) - x_{T'}(\theta_2)\|_{X(K)} \le C\eta e^{-cD}. \tag{V7N30}$$

The inverse majorant has a bounded exponentially weighted row sum, so there is no volume factor for the remote forcing region. The same conclusion holds on one torus for two targets agreeing on U. It also holds in the actually admitted mixed hierarchy of Section 6, using the adapted target norm and D_{ad}; the metric qualification following V7N26 continues to apply.

118.7.1 Physical and gauge meaning of the comparison

V7N30 compares the five unknowns in one fixed compatible reference frame. At either endpoint the proof in Section 2.2 removes ϕ and identifies the physical raw constrained value. The raw-target restoration in MC24 uses the actual node-prefix gauge factors and the prescribed disjoint local node bumps. On a protected carrier these are local analytic maps of the same nearby solution, with their complete prefixes included. Their bounded first derivatives therefore preserve V7N30, after a fixed carrier enlargement.

This gives locality for the resulting gauge-invariant local observables, and for covariant quantities with their common recorded transports. It does not identify arbitrary independently chosen raw axial gauges. A gauge-dependent coefficient requires its actual gauge record; the comparison cannot simply drop it.

In particular, take a fixed target reference and replace all target coordinates farther than R from K by zero. Denote the truncated target by $\theta^{[R]}$. It remains in the same target ball; its physical solution has the actual raw constraints. V7N30 shows convergence on K to the untruncated solution at rate $C\eta e^{-cR}$. A local analytic observable of these fields consequently has an exact telescoping expansion in finite target readsets, with exponential radius tails. The fixed reference is still a parameter of every summand. If it depends on the full varied target or on an outer complex pair, this construction does not make the summands independent of that outer data.

118.7.2 A local density for the full physical action

The multiplier identity provides a useful exact density without tracing the fine action Hessian. In current units define

$$f_{T,c}(\theta) = -\int_0^1 \langle \theta_c, \kappa t\theta_c + \bar{\mu}_c + \mu_T(t\theta; c)\rangle\, dt. \tag{V7N31}$$

Then $\sum_c f_{T,c}(\theta) = F_T(\theta) - F_T(0)$ with the actual current coordinate count. In the mixed case insert the physical weight r_c^4 and the original κ_c in each summand. If the two targets agree in the protected collar of c, V7N30 and the copied base multiplier give

$$|f_{T,c}(\theta_1) - f_{T',c}(\theta_2)| \le C\eta^2 e^{-cD}. \tag{V7N32}$$

Subtracting its first and second target Taylor terms gives a cubic density. Apply one-variable Cauchy to $f_{T,c}(t\theta_1) - f_{T',c}(t\theta_2)$ on the common complex target ball. Its cubic remainder difference is at most $C\eta_1^{-1}\eta^3 e^{-cD}$. Under the original current substitution V7N18a and action prefactor, it has the same $O(g_k)$ canonical nonlinear defect as V7N18, now per rooted current coordinate.

These formulas compare genuine physical functions on the admitted small-target domain. They do not yet establish the original polymer norm of every localized shell. A radius-R ball can contain $O(R^4)$ current cells. A tail e^{-cR} alone does not pay a weight exponential in that whole carrier volume or spanning-tree length. A connected carrier expansion, or a formation estimate with only polynomial costs in this readset, must be supplied before using such shells in that stronger norm.

Finally, none of this permits a change of base reference between the two sides without an estimate. Distinct physical base fields or distinct reference multipliers add coefficient and base-source defects to V7N29. If they are already controlled in the same local typed norm, the displayed resolvent gives the corresponding inhomogeneous comparison. Their control, and membership in MC's common curvature domain, remain obligations for arbitrary original R histories. They have not been proved by the target-locality argument itself.

118.7.3 The current and curvature differences are also physical

The weak field comparison can be upgraded on a further protected collar. This upgrade is useful when a carrying function reads the physical current as well as the connection.

Work first in current units, in one common weak analysis frame of the fixed reference. Write $W_i = e^{ab_i}\bar{U}$. Both endpoints are the genuine constrained critical fields just constructed, and both have MC's separate small complex curvature bound. Let $\mathcal{L}(\theta)$ denote the derivative of the target logarithm, acting on a physical left target tangent; thus $\delta\theta = \mathcal{L}(\theta)\,\delta V_{\mathrm{phys}}$. At the physical endpoint the exact first row is

$$J(W_i) = -\mathcal{Q}^{\mathsf{t}}_{\bar{U},W_i}\mathcal{L}(\theta_i)^{\mathsf{t}}(\kappa\theta_i + \bar{\mu} + \mu_i). \tag{V7N33}$$

All final logarithm and fine endpoint factors are included in this identity. On the complete target carriers of the observation collar, $\theta_1 = \theta_2$. CL17 bounds the compatible adjoint from current ℓ^∞ to fine L^∞; operator-valued Cauchy bounds its first change by the supremum change of the relative potential on that whole carrier. The reference is fixed. The physical multiplier is uniformly small, so V7N30 and $W^{1,p} \hookrightarrow L^\infty$ give

$$\|J(W_1) - J(W_2)\|_{\infty,K} \le C\big(\|\mu_1 - \mu_2\|_{\infty,K^+} + \epsilon\|b_1 - b_2\|_{W^{1,p}(K^+)}\big) \le C\eta e^{-cD}. \tag{V7N34}$$

Here the ceiling denoted by ϵ includes the fixed smaller target ball. The bound concerns the actual physical current, not a separately varied current coordinate.

For clarity the curvature upgrade can be checked directly without postulating an arbitrary high spatial jet. Let X_i be the holomorphic Wilson flux of W_i, let F_i be its logarithmic plaquette curvature, and put $\delta X = X_1 - X_2$, $\delta F = F_1 - F_2$. The analytic lattice curvature map is locally Lipschitz $W^{1,p} \to L^p$ on this common chart, with no inverse fine power. Thus V7N30 first controls δF and δX in L^p.

Subtract the two actual current equations and the two multiplicative cube identities used in LM13–16. A change of a one-edge covariant difference is a zero-order coefficient bounded by $C|\delta A|$, where $A_i = a^{-1}\log W_i$ in the common frame. In particular its apparent division by a is canceled by $W_1 - W_2 = O(a\,\delta A)$. The additive cube defect contains at least two plaquette deviations. Its difference is bounded by

$$|\delta B_{\mathrm{cube}}| \le Ca\epsilon\sum_{\mathrm{cube}}|\delta F| + Ca^2\epsilon^2\sum_{\mathrm{cube}}|\delta A|. \tag{V7N35}$$

Every transport in a cube has bounded length. Varying such a transport costs $a|\delta A|$, explaining the second term; varying either plaquette factor explains the first. The same calculation applies to the inverse plaquettes in the holomorphic flux. The separate endpoint curvature bound is used before any fine difference is estimated.

Apply the flat discrete L^p div–curl estimate with a smooth interior cutoff, and treat the common connection coefficients as the bounded lower-order terms in LM14. This estimate follows from the constant-coefficient discrete elliptic estimates already used in BI/MC; it applies componentwise to complex matrix entries with absolute norms. The subtracted identities and V7N35 give

$$\|\delta X\|_{W^{1,p}(K)} \le C\big[\|\delta J\|_{L^p(K^+)} + \|\delta F\|_{L^p(K^+)} + \epsilon\|\delta A\|_{L^p(K^+)}\big] \le C\eta e^{-cD}. \tag{V7N36}$$

The cutoff thickness is fixed in current units. Its derivative does not cost the microscopic inverse spacing.

On the small plaquette chart the holomorphic flux-to-curvature map is analytically invertible, uniformly in a. Individual endpoint curvature derivatives have the protected MC bound. The derivative of its divided difference produces a term bounded by $Ca^2(|DX_1| + |DX_2|)|\delta X|$, in addition to $C|D\delta X|$. Use $\delta X \in L^\infty$ from V7N36 and the protected covariant endpoint derivative bounds; the common frame's bounded connection changes covariant to ordinary first differences with only lower-order terms. No ordinary derivative is transported through independently chosen rough gauges. The product rule and $p > 4$ give

$$\|F_1 - F_2\|_{W^{1,p}(K)} + \|F_1 - F_2\|_{\infty,K} \le C\eta e^{-cD}. \tag{V7N37}$$

The mixed version is obtained by the constant tile rescaling MC3: multiply the current difference by ℓ^3, curvature by ℓ^2, and use D_{ad}. This keeps the actual own-scale powers.

These estimates justify physical paired readouts on a verified common protected core. They do not prove that every old coefficient is analytic on an arbitrary weak-field ball, nor that a synthesized rough interface has the needed curvature and current bounds. Source Cauchy estimates must still use the source's actual spare paired domain.

118.9 What these estimates do and do not supply to the complete state

V7N9–18 verify a full nonlinear physical primitive of the clean construction. V7N27–37 extend this to changes of remote target data in the same fixed reference and retain true physical current and curvature comparisons. V7N38–40 identifies the original finite relaxed reference and its denominator under a literal copy of its complete graph. These statements do not assume a one-step continuity theorem for an unspecified complete state. The actual equations, multipliers, constraint-Hessian correction, target-forcing rows, cutoff residuals, action prefactor and reference normalizations have been identified in their stated scopes.

They also explain why the fixed-selector Gaussian comparison is useful: when the current coordinate functions are the same, the selector never needs an anti-concentration estimate. V7N21 is a relative bound for a positive integral on that same support. A change in an internal old readout or its target support is a different problem.

The following items still have to be established before applying TD to S94.

1. **A complete inherited activity comparison.** Every original old E_j, lower coupling term, and generated regular activity in the next integral must have a uniform common-domain difference

bound. The nonlinear physical action estimate does not determine their ambient normalization or their off-graph analytic representations.

2. **The actual completed R reference bodies.** WCT27–28 controls the outer-parameter variation of one original component relative to its own fixed reference. It leaves $\log Z(z_0)$ in place. The references, internal retained variables, target deltas, and selector supports on different physical tori must still be compared.
3. **Uniform formation and the original vacuum increment.** One needs a rooted bound for the complete original nonlinear/R formation and the corresponding N_k scalar, with the actual component count. A derivative bound for an individual integrand or for its Gaussian marginal is not yet that bound.
4. **The specified running and terminal convention.** I (1.20)–(1.22) defines its beta function using the infinite-volume polarization tensor, and (2.15) uses that function. In this original scheme the common bare coupling determines one volume-independent running sequence. That convention can be retained literally; there is no need to invent a separate finite-torus coupling extraction. It still leaves the local representatives of the subtracted coupling rows, original N_k, and terminal vacuum convention to be identified in the complete comparison. If a different torus-dependent extraction is used, its discrepancy must instead be retained in the state and estimated.

If these estimates give TD2 or TD5 for the complete state and its vacuum increments, the already proved TD ultraviolet sum applies. No rarity estimate depending only on the coupling replaces the required torus-distance factor. In particular the bound $g_k^{-2} \asymp \log(1/\ell_k)$ makes a fixed inverse-coupling power at most a fixed logarithmic loss in distance, but it does not make an unmatched local vacuum term ultraviolet summable.

Accordingly this chapter advances the physical nonlinear part of the source-zero comparison, while leaving S94 explicitly unproved.

119.1 A comparison that is independent of a small reference mass

Let a single admissible original component, with a fixed internal history record, have the real positive measure

$$d\mu_0(\xi) = e^{-H_0(\xi)}\, d\lambda_\omega(\xi), \qquad 0 < Z_0 = \int d\mu_0 < \infty. \tag{RC1}$$

Here $d\lambda_\omega$ includes the original selectors and constraint measures. It need not have a nonsingular density with respect to an unconstrained product Haar measure. In particular a retained delta record is allowed; it has not been integrated away or replaced by independent variables. A local coordinate formula, when used, includes its original Haar/coarea and gauge factors.

Keep this record fixed and change the actual exponent by a measurable function W:

$$d\mu_1 = e^{W}\, d\mu_0, \qquad Z_1 = \int d\mu_1. \tag{RC2}$$

If W is real and $|W| \le E$ on the retained support, then

$$e^{-E} Z_0 \le Z_1 \le e^{E} Z_0, \qquad |\log Z_1 - \log Z_0| \le E. \tag{RC3}$$

This holds for arbitrary $E \ge 0$; there is no smallness condition on E, and no lower bound on Z_0 apart from positivity. The proof is pointwise comparison of the two positive integrands before integration. In particular the small rarity mass of an R component creates no inverse rarity loss in its logarithmic comparison.

For a scalar or Banach-valued carrying function F, the unnormalized comparison is

$$\left\| \int F\, d\mu_1 - \int F\, d\mu_0 \right\| \le (e^{E} - 1) \int \|F\|\, d\mu_0. \tag{RC4}$$

The same estimate holds for complex W with $|W| \le E$. It uses the one original integral of $\|F\|$, including its once-owned energy or history reserve. It introduces neither a separate integral for every source label nor a sourced inverse mass.

For real W, normalized probability kernels satisfy

$$\|Z_1^{-1}\mu_1 - Z_0^{-1}\mu_0\|_{\mathrm{TV}} \le e^{2E} - 1. \tag{RC5}$$

The total-variation convention is the full variation mass, so two probability measures have distance at most two. The right side can therefore be replaced by its minimum with two.

For complex W, put $q = e^{E} - 1$. If $q < 1$, the exact formula $Z_1/Z_0 = \mathbb{E}_{\mu_0/Z_0} e^{W}$ instead gives

$$\left| \frac{Z_1}{Z_0} - 1 \right| \le q, \qquad \left| \log \frac{Z_1}{Z_0} \right| \le -\log(1-q), \qquad \|Z_1^{-1}\mu_1 - Z_0^{-1}\mu_0\|_{\mathrm{TV}} \le \frac{2q}{1-q}. \tag{RC6}$$

The logarithm is the branch continued from $W = 0$. The last estimate follows by writing $e^{W}/(Z_1/Z_0) - 1$, subtracting one in the numerator, and using $|Z_1/Z_0| \ge 1 - q$. There is no analogous large-E zero-free assertion here. Real source-zero logarithms use RC3; complex raw activities can use RC4 without an inverse.

119.3 The actual distance profile can pay the old-scale counts

The counts in RC10 can be summed against the actual torus-distance profile. Let $\ell_j = L^j a$, $j \le k$, and put

$$d_j = T_*/\ell_j = L^{k-j} d_k, \qquad d_k \ge d_0 > 0. \tag{RC11}$$

Suppose the complete physical carrier at level k contains at most N_k current cells. Each original level has one actual array of retained coordinates of each declared type, so ordinary lattice counting gives

$$N_j \le C N_k L^{4(k-j)}. \tag{RC12}$$

If the implementation introduces additional independent copies of a Gaussian array or repeated reference factors, their multiplicity must multiply this bound. RC12 cannot be used to erase such a copy. Complete collars belong to N_k, and a fixed finite number of coordinate types is included in C.

Let $x_j = g_j^{-2}$, and retain the upper running-tube estimate $x_j \le x_k + b_+(k-j) + C_1$. For fixed nonnegative powers p, q, assume a particular bracket in RC10 is bounded by $A(1+x_j)^p(1+d_j)^q e^{-cd_j}$. Then

$$\begin{aligned} &\sum_{j\le k} N_j (1+x_j)^p (1+d_j)^q e^{-cd_j} \\ &\qquad \le C_{p,q,c,L,d_0}\, N_k (1+x_k)^p (1+d_k)^q e^{-c' d_k}, \qquad 0 < c' < c. \end{aligned} \tag{RC13}$$

All constants are independent of the fine cutoff and the number of stored levels.

To prove this, put $n = k - j$. The running tube gives $(1+x_j)^p \le C(1+x_k)^p(1+n)^p$, and $(1+d_j)^q \le L^{qn}(1+d_k)^q$. After factoring out the right side of RC13, the remaining series is bounded by

$$C \sum_{n\ge 0} L^{(4+q)n} (1+n)^p \exp[-(c-c')d_0 - c d_0 (L^n - 1)]. \tag{RC14}$$

The exponentially growing L^n in the negative exponent dominates the displayed geometric and polynomial factors. This proves uniform summability, rather than replacing the old arrays by a count of current-scale coordinates without justification.

The scale of the distance hypothesis cannot be weakened silently. The mixed inverse in MB, and V7N24–26, decay in the adaptive cell distance. An old row may connect to a distant region through newer, larger cells. That estimate alone supplies no lower bound of the form $D_{\mathrm{ad}} \ge cT_*/\ell_j$. Replacing

e^{-cd_j} in RC13 by a common e^{-cd_k} leaves the series $\sum_{n\ge 0} L^{4n}$ divergent. RC13 therefore applies to old coefficient defects already proved at their birth-level distance, or to a different estimate which explicitly restores the missing scale factors. It does not close the mixed reference-body comparison from adaptive locality alone.

There is a useful distinction in the original R formula. Its old compact Wilson terms remain functions of the same real retained variables when the outer regular pair changes; R-II (1.71)–(1.73) places that outer dependence in the current quadratic form and the displayed inherited analytic potential. With the common running sequence, one need not manufacture a new difference of every old compact action through a changed classical minimum. This removes that particular artificial fine-volume cost. It does not prove the remaining common-record identifications or the distance estimate for the inherited potential and original reference normalizations.

If the coupling at the fixed terminal physical scale is also bounded above in $x = g^{-2}$, the running tube bounds x_k by a constant plus a multiple of $\log(d_k/d_{\text{terminal}})$. Any remaining fixed power of $1 + x_k$ is then absorbed in a further reduction of the exponential rate. This last assertion uses that terminal normalization; it is not a consequence of a lower weak-coupling floor alone. Without it, keep $(1 + x_k)^p$ in RC13.

For a product of original components, positive logarithms add and RC10 is summed over the actual components. For a sum over internal records with a common pointwise bound E, the same positive comparison survives the sum. If the bound depends on the record, its exponential moment in the original positive measure is required; replacing it by an unproved typical-size bound is not allowed.

119.4 The connected scalar formation has an explicit Lipschitz bound

This section concerns a completed original scalar polymer expansion, such as R-II (1.90)–(1.98), after the operations which define its scalar activities have actually been performed. It does not replace a still-live noncommuting operator by a scalar activity. All statements can be applied pointwise in a common admitted set of remaining field arguments and then in their uniform coefficient norms.

Let $\mathcal{P}$ be the finite-cutoff polymer set, with incompatibility $X \not\sim Y$. A polymer is incompatible with itself. Let the two activities be z_0, z_1, and write $\delta z = z_1 - z_0$, $z_t = z_0 + t\delta z$. Choose a majorant $\bar z$ satisfying $|z_t(X)| \le \bar z(X)$ for $0 \le t \le 1$, and a nonnegative weight $a(X)$ such that

$$\sum_{Y \not\sim X} \bar z(Y) e^{a(Y)} \le a(X). \tag{RC15}$$

For example one can use $\bar z = |z_0| + |z_1|$ when the two activities each lie in a fixed half-sized convergence ball. The strict size and locality margins in the original formation estimates serve this purpose; matching the two actual activities to that same ball remains part of their domain check.

The standard connected expansion is

$$\log \Xi(z) = \sum_{n\ge 1} \frac{1}{n!} \sum_{X_1,\dots,X_n} \varphi^T(X_1,\dots,X_n) \prod_{i=1}^{n} z(X_i). \tag{RC16}$$

For hard-core incompatibility the absolute connected coefficient is bounded by the sum of spanning trees with incompatible edges. The rooted tree sum, with root activity omitted, is at most $e^{a(X)}$: truncate tree depth, start with one at the leaves, and use $\exp\{\sum_{Y\not\sim X} \bar z(Y) e^{a(Y)}\} \le e^{a(X)}$ at each root. Monotone convergence proves the full majorant. Hence

$$\sum_{n\ge 1} \frac{1}{(n-1)!} \sum_{X_2,\dots,X_n} |\varphi^T(X, X_2,\dots,X_n)| \prod_{i=2}^{n} \bar z(X_i) \le e^{a(X)}. \tag{RC17}$$

Differentiate the absolutely convergent series RC16 along z_t. Mark its one differentiated polymer and use RC17. Integrating in t gives the explicit estimate

$$|\log \Xi(z_1) - \log \Xi(z_0)| \le \sum_X |\delta z(X)| e^{a(X)}. \tag{RC18}$$

This includes the scalar vacuum contribution of the connected formation. It is not a statement only about nonconstant output activities. No contraction constant less than one is required.

In particular, if

$$\sup_x \sum_{X \ni x} |\delta z(X)| e^{a(X)+\mu d(X)} \le \delta, \tag{RC19}$$

then the vacuum difference per current cell is at most $C\delta$. This conclusion already follows by summing RC18 over possible roots.

A stronger statement in a nonconstant rooted output norm needs an explicit weighted convergence criterion. For example, replace $\bar{z}(Y)$ in RC15 by $\bar{z}(Y)e^{\mu d(Y)}$, with a strict common margin, and include any extra carrier-size moment required by the chosen rooting convention in that same majorant. Assign each connected term to its complete union carrier using the original fixed rooting rule. The union's tree length is at most the sum of the constituent tree lengths, with the fixed joining convention included. The rooted-tree proof then applies to the weighted activities. A carrier-distance moment consumes a specified part of the spare μ; the original coarsening inequality must restore that part at the next step, as in RG26–26a. RC15 without this extra weighted hypothesis proves the vacuum estimate, not an arbitrary locality moment.

For tori of different sizes, RC18 is first applied to matched lifted nonwrapping carriers with their original multiplicities. Wrapped or unmatched carriers are bounded by their original exponential tail, including all complete constraint and history supports. A difference of global pressures is not obtained by pretending the two finite polymer index sets are equal. Exact equality of local rules is sufficient, but the exponentially small input defects in RC19 are also sufficient. The ordinary four-dimensional root sum is then TD11–12.

119.5 What this settles, and what still needs an actual identification

There are three concrete new conclusions.

First, the original positive component has a coefficient-sensitivity bound RC10 with no inverse small-mass cost. Its old Wilson, Gaussian, potential, and normalization terms are separately accounted for. Sharp selectors and singular constraint measures cause no loss when they are literally the same in both integrals.

Second, RC13 proves that a complete stack of old-level coordinate counts can be included in the torus-distance error. One need not postulate an unrealistically current-sized record. The exponential distance dependence, not mere smallness in the coupling, pays this stack.

Third, the completed scalar connected formation has the explicit Lipschitz bound RC18, including its vacuum logarithm, on the original common convergence ball. Thus a missing source-zero comparison cannot be attributed simply to the absence of a generic continuity theorem for a convergent cluster series.

For S94, the remaining assertions are instead specific to the input record and primitive maps:

1. Put the two actual constructions in a common set of retained coordinates with identical constraint records, selectors, gauge conventions, and normalization multiplicities, or prove the required

comparison of their differences. A formal coarea chart alone does not preserve a sharp selector as a fixed outer-parameter function.

2. Establish the required torus defect for the actual mixed-background quadratic forms and nonlinear reconstructed inputs, not only the clean physical ones. The new clean nonlinear minimum comparison supplies one primitive portion of this task.
3. Include every inherited reference body and its absolute scalar normalization difference in ζ_j. WCT's relative logarithm at one fixed reference does not give that difference between two references.
4. Verify the component and old-record counting bounds in the literal matched hierarchy, and use the same coupling/cutoff convention or explicitly compare its changes. If the actual selector set changes, RC3 is not a comparison of those two integrals.

These assertions are not assumed in a claimed proof of S94. This chapter proves the quantitative formation and counting implications after their precise inputs have been identified. The overall four-dimensional interacting existence proof remains incomplete.

119.6 Primary sources

The original fixed real component integral, its positive reference body, and its outer analytic comparison are T. Bałaban, *Large field renormalization. II. Localization, exponentiation, and bounds for the R operation*, CMP 122 (1989), (1.71)–(1.75), printed pp.379–380. Its complete scalar polymer formation is (1.90)–(1.100), printed pp.388–390. The per-component complex comparison constant is retained explicitly immediately after (1.92). Primary paper.

The ordinary Gaussian connected formation and strict activity/locality margins are Bałaban, *Renormalization group approach to lattice gauge field theories II. Cluster expansions*, CMP 116 (1988), Lemma 3 and (2.38)–(2.41), together with the connected series referred to there. Primary paper.

122.2 An auxiliary hierarchy with an exact nonlinear restoration map

Work in current units, with fine spacing

$$a = L^{-N}, \qquad L \geq 3 \text{ odd}. \tag{M8.1}$$

Fix a current packet window Q, a fixed number of complete current collars, and a larger contractible box D. The exterior record contains all fine bonds read by any averaging prefix touching D, as well as the Wilson plaquette boundary. Its current diameter is fixed. The outer fine variables are prescribed data, not additional integration variables.

Choose nested aligned boxes Ω_j, $0 \leq j \leq N$, whose successive boundaries are separated by at least $H_0 L^j a$, with H_0 larger than the complete-prefix and MB parametrix constants. The innermost Ω_N contains the protected packet and all of its short read sets. This fits in a fixed current box because

$$\sum_{j=0}^{N} L^j a \leq (1 - L^{-1})^{-1}. \tag{M8.2}$$

An outer fine collar contains every stencil touching the literal action boundary. All of its dynamic variables have level-zero identity rows; fixed exterior variables are excluded from the unknowns. Use

exactly the MB determining geometry at each interface: finer rows outside the coarser region, coarse rows with at least one endpoint in that region, and any retained inward rows with their actual junction site rows. One may choose no unnecessary inward rows. The original full-prefix convention is retained on every row that is chosen.

Let M_m denote these *raw nonlinear* mixed determining data. Let C_N^D be the list of original current target rows whose complete prefix reads a dynamic variable. All such prefixes lie in the declared D^+ record. There is an exact restoration map from the mixed data and that prescribed exterior record to this entire list:

$$C_N^D = \mathcal{R}_m \circ M_m. \tag{M8.3}$$

Here identity exterior data are included in M_m when the formula is viewed on the complete local record. The other original current targets do not read a dynamic variable, so they are unchanged under every variation in this boundary problem. They need not be added to the source's read set.

To prove M8.3, fill the hierarchy recursively using the actual raw one-step means. At a given interface, a coarse row with an endpoint in the unrefined region is already retained. A coarse row with both endpoints outside that region reads only fine-level rows inside its two refined endpoint blocks and their actual connecting paths. Those rows are retained. Thus its original nonlinear mean can be computed. Repeat this operation through the levels. This is the nonlinear version of the block-filling proof of MB2; it uses the literal composition of the original raw means, not associativity of a newly defined mean. Junction nodes and endpoint gauges are not discarded. The restoration is made in raw variables, before a compatible tangent is estimated.

In particular

$$\ker DM_m(U) \subseteq \ker DC_N(U). \tag{M8.4}$$

A variation supported in the dynamic fine graph and preserving the mixed data preserves every original current target, including a target whose prefix crosses the action boundary. Those crossing prefixes read the prescribed exterior record, which remains fixed.

The auxiliary Wilson functional is the restriction of the same original functional to all plaquettes involving a dynamic variable, with its literal prescribed boundary. Terms involving only fixed variables are constant in this variational problem. No periodic action is substituted for this boundary problem.

122.5 A weak analytic root for a previously generated coefficient

Let $E_{j,X}$ be an actual already generated paired coefficient whose domain is the **full RG paired core**, with its original potential, curvature, independent-current and nested-readout inequalities and their specified gauge conventions. Retain every source anchor, birth coupling, engineering factor, extracted lower row and free companion. Its full determining read set is part of X. Analyticity on an unspecified open set is not enough for the domain claim below. Assume that its physical read set and all nested determining collars lie inside the protected current window. For a packet of short old carriers this is precisely the complete-relative-carrier convention of RG38, with one common auxiliary field for the entire packet.

Define

$$\widehat{E}^{(e)}_{j,X}(b) = E_{j,X}\big(\mathscr{P}_m(b), J_{\mathrm{phys}}(\mathscr{P}_m(b))\big), \qquad e \geq j. \tag{M8.12}$$

The arguments are converted to the original birth units. Write $r = L^{j-e}$. On the protected old read sets, the curvature and current scale as r^2F and r^3J, and a first-order potential norm is bounded by Crt. The finer old constraints have the same action and literal data there. Their complete read sets and boundary traces are protected. Their factorization through the current constraints makes this

field critical for those local old problems. The scalar-slice argument M8.10a–f, now in the old units on their same boundary records, identifies their small critical branches. Thus the nested physical readouts appearing in the stated core are this same field on their actual determining sets. No boundary trace is silently replaced by a periodic value.

Let ρ_j be a strict smaller common radius for those actual core inequalities after their specified fixed comparison constants are included. The potential inequality uses Crt, the curvature inequality uses Cr^2t, and the own-current inequality uses Cr^3t. If an original condition is stated on an allowed nested readout, the preceding identity supplies that same inequality there. The gauge-orbit convention is the original RG one, so the controlled common analysis gauge is an allowed estimate. With precisely this core and these complete protected records, M8.10 gives

$$C_D rt < \rho_j \quad \Longrightarrow \quad (\mathscr{P}_m(b), J_{\mathrm{phys}}(\mathscr{P}_m(b))) \in \mathfrak{C}_j^{\mathrm{spare}}. \tag{M8.13}$$

For separate curvature/current radii use their respective r^2 and r^3 factors; the displayed first-order bound is conservative. The constant includes the declared fixed old subdomain comparisons and every original separate radius ratio. It is independent of microscopic depth. The assertion does **not** cover an old domain with an extra unverified selector, boundary condition, topology restriction or independent-current condition. Such a condition must be checked separately; a single number ρ_j cannot encode an unstated admission theorem.

Therefore M8.12 is a bounded weak analytic function on the actual ball

$$t < t_{j,e} := c_D \min\{t_0, \rho_j/r\}. \tag{M8.14}$$

It is a finite-current-block root: it reads the fine field in the one declared current box and its complete boundary/prefix record. This is not finite dependence on the coarse variables determining that field. No weak extension of the old independent-current function is postulated. The old coefficient is evaluated only on the strong physical pair constructed in M8.9–10, where its existing domain bound applies.

At a physical current field, M8.11 and the genuine nested-minimum identity identify M8.12 with the original deterministic old tag. This holds for a previously generated analytic coefficient, not only for the two primitive H and C-sharp functions. On a nonphysical weak input it is a source-only extension. The unmarked independent-current kernel remains its original function.

The fixed-order derivatives satisfy the usual Banach Cauchy bound

$$\|D^q \widehat{E}^{(e)}_{j,X}(b)\| \le q!\,(2\mathrm{e}/t_{j,e})^q \sup_{\|z\|\le t_{j,e}} \|\widehat{E}^{(e)}_{j,X}(z)\| \quad (\|b\| \le t_{j,e}/2). \tag{M8.15}$$

This retains the actual birth radius. It does not replace it by a fixed clean radius, and it does not assert an absolute fine-link coordinate derivative sum. The latter still needs the actual current transport and contact norm.

The displayed norm is the full multilinear Banach norm; its factor e^q pays polarization. For repeated directions the ordinary one-variable bound has no such extra factor. If a homogeneous source row vanishes to field order m, composition with $\mathscr{P}_m(0) = I$ and its own current preserves that vanishing. Its q-th jet has the usual residual valuation $\max\{m-q, 0\}$. Neither the Cauchy estimate nor this composition changes its birth coupling or declared engineering normalization.

122.6 The actual shrinking radii admit the real current cutoff uniformly

For the completed regular R sector, LM31–34 gives a birth radius comparable to

$$\rho_j = c_j g_j (\log x_j)^q, \qquad x_j = g_j^{-2}, \qquad 0 < c_- \le c_j \le c_+, \tag{M8.16}$$

with the previously chosen fixed logarithmic exponent $q > p_0$. The current real cutoff has $t \le C_D g_e P_e$, $P_e \le C(\log x_e)^{p_0}$. Keep the same weak trajectory hypothesis used in FR and RG:

$$x_j \le x_e + b_+(e-j) + C_1, \qquad x_e \ge x_*. \tag{M8.17}$$

The couplings increase from fine to coarse, so $x_j \ge x_e$. If only a bounded error is allowed in that monotonicity, enlarging the fixed constants gives the same estimates below.

With $n = e - j$,

$$\begin{aligned} \frac{r g_e P_e}{\rho_j} &\le C L^{-n} \sqrt{1 + (b_+ n + C_1)/x_e}\, \frac{(\log x_e)^{p_0}}{(\log x_j)^q} \\ &\le C L^{-n} \sqrt{1+n}\, (\log x_e)^{p_0 - q}. \end{aligned} \tag{M8.18}$$

The supremum of $L^{-n}\sqrt{1+n}$ is finite. Hence increasing one weak floor x_* makes the right side smaller than the strict margin required in M8.13 for **every** birth $j \le e$. Also $g_e P_e \to 0$, so the weak-ball ceiling in M8.14 holds.

The same calculation admits a common current analytic radius $c\rho_e$: the quotient of logarithmic powers is now at most one, and a once-small fixed c pays the supremum of the geometric and relative-coupling factor. A radius is not divided by a response constant again at each age.

For completeness, canonical differentiation on this **constructed** weak root has no hidden inverse birth coupling beyond the old bound. For an order m derivative, define

$$B_{j,e} := \sup_{\|z\| \le t_{j,e}} \|\widehat{E}^{(e)}_{j,X}(z)\|.$$

M8.15 gives

$$g_e^m \|D^m \widehat{E}^{(e)}_{j,X}(b)\| \le m!(2\mathrm{e})^m \left(\frac{g_e}{t_{j,e}}\right)^m B_{j,e},$$

$$\begin{aligned} \frac{g_e}{t_{j,e}} &\le C_D \left\{ \frac{g_e}{t_0} + L^{-n} \frac{g_e/g_j}{(\log x_j)^q} \right\} \\ &\le C_D \left\{ \frac{g_e}{t_0} + L^{-n}\sqrt{1+n}\, (\log x_e)^{-q} \right\}. \end{aligned} \tag{M8.18a}$$

All old coupling powers already in $B_{j,e}$ remain visible. If that bound is a sum of proved graded rows, the estimate is applied to each row separately, retaining its own saved scale factor and derivative lowering. It supplies no new grade for a row whose original bound has not been proved.

This is a quantitative domain comparison for the explicit auxiliary readout M8.12. It is not an assertion that the original mixed R preparation has a weak norm $C g_e P_e$. That latter identification still has to be proved for the actual input of M8.12 if this extension is used there. Nor does M8.18 provide a weak trajectory; it retains M8.17 as the previously declared hypothesis.

122.7 Saved regular grades remain visible in this extension

For a completed contractible **pure-clean one-label scalar D6 packet in the scope of RG55**, use one common $\mathscr{P}_m(b)$ for all short old carriers, roots, path shifts and pointed descendants. The field hypotheses required by that calculation are the common controlled $W^{2+s,p}$ potential, $W^{1+s,p}$ curvature, $C^{1,\beta}$ potential for $s < \beta < 1 + s - 4/p$, and bounded physical current, all on complete

protected carriers; the actual current multiplier and its raw/compatible Ward conversion are also needed. M8.10 and MC18 give these properties, and M8.9 together with M8.10e gives the multiplier. The old coefficient must retain RG's physical and canonical paired bounds, its completed geometric symmetries, its exact dimension-four family, all pointed descendants, and the source-moment grading RG53–54.

On the common protected window, every current constraint has length one. The physical equation, finer criticality, prefix adjoints and boundary trace there are therefore the ones used in RG39–46. The auxiliary mixed outer boundary does not enter those short identities. In the comparison to old coarse kernels, perform the same fixed CE extension and remote residual construction RG45–51 on the **same old global graph** as the coefficient. Its transition is outside all protected short read sets; it does not copy the new auxiliary boundary into the old graph. Hence the old comparison uses exactly the established weak seam bounds and exponential remote-residual estimate. The entire short kernel family sees one field. Long original carriers and long derivative kernels retain their separate tail estimates and original admission conditions. No new coefficient is substituted.

Consequently the already established RG55 bound becomes the genuine weak-root bound

$$\|\widehat{R}_{j\to e}(b)\|_{\mathfrak{T}} \le C_D A_j \left\{ r^s a_2^{(d)}(j) t^2 + r^\beta a_3^{(d)}(j) t^3 + r^2 \alpha^{d-4} g_j^{-d} t^4 \right\}, \qquad d = 2, 3. \tag{M8.19}$$

Here α is the actual fixed clean radius in the established RG55 hypothesis. M8.18 also proves admission for the stated shrinking R birth radii, but that domain result does not prove an RG55 grade for an arbitrary R-generated or contact coefficient. If a separate graded theorem is established on a shrinking radius α_j, its remainder must keep α_j^{d-4}; it cannot borrow the fixed α in M8.19. All coupling ratios, old-root counts and engineering factors in the proved displayed estimate are those in RG55. The source norm is the stated test-functional norm, not a fictitious strict-age anchor ℓ^1 norm.

Each homogeneous row in M8.19 is estimated before the one joint fluctuation reserve is used. The old label j remains fixed. The factor r^s, r^β, or r^2 was not paid by a new failure event and is not restarted at entry. A finite field jet can be extracted from M8.19 on the actual physical or canonical ball; its derivative lowering and its literal radius power remain in its own row. An old projected coefficient with vanishing field jets keeps that vanishing under composition with a map fixing the flat point. Every removed lower term is still a separate exact coefficient.

For the free companion, linearize the **same** auxiliary problem. On a free physical current minimum, its finer mixed constraints again factor the current constraint, so the free local retraction is the identity there. Thus composition with the full physical free target map has the original free companion exactly. Away from that graph, the weak extension uses its own linearized auxiliary readout. No new $O(g)$ discrepancy estimate for arbitrary old independent-current extensions is inferred merely from this identity.

The weak root can now be substituted into the ME target-tree algebra on any already admitted common regular chart. The root has finite current carrier and bounded Banach Taylor coefficients. Applying the positive majorant separately to the three rows in M8.19 preserves their old factors. The finite current-coordinate leaf bound is used only for target derivatives; it does not turn M8.15 into fine Haar derivative control.

This closes the weak-root interface for the **previously generated completed pure-clean scalar packet with its already proved RG grade**. It does not close the higher-contact grade, a selected nonsymmetric history class, or an original mixed input not admitted to this chart.

123.2 The field to be tested is the prepared composite

R-II, printed pp. 373–375, (1.61)–(1.65), prepares a nested field retaining u_k, the determining maps, $M'(U)$, $M''(U)$ and the relaxed target function of $M_k(U)$. Its range U_k' is defined in the uninspected R-I (1.64)–(1.69). Formula (1.71), p. 379, compares density factors; it states no current-unit weak norm for our marked readout.

Source: T. Bałaban, *Large field renormalization II*, CMP **122** (1989), 355–392, original PDF.

Let

$$\Gamma^{R}_{\omega,\mathbf{s}}(z,\xi) \tag{AE8.1}$$

denote precisely that composite for a fixed original history record ω. Here z is the outer pair, ξ denotes the original retained/integrated variables, and $\mathbf{s}$ contains the actual preparation/localization walls.

This notation is a name for the original composite, not a new definition of it.

The completed old physical source readout in LM47 is a further map,

$$W^{\mathrm{old}}_{j,\omega,\mathbf{s}} = P_{\mathrm{old}}\Big(q_{\mathrm{old}}(\Gamma^{R}_{\omega,\mathbf{s}})\Big)\,. \tag{AE8.2}$$

Its physical current, boundary traces, lower projections and source partners remain part of the record. Estimates for a raw variable, for Γ^{R}, and for W^{old} are three different statements. A valid use of M8 must specify which actual field is its weak input and retain the exact naturality identity used to change the marked extension.

The exact ME retention map does not decide this analytic issue. If retained coordinates are written as $M_H(v) = (h, z)$ and the original inverse chart is Ψ_H, an old selector is still $\chi_\omega(\Psi_H(h, z, w))$. Its support has not become a product domain, and adjoining an exact boundary trace does not prove that trace has a uniform current-unit weak norm.

123.3 A necessary curvature estimate for any proposed M8 input

Fix a current-size contractible record D^{+} of the size used in M8, with complete prefixes and prescribed boundary data. Let Ξ be the specific actual prepared field proposed as input; for example $\Xi = \Gamma^{R}_{\omega,\mathbf{s}}(z, \xi)$, if that is the source composition being justified.

Suppose one has an allowed common gauge in which

$$\Xi^{h} = e^{aA}, \qquad \|A\|_{W^{1,p}_{a}(D^{+})} \le t, \qquad p > 8, \quad t < t_0. \tag{AE8.3}$$

For complex fields the gauge and inverse must have the uniform adjoint bounds of the declared chart. All original records are transformed jointly; a prescribed exterior is not set to the identity by fiat.

The actual finite plaquette product gives the uniform weak bound

$$\|F_a(e^{aA})\|_{L^{p}_{a}(D)} \le C\left(\|D_a A\|_{L^{p}_{a}(D^{+})} + \|A\|_{\infty}\|A\|_{L^{p}_{a}(D^{+})}\right) \le C_D(t + t^2). \tag{AE8.4}$$

Here F_a is the original small-branch plaquette curvature. For the first inequality, telescope the four exponential factors and retain their first lattice differences. The linear part is the discrete curl; every remaining term has at least two field factors, or a bounded analytic coefficient multiplying one difference. The small bound on aA, obtained from discrete Sobolev embedding, makes the logarithm branch common. Thus the estimate has no inverse fine-mesh constant. This is the same finite-word weak estimate used in the classical reconstruction.

For a real field and a compact gauge, the curvature norm is unchanged. On the stated complex gauge chart it changes by at most its bounded adjoint constant. Consequently a necessary condition for AE8.3 is

$$\|F_a(\Xi)\|_{L^{p}_{a}(D)} \le C_D t. \tag{AE8.5}$$

This condition is weaker than M8 admission. In particular it does not supply a common holomorphic gauge or its compatibility with the boundary and junction records.

123.5 What own-scale regularity would have to contribute

Suppose an original range theorem supplies only estimates $c_\Delta(\Xi) \le \epsilon_\Delta$. Even granting precisely those bounds, their consequence is only

$$\|F_a(\Xi)\|_{L^p_a(D)} \le \left(\sum_{\Delta \subset D} \sigma_\Delta^{4-2p} \epsilon_\Delta^p \right)^{1/p}. \tag{AE8.9}$$

The right side is not a uniform current-unit smallness estimate until this actual weighted sum is controlled. For one cell of length $\sigma = L^{-n}$, its contribution to the upper bound is $\epsilon_\Delta L^{(2-4/p)n}$. For $p > 8$, the power $2 - 4/p$ is strictly positive. Fixed logarithmic improvements or a relative coupling factor with polynomial growth in n do not cancel that geometric conversion cost.

Two concrete ways an original first-entry estimate could close this test are now visible:

1. Prove from the original history, old selectors and complete source collar that every cell read in D^+ has $\sigma_\Delta \ge c_* > 0$, with a constant independent of the history depth, and that its actual own-scale moments have size $C g_e P_e$.
2. Prove an actual additional estimate $c_\Delta(\Xi) \le g_e P_e \sigma_\Delta^{2-4/p} w_\Delta$, with $\sum_\Delta w_\Delta^p \le C_D$, or an estimate with the same quantitative effect.

The second alternative needs actual decay or compatibility of the prepared field; norm terminology alone cannot supply it. The first alternative is a statement about the **complete** source record, including

the new auxiliary boundary and every old prefix. Shortness of a coefficient's original microscopic support does not establish it.

Neither alternative has been derived for the whole original first-entry class from the inspected formulas. Conversely, the scaling calculation does not prove that an actual permitted prepared readout violates AE8.8. No off-shell rough field is being offered as a counterexample.